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7

Responses of Musculoskeletal Tissues to Disuse and Remobilization

Darryl L. Millis

Patients afflicted with neurologic, orthopedic, metabolic, and other diseases frequently have reduced weight bearing and use of limbs. In addition, some conditions may be treated with splints, casts, or other coaptation devices, resulting in decreased limb use. The tissues most affected by disuse and immobilization are cartilage, joint capsule, muscle, ligament, tendon, and bone. Knowledge of how these tissues respond to disuse and immobilization, the types of deleterious tissue changes, and the time frame over which these changes occur, are important in understanding the need for physical rehabilitation. For example, stress deprivation of joints results in proliferation of connective tissue within the joint space, adhesions between synovial folds, adherence of connective tissue to cartilage surfaces, atrophy of cartilage, reduced proteoglycan (PG) content, ulceration at points of cartilage to cartilage contact, disorganization of cellular and fibrillar ligament alignment, reduced collagen mass, increased ligament compliance, weakened ligament insertion sites, reduced load-to-failure and energy-absorbing capacity of the bone-ligament-bone complex, and osteoporosis of the involved extremity.¹

Rehabilitation of patients with acute or chronic neurologic or orthopedic conditions involves the application of controlled challenges to tissues to improve strength, condition, and function. It is important to understand how to safely remobilize tissues after injury and a period of immobilization. Rehabilitation must sufficiently challenge tissues to enhance and positively influence their recovery and healing. However, if tissues are overchallenged, they may be damaged, ultimately delaying recovery or cause additional injury. There must be a balance between the simultaneous demands for protection against undue stress to facilitate healing, and the need for stress to attenuate atrophy of musculoskeletal tissues.² A review of these concepts forms the basis for a rational approach to developing rehabilitation programs.

Cartilage

Disorders of articular cartilage and joints are some of the most common and debilitating diseases encountered in veterinary practice. Understanding the normal structure and

function of articular cartilage is a prerequisite to understanding the pathologic processes. The mechanical properties of articular cartilage arise from the complex structure and interactions of its biochemical constituents. The viscoelastic properties of cartilage, primarily resulting from fluid flow through the solid matrix, explain deformational responses observed during loading. For example, cartilage that is loaded rapidly is stiffer than the same cartilage that is loaded slowly. Degenerative processes include the breakdown of normal constituents of cartilage, especially the collagen network and PGs, which affects the mechanics of fluid flow through cartilage. Factors contributing to such breakdown include direct trauma, obesity, immobilization, and excessive repetitive loading of the cartilage. Training activities, without traumatic injury, do not appear to be a risk factor for developing osteoarthritis (OA) in normal joints, but such activity may be harmful to an abnormal joint.³

Normal Articular Cartilage

Articular cartilage covers the ends of long bones and is composed of chondrocytes, extracellular matrix, and water. Articular cartilage is avascular, aneural, and alymphatic. Synovial fluid is present in joints and provides a near-frictionless, wear-resistant, weight-bearing layer in contact with articular surfaces. Articular cartilage dissipates contact stresses during loading, distributing compressive and shearing forces, which are transmitted through subchondral bone to the stiffer cortical bone.

Chondrocytes are metabolically active and make up less than 5% of the cartilage volume. Metabolism of the cells is modulated by their location within the cartilage, by aging, and by biomechanical stresses. Chondrocytes produce and maintain the extracellular matrix and pericellular environment. The substrates for these processes come mainly from the synovial fluid. Weight bearing results in a synovial fluid “pumping” mechanism that facilitates the transfer of nutrients and waste exchange by diffusion from the cartilage surface.

The extracellular matrix is composed of collagen, PGs, and water. The orientation of collagen and PGs functions to distribute forces over the subchondral bone and provides a smooth surface that allows movement of joints

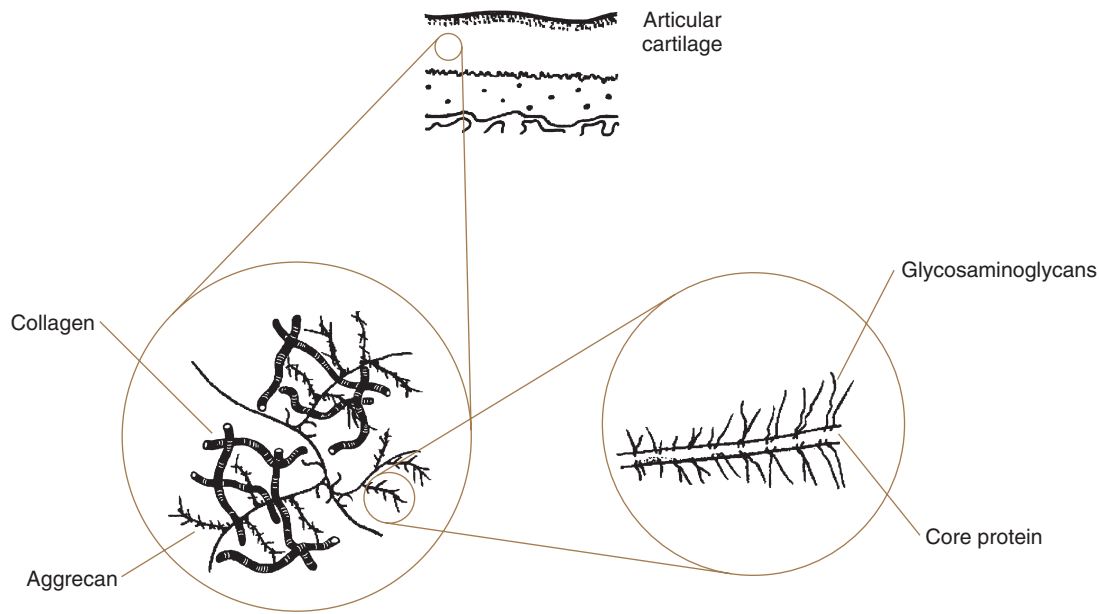


Figure 7-1 Diagram of articular cartilage, including collagen, proteoglycans, and water.

(Figure 7-1).³ Collagen provides tensile strength to the cartilage, and structural support for the extracellular matrix. Type II collagen is the primary form in articular cartilage and is unique to cartilage. Type IX collagen is also unique to articular cartilage and links type II fibrils together, limits the separation of collagen fibrils, limits fibril diameter, and binds PGs to collagen. Processes resulting in articular cartilage loss may cause collagen degradation, either enzymatically, mechanically, or both. Type IX collagen is especially susceptible to enzymatic degradation.

PGs make up most of the extracellular matrix that is not collagen, and compose 22% to 38% of the dry weight of articular cartilage. PGs are highly hydrophilic, which results in water retention, creating swelling pressure and turgidity that is essential to articular cartilage function. A PG monomer consists of a core protein with one or more types of glycosaminoglycan (GAG) chains attached. The core protein has a hyaluronan binding region, GAG binding regions, and a carboxy-terminal end. A link protein helps to bind the PG chain to hyaluronan (Figure 7-2). Hyaluronan is a nonsulfated GAG located in the extracellular matrix and forms the backbone to which the PG monomers are covalently bound. *Aggrecan* refers to a PG monomer that combines with hyaluronan.

GAGs are structures with varying lengths of repeating disaccharides. Chondroitin sulfate and keratan sulfate are the major GAGs, composing almost 90% of total articular cartilage GAGs. GAGs are negatively charged, causing them to repel one another when attached to a core protein and occupy a large area.

Water composes 65-80% of total cartilage weight.³ Although PGs are highly hydrophilic, enclosure in the

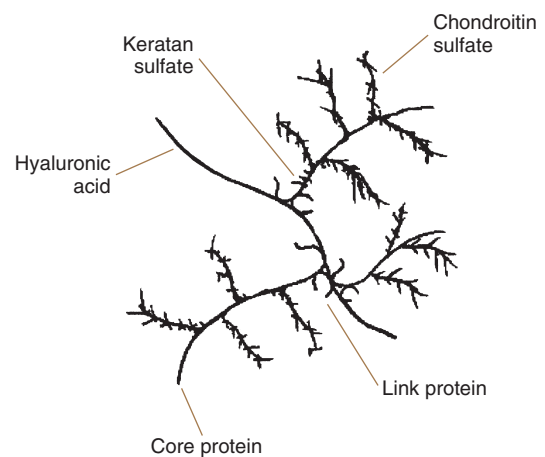


Figure 7-2 Proteoglycan monomer consisting of a core protein, glycosaminoglycans, hyaluronan backbone, and link protein.

collagen meshwork limits their ability to expand. This keeps cartilage turgid, resisting deformation. Water moves slowly within the cartilage matrix during loading as a result of resistance of flow through the collagen meshwork and the hydrophilic nature of the PGs. Water is forced out of articular cartilage during loading and some water weeps onto the articular surface, allowing hydrostatic lubrication. When the load ceases, water is reabsorbed. With rapid loading, cartilage is stiffer because water distributes more slowly, resulting in less compression of the cartilage. Cartilage is more compliant with slow loading because there is additional time for fluid movement to the cartilage surface.

Histologically, cartilage may be divided into various zones based on their depth within the cartilage and their characteristics. These zones include a relatively thin superficial layer; a middle layer with the cells arranged in vertical columns; and a tidemark, which is the upper limit of the zone of calcified cartilage, which in turn interdigitates with the subchondral bone plate.

The subchondral bone is the thin layer of bone beneath the calcified cartilage layer. It is 10 times more deformable than cortical bone and is important in distributing forces during loading. Its relatively high compliance helps to diminish the peak load placed on the cartilage, decreasing damage to the articular surface. The subchondral bone becomes stiffer with OA. As the bone becomes more dense and sclerotic, the cartilage and bone do not deform normally, and the chondrocytes and matrix are more likely to be damaged with loading.

Cartilage Changes Caused by Disuse and Immobilization

Clinicians who attempt to restore joint function should understand the effects of immobilization on articular cartilage (Box 7-1). Disuse of cartilage may result in atrophic or degenerative changes, and these changes are more marked and appear sooner in areas of contact.⁴ Immobilization leads to a decrease in matrix and cellular components, disorganization of cartilage, and potential irreversible damage if immobilization is prolonged.⁵ In addition to a reduction in synovial fluid production with immobilization, nutrition to cartilage is diminished as a result of reduced synovial pumping and nutrient diffusion. The age of the dog, as well as the form of immobilization used, may affect results. Several models of joint immobilization and disuse have been studied, including casting with the limb in flexion or extension, external skeletal fixation, and non-weight-bearing models (such as sling application). The changes in mechanical behavior and biochemical composition of articular cartilage appear to be less dramatic in limbs immobilized in a sling than in limbs immobilized in a cast. Adult greyhound dogs had stifles placed in a sling at 90 degrees of flexion for 4 or 8 weeks, which allowed some limited motion of the stifle but no weight bearing. There were no changes in the compressive properties of cartilage, PG content, collagen content, or cartilage thickness.⁶ This suggests that immobilization with a sling may be less deleterious to articular cartilage as compared with other models of joint disuse in which cartilage changes are both progressive and degenerative.⁶ Immobilization with a joint in flexion does not generally lead to arthritic changes in the short term, but cartilage atrophy may occur. Immobilization of a joint in extension results in increased muscle contraction against the immobilization device and changes in articular cartilage similar to those seen in OA.

The cartilage of limbs of young dogs immobilized in flexion for 3-11 weeks was grossly normal with no

Box 7-1 Summary of the Effects of Disuse and Immobilization on Articular Cartilage

- Atrophic or degenerative changes
 - Changes more severe and appear sooner in areas of contact
- Decrease in matrix and cellular components
- Disorganization of cartilage and potential irreversible damage if immobilization is prolonged
- Reduced synovial fluid production
- Diminished nutrition to cartilage because of reduced synovial pumping and nutrient diffusion
- Immobilization of joints in extension results in arthritic changes to cartilage
- Immobilization of stifle joints in flexion results in cartilage atrophy in young dogs, but less atrophy in older dogs
 - 3 to 11 weeks of immobilization in flexion results in 13% to 60% reduction of proteoglycan content in young dogs
 - 40% to 60% reduction in proteoglycan synthesis with immobilization in flexion
 - Cartilage thickness reduced 9% to 50%
 - Up to 30% decrease in chondrocytes
 - Atrophy of subchondral bone
 - Elevated MMP-2 and decreased TIMP with 4 weeks of immobilization
 - Up to 30% decreased stiffness of cartilage
- Active stifle joint motion may be more important than weight bearing in meniscal cartilage
 - Joint inflammation with reduced weight bearing following cranial cruciate ligament transection and stifle stabilization may result in decreased meniscal stiffness (30%)
 - Joint motion and weight bearing may improve regeneration of tissue and maintain articular cartilage following partial meniscectomy as compared with immobilization
- Some joint motion during immobilization (as opposed to rigid immobilization) reduces deleterious effects
- If joint instability exists, such as cranial cruciate ligament rupture, temporary immobilization in flexion (non-weight-bearing position) may result in less osteoarthritis until stabilization procedure performed

MMP-2, Metalloproteinase 2; TIMP, tissue inhibitor of metalloproteinases.

osteophytes.^{7,8} However, there was a 13% to 60% early and progressive reduction of PG content following immobilization, with the depletion being most severe in the superficial zones.^{4,8-10} Immobilization for as little as 6 days may result in a 40% reduction in cartilage PG synthesis, and up to 60% for joints immobilized for 8 weeks.⁴ In addition, there is a 10-20% increase in water content. The accelerated PG loss appears to be caused by a combination of decreased synthesis and increased proteolysis of PGs.¹¹ The loss of PGs is somewhat selective, with preferential loss of

chondroitin-6-sulfate.⁹ The levels of synovial fluid PG components, including link protein, keratan sulfate, and total sulfated GAG, were determined in dogs after a 3-week recovery period following either 4 or 8 weeks of immobilization with the stifle in flexion.^{12,13} The concentrations of keratan sulfate and sulfated GAG in the synovial fluid lavages were increased after disuse and after disuse with recovery compared with controls, but the levels of link protein remained low. This pattern of catabolism differs from that found in OA, in which link protein is elevated in synovial fluid lavages, along with keratan sulfate and sulfated GAG.

After 3 weeks of immobilization, PG aggregation is poor. These changes are reversible, however, because PG aggregates were normal 2 weeks after removal of a cast that had been worn for 6 weeks.⁴ The interaction of PGs with hyaluronic acid (HA) may be reduced in older dogs because of an abnormality in the HA binding region of the PG core protein,⁴ but others have shown no abnormality in aggregation in young dogs.^{9,11}

Young dogs subjected to 11 weeks of immobilization by splinting had a significant reduction in hyaluronan concentration in the tibial and femoral condyles, and the patellar surface of the femur, with a concurrent decrease in aggrecan, but the ratio of hyaluronan to aggrecan was unchanged.¹⁴ Young dogs may respond differently, because there was no change in hyaluronan content in skeletally mature dogs following 4 to 8 weeks of immobilization, as compared with an 80% decrease during early OA.¹⁰ This difference supports the concept that the mechanism of cartilage degradation in disuse and OA is different, and immature dogs may respond differently than older dogs to immobilization. It may also help explain why the changes in cartilage following immobilization may be reversible, whereas the changes in osteoarthritic cartilage are not reversible.

Cartilage thickness may be reduced 30-50% with immobilization, but the location of thinning within the cartilage is not consistent. In mature dogs, thinning occurs primarily in the uncalcified cartilage, whereas in young dogs it tends to occur in the calcified cartilage under the tidemark.^{4,8} Immobilization may result in 30% fewer chondrocytes per unit area in cartilage.⁴ The subchondral bone also atrophies.

Sling immobilization of the stifle for 4 weeks results in elevation of gelatinase (neutral metalloproteinase 2 [MMP-2]) and almost complete suppression of tissue inhibitor of metalloproteinases (TIMP), especially in the superficial zone, as well as reduced protein synthesis by chondrocytes.¹⁵ In contrast, collagenase activity is normal. The return of MMP-2 and TIMP levels to normal was observed with a 2-week remobilization period or treatment with insulin-like growth factor-1 (IGF-1) and pentosan polysulfate. This suggests that cartilage is remodeling, similar to bone remodeling in disuse atrophy, but the material

properties of the articular surface remain relatively intact following a period of immobilization with a sling.

Contact forces between articulating surfaces appear to be necessary to maintain normal PG content of articular cartilage.⁹ The greatest depletion of GAGs (64%) occurred at the cranial and caudal extremes of the femoral condyles in the stifle joint cartilage of young beagle dogs following cast immobilization for 11 weeks in 90 degrees of flexion. These were locations where the immobilized cartilage lost contact with the opposing cartilage.⁹

The biochemical changes that occur with immobilization also affect the biomechanical properties of articular cartilage. Splint immobilization of stifle joints of young dogs for 11 weeks caused significant softening of femoral and tibial cartilage with no visible changes of the cartilage surface.¹⁶ The rate of deformation during loading increased 42% and the average thickness of the cartilage decreased 9% as compared with controls. Loading of cartilage may be necessary to maintain cartilage stiffness, because stiffness was maintained in the contact area between the patella and patellar surface of the femur as a result of sustained loading between the femur and the patella produced by the flexion of the knee joint, whereas cartilage softening was greatest in nonloaded areas. There is correlation between cartilage stiffness and PG concentration. Therefore loss of PGs during immobilization may also affect biomechanical properties. Immobilization of stifles with a cast caused significant softening of up to 30% in the lateral femoral and tibial cartilages.¹⁷ The GAG content of the cartilage was slightly decreased after immobilization, especially in the superficial zone of cartilage. The changes in stiffness correlated with alterations in GAG content of the superficial and deep zones. This confirms the key role of PGs in the regulation of cartilage stiffness. There may also be changes in the compressive and shear stiffness of cartilage from nonimmobilized contralateral limbs, possibly as a result of increased loading.¹⁸

The changes that occur to articular cartilage during immobilization are probably caused by a combination of decreased joint motion and reduced loading. Although both are important in maintaining cartilage integrity, normal loading is particularly important with contraction of muscles that span the joint and stabilize the limb during weight bearing.⁶ In a non-weight-bearing model that allowed joint motion of the stifle joint, cartilage thickness, PG content, and PG synthesis were reduced, whereas water content was increased.¹⁹ These changes were similar to those that occur with immobilization in a flexed position and suggest that weight bearing is necessary to prevent the bulk of changes.

Some joint motion during immobilization is beneficial. Stifles immobilized for 6 weeks with a cast, allowing 8 degrees to 15 degrees of motion, had less PG loss and a smaller decrease in PG synthesis than occurred with more rigid fixation. The limited motion provided by casting was

Box 7-2 Summary of Effects of Immobilizing Stifle Joints in Flexion on Articular Cartilage

- PG content reduced 25-60%
- Cartilage thickness reduced 30-50%
- Cells per unit area 30% fewer
- Water content increased 10-20%
- Net PG synthesis decreased 40-60%
- Atrophic subchondral bone

PG, Proteoglycan.

protective.¹¹ Elimination of motion by rigid fixation induces even more pronounced atrophic changes to articular cartilage. To study the effects of immobilization on the ultrastructure and surface contour of articular cartilage, knee joints of adult dogs were immobilized with circular external fixation for two to four weeks.²⁰ The degree of cartilage changes increased with the length of immobilization, and consisted of destruction of the superficial cartilage layer, cleft formation in the surface, and erosion of the articular cartilage. In another study, knees of mature dogs were immobilized for 6 weeks with external fixators. There was a 7% increase in water content compared with normal knee cartilage, and a 28% decrease in GAG content, which was greater than if some motion was allowed.¹¹ More importantly, GAG content did not return to normal after a 1 week remobilization period as it did in dogs that were not rigidly immobilized. Spinal cord injury, which may result in reduced mobility and decreased joint motion, may also result in cartilage thinning.²¹

Immobilization of the stifle in flexion may be beneficial when joint instability exists, such as occurs with a ruptured cranial cruciate ligament (CCL), because bearing weight on an unstable stifle results in OA. Dogs undergoing transection of the CCL had their stifle joints maintained in flexion or were permitted ad libitum ambulation for 12 weeks.²² Dogs allowed to bear weight had osteophytes, fibrillation, and decreased cartilage PG content, although cartilage thickness was normal. PG synthesis was 80% greater than that in cartilage from the contralateral knee. In contrast, osteophytes were not seen when the limb was immobilized in flexion immediately after transection of the ligament, and the articular surface remained intact. The cartilage became atrophic, however, and PG content and PG synthesis were decreased compared with cartilage from the contralateral knee. Knee cartilage from immobilized limbs after cruciate ligament transection, therefore, resembled that from dogs whose limbs were immobilized without ligament transection (Box 7-2).

The effect of position of the limb during immobilization on blood flow to the joint must also be considered. For example, forced abduction of experimentally induced

Box 7-3 Considerations for Remobilization of Cartilage after Immobilization

- The length of immobilization, condition of cartilage, and the length and magnitude of weight bearing after immobilization affect cartilage recovery.
- After a 6-week immobilization period, 3 weeks of free, low-intensity activity results in normal cartilage.
- Longer periods of immobilization require longer recovery times.
 - Immobilization longer than 15 weeks may not result in complete recovery, even with 50 weeks of remobilization, in young dogs.
- Vigorous exercise after immobilization may be deleterious to cartilage.
 - Jogging young dogs 9.5 km/d at 5 km/h after immobilization for 3 weeks resulted in continued decreases in cartilage thickness (20%) and PG content (35%), even though PG synthesis increased (16%).

PG, Proteoglycan.

hip dysplasia with secondary OA results in significant reduction of blood flow, whereas immobilization in flexion results in the highest blood flow to the femoral head.²³

The Effects of Remobilization on Cartilage

The response of cartilage to remobilization depends on the biomechanical demand that the joint is exposed to, the condition of the cartilage, and the length of the immobilization and remobilization period (Box 7-3). If the joint is subjected to high stresses and repeated loading immediately after immobilization, cartilage may not be able to resist the stresses and may become damaged because of injury to the softened matrix. The duration and the degree of load bearing after immobilization are factors that determine the cartilage response. Also, the ability to restore biomechanical properties may depend on an intact collagen network.

The atrophic changes of cartilage that occur with a limb casted in flexion are somewhat reversible. Remobilization of a normal limb casted in flexion for 6 weeks by allowing ambulation for 3 weeks after cast removal resulted in the cartilage becoming normal.⁴

Longer periods of immobilization in young dogs may result in more long-standing effects, even with a gentle remobilization period. Skeletally immature female beagle dogs were immobilized in a cast for 11 weeks in 90-degree flexion followed by remobilization for 15 weeks.⁹ After remobilization, PG content remained an average 18% lower than controls, with content at the minimum contact sites 33% lower. GAG concentration was restored in the more loaded regions of the patellofemoral region and tibial condyles, but remained lower than the control values in the

less loaded peripheral regions of the femoral condyles.²⁴ Uncalcified cartilage thickness of the femoral condyles was 15% less than in the controls, and the thickness of the calcified zone was reduced in most regions of the remobilized stifles. The thinness of the calcified zone may be due to advanced bone growth from the subchondral bone during remobilization. Softening of the articular cartilage during immobilization may expose the calcified zone to increased mechanical loading and result in mineralization and thickening of subchondral bone. The changes induced by unloading are reversible to a great extent, but full restoration of articular cartilage may not occur at all sites following 15 weeks of remobilization. Immobilization of the skeletally immature joint therefore may affect the development of articular cartilage in such a way that very slow recovery or permanent alterations are induced.

Longer periods of remobilization may be more beneficial. Hindlimbs of immature dogs were immobilized for 11 weeks and then remobilized for 50 weeks.^{17,25,26} Immediately after the immobilization period, cartilage GAG concentration was reduced similar to that found in other studies. The amount of collagen crosslinks was also reduced during immobilization. After 50 weeks of remobilization, GAG concentration was restored at most sites, but remained 9-28% lower than controls in the patella, medial femoral condyle, tibial condyles, and the proximal femoropatellar surface. Collagen crosslinks were restored to control levels, indicating that collagen is resistant to reduced joint loading and slow remobilization. Although the articular cartilage of immobilized limbs became similar to control cartilage following long periods of remobilization, full restoration of articular cartilage GAG concentration was not obtained in all sites, even after remobilization for 50 weeks. This suggests that joint immobilization of young dogs can cause long-lasting articular cartilage PG alterations.

Because GAGs exert a strong influence on the biomechanical properties of cartilage, the biochemical changes that occur in cartilage during remobilization may also affect the biomechanical properties. The restoration of the biomechanical properties of articular cartilage was studied after 15 weeks of remobilization of the knee joint in mature beagles previously immobilized with a cast for 11 weeks.²⁷ Remobilization improved the decreased stiffness and cartilage thickness in femoral, tibial, and patellar cartilage created by immobilization, and approached values of nonimmobilized controls. However, stiffness of the femoral condylar cartilage remained lower and cartilage permeability was higher compared with the controls. A study with the same period of immobilization but with a remobilization period of 50 weeks resulted in restoration of the biomechanical properties of cartilage in the lateral condyle of tibia, but not in the lateral condyle of femur, where stiffness remained 15% below the control level following remobilization.¹⁷ Similar to biochemical changes,

articular cartilage biomechanical properties return toward normal during remobilization, although the recovery is not complete in all parts of the knee even after 50 weeks of remobilization. Immobilization of joints of young dogs may cause long-term, if not permanent, alterations of cartilage biomechanical properties, similar to changes observed in biochemical components of articular cartilage. This may predispose joints to degenerative changes later in life.

The atrophic changes seen in cartilage following immobilization are less reversible following immobilization with a rigid external fixator. However, in one study, 2 to 4 weeks of remobilization following 2 to 4 weeks of immobilization with an Ilizarov external fixator demonstrated that the longer the period of remobilization, the greater the reparative processes.²⁰

In contrast to the improvements seen with mild remobilization strategies, vigorous exercise after a period of immobilization may have deleterious effects on cartilage.²⁸ Dogs that ran daily (9.5 km/d, 5 km/h) on a treadmill for 3 weeks after cast removal had continued decreases in cartilage thickness (20%) and PG content (35%), even though net PG synthesis was 16% greater than in cartilage from contralateral nonimmobilized stifles. Therefore vigorous loading of the joint following a period of immobilization may prevent reversal of cartilage damage and ultimately be detrimental to cartilage. It appears that the atrophic changes are more reversible with gentle spontaneous reloading.

Meniscal Cartilage Changes

Various experimental models for producing atrophy of meniscal tissue have yielded contradictory results. These models have included denervation of the stifle, immobilization (both internal and external), and disarticulation. Active motion of the stifle may play a more important role than weight bearing in preventing meniscal atrophy. In one study, internal skeletal fixation, which did not allow motion of the stifle joint, was applied for 12 weeks. There was significant atrophy of the lateral meniscus, despite the opportunity for weightbearing.²⁹ Another study evaluated the effect of active joint motion on the maintenance of meniscus mass in a unilateral ankle disarticulation non-weight-bearing model of disuse. This model maintained active knee motion without weight bearing. Despite the loss of collagen and calcium mass in the femur and tibia because of decreased weight bearing, there was no loss of meniscal mass.³⁰

Although meniscal mass may be maintained when motion is allowed in stifles with limited weight bearing, there may be profound changes in biomechanical and biochemical properties. Meniscal cartilage stiffness decreased by 30% 8 weeks following transection of the CCL and immediate stabilization of the stifle in one study, despite the normal appearance of the meniscus.³¹ Although weight

bearing and joint motion were not restricted following surgery, limb use was reduced and it is possible that a combination of inflammation and decreased limb use after surgery resulted in decreased stiffness. Chronic stifle instability may result in further deterioration of the meniscus because clinical cases of cranial cruciate ligament rupture (CCLR) undergoing partial meniscectomy had a 60% decrease in meniscal stiffness.

The effect of early motion on healing meniscal repairs has been evaluated in humans at the time of anterior cruciate ligament (ACL) reconstruction. In one study, 80% of patients undergoing immediate postoperative range of motion (ROM) and early partial weight bearing had complete meniscal healing with no deleterious effects of immediate knee motion on meniscal repairs.³² Although meniscal repair is not generally performed in dogs, studies of healing medial meniscal incisions demonstrated that dogs with immediate limb mobilization had greater collagen content in the healing meniscus than those that were immobilized in a cast.³³ There were no significant differences in collagen between the repaired menisci with mobilization and the control menisci. Prolonged immobilization resulted in decreased collagen formation in healing tissue. The vascular response during meniscal healing is also affected by immobilization.³⁴ In a study of rabbits with meniscal injury, meniscal blood flow was increased five-fold in animals in which the stifle was allowed motion, but this increase was prevented by immobilization 4 weeks after injury. Furthermore, healing was diminished in immobilized knees.

Partial or complete meniscectomy is more commonly used to treat meniscal damage in dogs. In general, meniscal removal results in degeneration of articular cartilage, and cartilage stiffness is reduced.³⁵ Less severe changes occur with partial meniscal resection, but the changes are greater on the tibial surface as compared with the femoral surface.^{36,37} Early joint motion may be beneficial to these patients. In studies of bilateral medial meniscectomy in dogs, one group was allowed immediate joint movement, and another group underwent knee joint immobilization with an external skeletal fixator (ESF) for 5 weeks immediately following meniscectomy. After removal of the fixation, the animals were allowed free movement for 21 weeks.^{38,39} The normal menisci and the regenerated tissue at the meniscectomy sites were examined for collagen content, collagen assembly, and GAG distribution. Although the collagen content of the regenerated tissue was not different from normal menisci, collagen fiber development and GAGs were not normal. Movement following meniscectomy is likely beneficial to matrix formation in the regenerating tissue. Regarding articular cartilage, twice as much PG was present in femoral condyles of meniscectomized animals encouraged to bear weight on joints immediately after surgery compared with controls or joints immobilized after surgery.

Adaptations of Cartilage to Increased Mechanical Stress

Many reports have suggested an association between the biochemical and biomechanical properties of cartilage and the mechanical stress to which it is subjected. For example, the increased weight bearing in the limb opposite a limb casted in flexion resulted in a 19% increase in uncalcified cartilage thickness and a 25-35% increase in GAG concentration in the intermediate, deep, and calcified zones of the femoral condyles.⁸ It appears that increased weight bearing augments local PG content of the articular cartilage matrix. In fact, PG content and cartilage thickness are generally greater in central weight-bearing areas than in peripheral areas, and contact forces help maintain normal cartilage matrix.⁸ In general, chondroitin sulfate predominates in pressure-bearing regions, whereas dermatan sulfate predominates in tension-transmitting regions of connective tissues.⁴⁰

The influence of static and intermittent stress on articular cartilage metabolism was examined in vitro using full-thickness plugs of cartilage from femoral condyles of normal adult dogs.⁴⁰ When cartilage was exposed to a static stress equivalent to three times body weight, or to cyclic stresses at a duty cycle of 60 seconds on–60 seconds off, net GAG synthesis was suppressed to 30-60% of that in controls. Protein synthesis also decreased, and water content increased. In contrast, when a duty cycle of 4 seconds on–11 seconds off was used, GAG synthesis was increased by 34%, but protein synthesis or water content was not affected. The changes in GAG synthesis do not appear to be due to changes in diffusion of nutrients during loading. These results suggest that regular, intermittent stresses, such as those obtained with walking, are important to maintain normal articular cartilage.

Continuous strong compression or heavy-impact loading to joints may cause deterioration of articular cartilage.⁴¹ Blunt trauma applied to cartilage results in changes in the zone of calcified cartilage, with an increase in cellular clones, bony bar formation in the calcified cartilage, vascular invasion, and decreased PG content, thickness of the calcified cartilage, and the number of chondrocytes.^{42,43} Blunt trauma to articular cartilage causes significant alterations in the deeper layers of cartilage, without disruption of the articular surface.

Effects of Training on Cartilage

The adaptation that occurs in functioning joints is known as *Leed's hypothesis*, which states that cartilage becomes conditioned to transmit, without sustaining damage, the stresses to which it is most regularly subjected.²⁷ Exercise places biomechanical and physiologic demands on articular cartilage.⁴⁴ Mild to moderate levels of running in dogs may stimulate adaptation, but strenuous levels can cause detrimental articular cartilage changes (Box 7-4). There

Box 7-4 Summary of Effects of Increased Mechanical Stress and Training on Cartilage

- Increased weight bearing of limb contralateral to a limb maintained in a sling may result in up to 19% increase in uncalcified cartilage thickness and 25-35% increase in PG concentration.
- Blunt trauma to cartilage may cause changes in the zone of calcified cartilage (deeper zone) without disruption of the articular cartilage.
- Mild to moderate training (running 4 to 20 km/d for 15 weeks) has no apparent deleterious effects on cartilage, but increases cartilage stiffness (6%), cartilage thickness (11-23%), and PG content (28%).
- Severe training (young dogs running 40 km/d for 1 year, or older dogs running on a treadmill at 9.6-12.8 km/h for 1 h/d for 8 months) may have adverse effects on cartilage.
 - Decreased cartilage stiffness (12-14% in some parts of the joint)
 - Decreased PG content in some parts of the joint superficial cartilage
 - Irreversible changes in collagen of cartilage
 - Increased matrix metalloproteinases
 - Potential cartilage injury, indicated by biomarkers, in the first 2 to 6 weeks of intense training, followed by repair
- Lifelong, low-impact training apparently has no deleterious effects on articular cartilage.

PG, Proteoglycan.

may be activity thresholds below and above which articular cartilage is maintained or destroyed. Between these thresholds, cartilage seems to adapt to the loads placed on it.

Intermittent compression tends to stimulate chondrocyte biosynthesis. Joint motion without compression results in articular cartilage thinning, and static loading causes decreased chondrocyte biosynthesis.⁴⁴ Cartilage subjected to high stress has higher PG content and is stiffer than cartilage exposed to low stress levels. Most studies of moderate running indicate no injury to articular cartilage, assuming there are no abnormal biomechanical stresses acting on the joints.

Mild to Moderate Training

Young beagle dogs running 4 km/d, at a speed of 4 km/h at a 15-degree incline on a treadmill for 15 weeks had no macroscopic damage to the cartilage surface and a 6% increase in mean cartilage stiffness, with most of the increase occurring on the patellar surface of the femur and the tibial condyles.⁴⁵ Cartilage thickness also increased by a mean of 11%, with increases of 19-23% in the lateral condyle and patella surface of the femur. In addition, there was a 28% increase in GAG content of the femoral condyles, located mainly in the intermediate and deep zones.⁴⁶

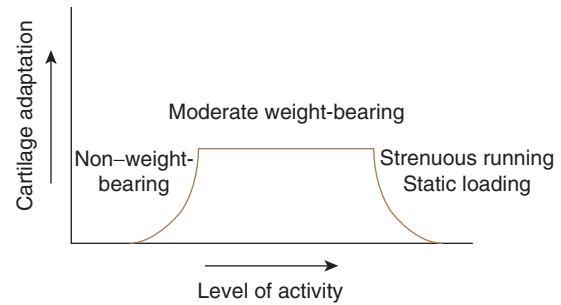


Figure 7-3 Adaptive responses of articular cartilage to activity. (Based on a diagram from Hallett and Andrich.⁴⁴)

This degree of exercise had a mildly anabolic effect on properties of cartilage. Running 20 km/d on a treadmill for 15 weeks did not further improve mechanical properties (Figure 7-3).⁴⁷

Severe Training

Skeletally immature dogs subjected to 15 weeks of running exercise at a rate of 40 km/d had no change in cartilage total hyaluronan or PGs.¹⁴ However, running 20 km/d for nearly 1 year resulted in a 6% reduction in thickness of the uncalcified cartilage of the medial femoral condyle, with an 11% reduction in GAG content.⁴⁸ Although tibial cartilage was stiffer, femoral condyle stiffness was unchanged with this degree of training.⁴⁷

In similar studies of running 40 km/d, the effects of training for 1 year on young canine articular cartilage were investigated and found that the cartilage response to running training was site dependent.^{49,50} Cartilage stiffness decreased 12-14% in the lateral, but not in the medial condyles of the femur and tibia, perhaps because of excessive impact loading and damage to the lateral region.⁴⁹ GAG content was depleted in the superficial weight-bearing areas of the joint, including the lateral femoral condyle.⁵⁰ There were slight increases in the thickness of the uncalcified and calcified cartilage, and the subchondral bone.⁵¹ There also appeared to be reorientation of the superficial zone collagen network.⁵² Although no overt cartilage damage was seen, softening of the cartilage may jeopardize the ability of cartilage to maintain its normal structural and functional properties over time. The response to long-term exercise may be different in other joints. Although there was a decrease in GAG content and cartilage thickness in the head of the humerus following training, cartilage stiffness was not affected.^{49,50} The articular cartilage of the femoral head appears to adapt very well to long-term distance training.⁵³

Strenuous training protocols in older dogs may result in deleterious changes to cartilage. Running aged dogs on a treadmill at 9.6-12.8 km/h for 1 h/d, 6 d/week for 8 months led to matrix degradation in the femoral head. PG content was decreased, and there was irreversible destruction of

collagen fibrils, with erosion and fibrillation of the cartilage surface. Older dogs may be more vulnerable to mechanical stress changes than younger animals.⁵⁴

A study attempted to use magnetic resonance imaging (MRI) of cartilage and serum and synovial fluid markers of cartilage damage to determine if training injury to cartilage could be detected and monitored in dogs training at different intensities.⁵⁵ Three groups were evaluated: (1) a training group that ran on a slope of 10 degrees at a speed of 200 m/min, (2) an intensified training group that ran on a slope of 10 degrees at a speed of 300 m/min, and (3) a control group that was allowed unrestricted cage activity. Dogs in the training and intensified groups were trained daily, running for an hour in the morning and in the afternoon for 10 weeks. Changes in the MRI examination suggestive of early injury of articular cartilage were noted in the training and intensified groups as early as 2 weeks of training. Damage was most severe at 4 to 6 weeks, and then gradually improved. There were no differences between the two training groups. Elevations of cartilage oligomeric matrix protein, matrix metalloproteinases 1 and 3, and tissue inhibitor of matrix metalloproteinase 1 in serum and synovial fluid were seen in both training groups. Histologic examinations at the end of training demonstrated cartilage damage and repair in the stifle joints in the training and intensified groups; furthermore, the intensified group indicated more remodeling of articular cartilage. The authors concluded that high-intensity training may induce injury, followed by a repair process. In addition, changes in biomarkers reflected pathologic changes in the cartilage better than MRI.

Lifelong Training

One study evaluated the effect of lifelong low-impact exercise on articular cartilage.⁵⁶ Dogs were exercised on a treadmill at 3 km/h for 75 min, 5 d/wk for 527 weeks while wearing jackets so that the total weight carried was 130% of body weight. There were no ligament, meniscal, or cartilage injuries, or osteophytes found on necropsy. Biochemical, cartilage thickness, and biomechanical properties were not affected by training. These results suggest that a lifetime of relatively low-impact weight-bearing exercise in dogs with normal joints does not cause alterations in the structure and mechanical properties of articular cartilage that might lead to joint degeneration.⁵⁶

Effects of Exercise on Cartilage during Acute Inflammation

Excessive exercise during periods of acute joint inflammation may exacerbate synovitis. Passively exercising the stifle joints of dogs in a model of acute inflammation for 5 seconds every 15 minutes for 4 hours resulted in a twofold increase in the synovial fluid leukocyte count, whereas exercise for 5 minutes every 15 minutes for 4 hours produced a ninefold increase in inflamed joints.⁵⁷

Forms of acute inflammatory arthritis may require periods of physical inactivity for recovery. Observations of humans with rheumatoid arthritis suggest that excessive activity may increase the severity of the disease.⁵⁸ Immobilization was chondroprotective to guinea pig stifles following an intraarticular injection of iodoacetate. Animals that were not immobilized had decreased GAG content and a 10-20% reduction in the number of chondrocytes 1 week after injection. Three weeks after injection, cell death and loss of GAG progressed, and surface fibrillation and osteophytes developed. Although the loss of GAGs was not altered by immobilization after iodoacetate injection, the depletion of chondrocytes was reduced. Furthermore, neither osteophytes nor fibrillation developed.⁵⁹ Therefore immobilization may be protective against exacerbation of synovitis during acute bouts of inflammatory joint disease.

However, isometric exercise does not appear to increase joint inflammation.⁶⁰ The effects of passive range of motion and isometric exercises on experimental inflammatory arthritis in a rabbit model were assessed. Passive range of motion significantly increased synovial leukocyte counts over control values of the nonexercised joints, but isometric exercises across a joint with acute synovitis did not affect synovial leukocyte count. Another factor to consider is the increased synovial cell count in joint motion as compared to the overall deleterious effects of immobilization on articular cartilage. A study of acute crystal-induced joint inflammation in rabbit knees compared the effects of joint movement with immobilization.⁶¹ Over 42 days, immobilized knees had greater cartilage histologic abnormalities and loss of GAGs compared with exercised joints. Overall, joint motion enhanced the synovitis, but joint immobilization had more adverse effects on articular cartilage. Another study of a rabbit antigen-induced synovitis model had similar findings, with less articular cartilage damage at the end of the 6-week study in those rabbits that had early continuous passive range of motion as compared to those with immobilized knees.⁶² Therefore, it seems prudent to allow active motion of inflamed joints, with reasonable caution. Aquatic exercise may be appropriate to reduce weight-bearing stresses on articular cartilage while allowing active joint motion.

Abnormal Biomechanical Stresses

Although moderate training programs appear to have no detrimental effects on articular cartilage in animals with normal joints, training program may be deleterious to articular cartilage in dogs with abnormal biomechanical stresses acting on joints. Transection of the CCL is a common model to produce OA. OA occurs as a result of joint instability during weight bearing. Joint immobilization may be somewhat protective against OA development. In a CCL transection model, the stifle joint of one group was immobilized in a cast for 12 weeks, and the other group had no immobilization for 12 weeks.²² Stifles of

dogs with CCL transection without immobilization had changes typical of OA, including increased water content and PG synthesis, and decreased PG content, but the cartilage thickness was normal. Osteophytes were not seen with CCL transection and immobilization, but the cartilage was atrophic (40% thinner), water content was increased, and PG content and synthesis were diminished. These data suggest that OA develops with cruciate ligament transection as a result of mechanical instability. If the knee is immobilized after cruciate transection so that loading is reduced, OA does not develop in the short term.

Effects of Medications on Immobilized Cartilage

Medications are commonly prescribed for dogs with joint injuries. Many of these drugs may have deleterious effects on articular cartilage metabolism, especially if the joint is immobilized. For example, fluoroquinolones, such as enrofloxacin and ciprofloxacin, may be toxic to chondrocytes, resulting in cellular changes that alter cell adhesion.⁶³ Knowledge of these effects may result in more effective rehabilitation strategies to minimize damage to cartilage. Alternatively, some drugs may provide beneficial effects to immobilized cartilage.

Nonsteroidal antiinflammatory drugs (NSAIDs), such as aspirin, are commonly used in the management of OA. Aspirin may suppress PG synthesis in normal canine knee cartilage, as well as in early OA cartilage, such as that following CCL transection.^{64,68} Aspirin may affect GAG synthesis by inhibiting uridine diphosphate-glucose dehydrogenase, an enzyme important in the synthesis of chondroitin sulfate.⁶⁴ Uptake of acetylsalicylic acid increases by 35% in osteoarthritic cartilage, suggesting that the drug permeates it more readily than in normal cartilage. The lower PG content of OA cartilage may be more important than fibrillation or surface disruption in rendering it vulnerable to aspirin.⁶⁹ In addition, aspirin suppresses GAG synthesis in unloaded cartilage to a much greater extent *in vitro* than it does in loaded cartilage.^{70,71} Aspirin also markedly reduces prostaglandin E₂ synthesis by the cartilage, and drug concentrations in cartilage from loaded zones are lower than those in cartilage from unloaded sites.

Aspirin administration aggravates the changes that occur in cartilage during limb immobilization. In one study, a high dose of aspirin (40 mg/kg tid) was administered daily for 6 weeks while one hindlimb was immobilized in a cast.⁶⁹ Whereas the cartilage of contralateral nonimmobilized limbs in dogs receiving aspirin had normal GAG content, the decreases in GAG content and net PG synthesis were significantly greater in immobilized stifles receiving aspirin as compared with immobilized stifles alone. Aspirin also apparently causes synthesis of type I rather than type II collagen by atrophic articular cartilage during immobilization.⁷⁰ Although these results indicate that aspirin has an adverse effect on articular cartilage of

immobilized joints, aspirin administration did not preclude reversal of these changes if the dog was allowed to walk freely for 3 weeks after cast removal.⁶⁹ However, if dogs ran daily on a treadmill for 3 weeks after cast removal, the decrease in GAG content of immobilized knees persisted and actually worsened in dogs receiving aspirin.

Although high doses of aspirin may have detrimental effects on cartilage, other NSAIDs apparently have little or no negative effects at doses that achieve expected levels in synovial fluid. In fact some NSAIDs may be somewhat chondroprotective, including carprofen,^{71,72} deracoxib,^{73,74} and etodolac.^{75,76}

Treatment with corticosteroids may also have detrimental effects on normal cartilage and cartilage of immobilized limbs. Prednisone given to normal dogs with immobilized stifle joints resulted in a 58% loss of GAG content in one study.⁷⁷ In nonimmobilized joints of dogs receiving prednisone, GAG content decreased 11-31%, compared with controls.

The use of HA in dogs may have some benefit to articular cartilage following immobilization. In one study, stifles were immobilized for 4 weeks using transarticular external skeletal fixation.⁷⁸ Treated dogs received HA. The PG content was reduced after immobilization in all dogs. Remobilization of the stifles was associated with damage to the surface and tangential layers of cartilage. However, remobilization with HA improved PG content and reduced structural damage. In another study, casts were placed on the hindlimbs of dogs for 92 days. Beginning on day 56, dogs received intraarticular injections of HA. Injections were repeated at 4-day intervals until the end of the study. Although femoral condylar articular cartilage had decreased PG content in all cast groups, the decrease was less in the HA-treated group. HA may inhibit inflammatory mediators, such as tumor necrosis factor- α (TNF- α) and stromelysin.⁷⁹ Other studies have indicated that HA may not be beneficial for treatment of OA, however.⁸⁰

Joint Capsule

Changes in Joint Capsule with Immobilization

Joint capsules are also affected by immobilization and remobilization. The functional and structural consequences of immobilization and remobilization of the glenohumeral joint after 12 weeks of immobilization were studied in 10 beagle dogs.⁸¹ One forelimb was immobilized in a cast. The cast was removed and remobilization was allowed for 4, 8, and 12 weeks. After 12 weeks of immobilization, the passive ROM was markedly impaired, intraarticular pressure was increased during movement, and the filling volume of the joint cavity was reduced. There was synovial lining hyperplasia and vascular proliferation in the wall of the joint capsule, but there was no

increase of fibrous collagen in the capsular wall. Both the functional and structural changes were unaltered after 4 weeks of remobilization, but after 8 weeks they began to reverse, and they returned to normal levels after 12 weeks. Both functional and structural changes after 12 weeks of immobilization of a normal glenohumeral joint are reversible by remobilization. The collagen composition of the capsule seems unrelated to the degree of capsular contraction that occurs during 12 weeks of immobilization.

Similar changes were observed in the carpus and elbow in dogs in which normal forelimbs were immobilized for 16 weeks.⁸² All dogs had a 20-30% decrease in ROM in the elbow and carpal joints, which returned to normal within 6 weeks after remobilization. In addition, all dogs were significantly lame immediately following immobilization, but returned to soundness within 6 weeks after remobilization began. Limbs casted for 32 weeks had severe joint stiffness and muscle atrophy, and recovery of normal limb use was delayed by several weeks.⁸³ Deficits were also present after only 12 weeks of immobilization, but full function resumed much sooner.

It is helpful to know which tissues contribute to joint contractures following prolonged immobilization. One study evaluated contractures in a rat model by immobilizing one knee in flexion.⁸⁴ After 8 weeks, the animals were allowed to resume unrestricted activity for 4 weeks. Interestingly, 88% of the contractures remained in the immobilized group after dividing skin and muscle, suggesting an important contribution of the caudal knee capsule in limiting knee mobility. In fact, another model in rats confirmed the role of the joint capsule in contracture during knee joint immobilization.⁸⁵ Joint contracture rapidly progressed after joint immobilization until 8 weeks and advanced more slowly after 8 weeks, suggesting that joint contracture develops early in immobilization and progresses over time. After releasing the caudal capsule, the immobilized group regained knee extension.

The contribution of collagen content as a specific cause of joint capsule contracture is somewhat controversial. Type I collagen composes approximately 85% of the collagen content of the joint capsule, and provides mechanical strength, contributes to tissue elasticity, and is produced in sites undergoing fibrosis.⁸⁶ Collagen type III may be present in tissues requiring high levels of mechanical compliance. It is a constituent of normal synovial tissue, and is also present in inflamed and proliferative rheumatoid synovium. Some studies have shown an increase in joint capsule collagen content, especially with concurrent joint injury,⁸⁷⁻⁸⁹ whereas others have shown no change or even a decrease in collagen content with joint immobilization. In a rat knee immobilization model, mRNA of type I collagen decreased rapidly after immobilization, but the immunoreactivity of the capsule was not changed in the immobilized or the control groups at any time points.⁹⁰ In another study, the collagen of control and immobilized

periarticular connective tissue was type I collagen, whereas type III collagen, which occurs in response to inflammation, was not present.⁹¹ The periarticular connective tissue contracture process is apparently different from the intraarticular process in which fibrofatty connective tissue and type III collagen proliferation are prominent. A later study confirmed that types I and III collagen were not increased in the joint capsule following immobilization of rat knees, suggesting that another process, such as synovial or capsular adhesions, was responsible for joint contracture.⁸⁹ In addition to potential changes in collagen content, GAG and water content decrease in the joint capsule and synovium with immobilization.⁹¹ This may reduce the plasticity and lubrication efficiency of these tissues.

Changes of the joint capsule undergoing contracture may also be related to growth factors. Joint capsule may produce transforming growth factor- β 1 (TGF- β ₁) and connective tissue growth factor (CTGF) after immobilization, and these growth factors may play a role in causing and maintaining capsular changes. Collagen synthesis and myoblast induction are related to TGF- β , and CTGF is regulated by TGF- β .^{92,93} CTGF also induces type I collagen formation. In a flexion knee immobilization model in rats, TGF- β ₁, CTGF, and fibrotic change were greater in the caudal capsule and may have resulted in limited motion in extension.⁹² Another study of joint injury followed by immobilization indicated similar increases in mRNA and protein levels for collagen types I and III, matrix metalloproteinases 1 and 13, and TGF- β ₁ were significantly greater in immobilized capsules compared with controls, whereas tissue inhibitor of matrix metalloproteinases were significantly less in experimental capsules.⁹³ It may be possible to prevent joint contractures by somehow blocking these growth factors. In another study, although both TGF- β ₁ and CTGF were increased in the joint capsule after immobilization, collagen type I was not increased.⁹³

The specific cause of joint capsule contracture may be associated with the proliferation of myofibroblasts in the joint capsule following immobilization. In one study rabbit knee joints were rigidly immobilized in maximum flexion.⁹³ After 8 weeks groups were allowed periods of remobilization for 0, 8, 16, or 32 weeks. The average flexion contracture of knees was greater in the remobilization groups as compared with nonimmobilized controls, especially during early mobilization. Myofibroblast numbers in the posterior joint capsules were elevated four- to fivefold in the knees with contractures when compared with the contralateral knees, suggesting that these cells may be involved with joint contracture. A follow-up study indicated that these changes occur relatively soon after immobilization, and the changes persist for some time.⁹³ Differentiated myofibroblasts are characterized by the expression of α -smooth muscle actin (α -SMA), a protein associated with the contractile properties of this cell.⁹⁴

Mast cells and neuropeptide nerve fiber numbers may also be involved in joint capsule contracture because elevated numbers of these cells have been implicated in other fibrotic and contracture conditions. In a posttraumatic contracture model in rabbits, the knee joint was immobilized.⁹⁵ Rabbits were sacrificed 4 or 40 weeks after surgery (including 32 weeks of remobilization). Myofibroblasts, mast cells, and nerve fibers were four to five times greater in the contracture capsules when compared with the control capsules at all time points, suggesting myofibroblasts, mast cells, or neuropeptide influences may underlie some of the pathologic changes in the joint capsule following trauma. In addition, the neuropeptides substance P and calcitonin gene-related peptide can cause mast cell degranulation, and this may contribute to neurogenic inflammation, pain, and swelling.⁹⁶

Remobilization of Joint Capsule Immobilization

The form of joint mobilization may be important in restoring joint motion after a period of rehabilitation. The effect of low torque–long duration stretching to increase the ROM and to restore the mechanical properties of contracted joints was studied in rat knees immobilized for 40 days.⁹⁷ After remobilization, the contracted knees were placed in one of four treatment groups treated with repeated stretches of diverse torques and duration: stretching with low torque (0.02 Nm) and long duration (40 minutes), high torque (0.045 Nm) and short duration (20 minutes), high torque and long duration, and low torque and short duration. Animals received three sessions a week for 4 weeks, for a total of 12 treatment sessions. Low-torque and long-duration repeated stretching resulted in greater restoration of ROM with more normal mechanical properties compared with high-torque and short-duration stretching.

Heat may be useful to improve motion in joints that have contracture after immobilization. Infrared and ultrasonic heat therapies were compared in a rat knee flexion contracture model.⁹⁸ Six treatment sessions of stretching with infrared or ultrasound over a 2-week period were more effective than stretching without heat, but there was no significant differences between the two heat treatment groups.

Regular therapeutic exercise is also important to recover joint motion after immobilization. Joint ROM was reduced by 46% in a 2-week ankle joint immobilization model in rats.⁹⁹ Immobilization was followed by free cage activity and free remobilization or treadmill running once a week, three times a week, or six times a week. Ankle joint motion was improved in all three treadmill exercise groups in comparison with the free remobilization group, with running three and six times a week being more beneficial for recovery of joint contracture compared with no running or once-a-week running (Box 7-5).

Box 7-5 Summary of Effects of Disuse and Remobilization on Joint Capsule

- Immobilization may result in reduced (20-30%) passive range of motion, lameness, muscle atrophy, increased intraarticular pressure during joint motion, reduced filling volume of the joint, synovial hyperplasia, and vascular proliferation of the joint capsule.
- Improvement may occur 4 to 8 weeks after remobilization.
- Contracture of joints during immobilization occurs relatively rapidly in the first 4 to 8 weeks and involves the flexor side of the joint capsule.
- The role of collagen in joint capsule contracture is controversial.
- TGF- β_1 , CTGF, and matrix metalloproteinases may be involved in joint contracture.
- Myofibroblasts may be the cells that result in joint capsule contracture (may increase four- to five-fold)
 - Changes occur relatively soon after immobilization
 - α -SMA, a protein associated with the contractile properties of this cell, is expressed.
- Mast cells and neuropeptide fibers are also increased four- to five-fold with joint capsule contracture.
- Low torque, long duration stretching over time may be more effective in improving joint capsule contracture.
 - Stretching with heat application may be more effective.
- Frequent, regular therapeutic exercises are more effective in improving joint mobility than free activity.
- The effects of immobilization on joint capsule may be improved by the use of intraarticular sodium hyaluronate, cyclooxygenase-2 inhibitors (reduces the expression of α -SMA), and medications that inhibit mast cell degranulation.

α -SMA, Alpha-smooth muscle actin; CTGF, connective tissue growth factor; TGF, transforming growth factor.

Medications for Joint Capsule Immobilization

Although evidence exists for improving joint capsule stiffness using physical modalities, little is known about pharmacologic intervention to improve joint capsule changes as a result of immobilization. However, intraarticular sodium hyaluronate may be of benefit. Rabbit knee joints were immobilized in one study.¹⁰⁰ A 1% solution of sodium HA was administered intraarticularly at a dosage of 0.1 mL/kg body weight twice weekly. Changes in joint contracture, articular cartilage, and synovial membrane were compared with a saline control group during a 6-week immobilization period. In all groups, joint contracture and articular cartilage degeneration progressed, but were inhibited by HA. In addition, the effects on decreasing joint contracture and articular cartilage degeneration were more potent with higher-molecular-weight HA. Inhibition of cyclooxygenase-2 (COX-2) with NSAIDs may downregulate the expression of α -SMA, a protein associated with the contractile function of myofibroblasts.⁹³

Models of joint contracture have shown an increase in mast cells in fibrotic tissue, suggesting that mast cells may be important in joint capsule stiffness. Mast cells synthesize and secrete proteases, growth factors, and cytokines, many of which have mitogenic effects on fibroblasts and myofibroblasts.¹⁰¹ Medications that inhibit mast cell activation, such as ketotifen fumarate, have antiallergic and antihistamine properties. In particular, ketotifen noncompetitively blocks the binding of histamine to its H1 receptor and inhibits calcium-dependent vesicle degranulation of activated mast cells (mast cell stabilization).¹⁰² In a rabbit posttraumatic joint contracture model, animals treated with ketotifen had reductions in the severity of joint contractures and the degree of joint capsule myofibroblast and mast cell hyperplasia.¹⁰² However, contractures in the ketotifen-treated groups were not completely reversed as compared with nonoperative controls. Substance P, a neuropeptide that is an important mediator in the transmission of pain, is released into joints and is an important proinflammatory agent. Substance P is also a potent mast cell activator and is secreted in response to pain and injury. One study has documented elevated substance P in nerve fibers in contracted joint capsules of humans and rabbits. Increased expression of substance P has also been documented in other models of fibrosis, suggesting that this may be another target to modulate capsular stiffness.⁹⁵ Due consideration for the potential for delayed wound healing, including bone healing, should be made if considering treatments that inhibit mast cell function.

Muscle

Canine Muscle Fiber Types

Several types of muscle fibers exist, which perform somewhat specialized functions. Traditionally, they have been classified as two general types: type I (slow twitch) and type II (fast twitch) fibers (Table 7-1). Type I fibers are more adapted for oxidative metabolism and functions, such as maintaining posture, whereas type II fibers are associated with glycolytic metabolism and the generation of power and speed during muscle contraction. Fiber types previously identified in immature and mature canine skeletal muscles include types I, IIA, and IIC, with the latter representing less than 10% of the total fiber population.¹⁰³⁻¹⁰⁵ Type IIC fibers likely represent a transitional fiber between types I and II and are precursors to type IIA and IIB fibers in neonatal animals. Type IIA fibers have more oxidative capacity, but they still function in the generation of speed and power. Type IIB fibers have less oxidative capacity. The presence of true type IIB fibers has been debated in the dog, and type IIDog fibers, which are peculiar to the dog, or type IIX fibers have been suggested as a fiber type that is similar to but does not correspond exactly with classical type IIB fibers of other species.^{105,106} True type IIB fibers in the dog may only be present in the laryngeal muscles. The interpretation of fiber type composition depends on the staining and incubation methods used.¹⁰⁵ Type IIB or type IIDog fibers are not found in all muscles

Table 7-1 Canine Muscle Fiber Types

SLOW TWITCH OXIDATIVE FIBERS

Type I Greatest oxidative capacity

FAST TWITCH GLYCOLYTIC FIBERS

Type II

Type IIA	Greater oxidative power than IIB, but function in generation of speed and power
Type IIB	Classic fiber type in other animals, least oxidative. True type IIB fibers may not exist in dogs except in the laryngeal muscles.
Type IIC	May be transitional fiber between types I and II, precursors to type IIA and IIB in neonatal animals
Type IIDog	System that recognizes this as a peculiar fiber type to dogs, similar to type IIB in other species
Type IIX	System that recognizes this as a fiber type in dogs similar to, but not exactly corresponding with, type IIB fibers in other species. Rapidly contracting, greatest cross-sectional area, lowest concentration of capillaries and muscle nuclei. Still has high oxidative capacity compared to other species (intermediate between types I and IIA of other species), but is the least oxidative muscle fiber type in dogs.

HYBRID FIBERS

Type I+IIA	Second greatest oxidative capacity
Type IIAX	Intermediate between fiber types IIA and IIX

and were biologically significant only in the semitendinosus muscle in one study.¹⁰⁴ The metabolic potential of these fibers is fairly similar to that of IIA fibers, but significantly different to that of IIB fibers in other mammals, suggesting that they may be designed to play a different role during locomotion. All canine muscle fibers have an abundant capillary supply and moderate to high oxidative capacity, which may be related to the extraordinary athletic capability of dogs.

Recent methods of assessing muscle fiber types have incorporated immunohistochemical analysis of myosin heavy chain content of muscle fiber types rather than relying strictly on enzymatic staining. This technology has suggested that there are several muscle fiber types in dogs, with some of these being hybrid muscle fiber types.¹⁰⁷ It is believed that these hybrid fibers allow fine-tuning activity to allow for different contraction forces, contraction velocities, and resistance to fatigue. The main fiber type isoforms are type I, IIA, and IIX, which have been identified in previous studies. Hybrid fiber types include type I + IIA and IIX. Small amounts of type IIB fibers have been found in laryngeal muscles. Type IIX fibers are rapidly contracting fibers and have the greatest cross-sectional area and lowest concentration of capillaries and muscle nuclei. Nevertheless, type IIX fibers are more oxidative than other species, and it is this unusually high oxidative capacity that is probably responsible for the high endurance ability of dogs. Oxidative capacity decreases significantly in the order I > I + IIA > IIA > IIX. Unlike other species, IIX fibers appear to be intermediate between type I and IIA fibers regarding their contractile, metabolic, and morphologic features.¹⁰⁷

One study of the semitendinosus muscle found that hound-type dogs had the largest muscle fibers, but the fiber-type percentages were similar compared with mixed-breed and beagle dogs with similar activity levels.¹⁰⁸ In general, there are no differences between the left and right limb muscles, but there may be significant differences in

the percentages of muscle fiber types between individuals of the same breed for the same muscle.¹⁰⁹ Other studies have shown that greyhounds have greater fiber size and a higher percentage of type II fibers than mixed-breed dogs, presumably because of the demand for intense physical activity.¹¹⁰ However, there were no differences in fiber types in trained versus untrained Greyhounds, indicating that the distribution of fiber types may depend more on breed and genetics than level of training for this breed.

The size of muscles may be genetically related to a negative regulator of skeletal muscle growth, myostatin, a member of the TGF- β family of growth and differentiation factors.¹¹¹ Decreased expression of myostatin as a result of genetic mutation results in larger muscle mass in knockout mice and double-muscling in cattle. Recently, a mutation in the myostatin gene was found in some whippet dogs.¹¹² Dogs with this mutation, known as “bully” whippets, are more muscular and faster in racing events. However, some dogs with this trait have muscle cramping in the shoulder and thigh and do not conform to the American Kennel Club breed standards; approximately 50% of affected dogs have an overbite.¹¹² The role of myostatin in resistance training and subsequent detraining has been studied in men.¹¹³ After 30 or 90 days of resistance training, followed by 3 to 90 days of detraining, myostatin RNA levels in muscle biopsies significantly increased. In addition, type II muscle fibers began to atrophy after 3 days, with half of the acquired fiber area lost after 10 days of detraining. Myostatin may play a role in muscle atrophy with immobilization. In addition, there is a different amount of myostatin in different muscles. In people, the greatest amount of myostatin is found in quadriceps muscle fibers containing 100% type IIB fibers, whereas the soleus muscle does not have any appreciable type IIB muscle fibers or myostatin mRNA.¹¹⁴

Different muscle groups have variable muscle fiber type patterns that may be due to function and use (Table 7-2).^{110,115} In fact, muscle fiber type may change if muscle

Table 7-2 Muscle Fiber Types in Various Muscles of Adult, Mixed-Breed Dogs

Muscle	%	Muscle	Fiber	Types	
	I	I+IIA	IIA	IIAX	IIX
Semitendinosus	11.6	5.8	36.1	38.4	8.1
Latissimus dorsi	18.1	2.3	31.4	32.9	15.2
Gluteus medius	19.9	3.8	48.1	22.9	5.3
Triceps brachii	27.1	4.2	48.3	20.3	0
Longissimus lumborum	38.4	5.1	46.5	10.1	0
Rectus femoris	48.1	2.6	32.5	16.9	0
Vastus intermedius	65.3	1.0	31.6	2.0	0
All muscles	27.1	3.1	37.1	24.8	7.9

Adapted from Acevedo LM, Rivero JL: New insights into skeletal muscle fibre types in the dog with particular focus towards hybrid myosin phenotypes. *Cell Tissue Res* 323:283-303, 2006.

function is altered by transposing muscles. In one study, the fibularis longus tendon was severed and surgically transferred to the tendon of the cranial tibial muscle, a functionally different muscle.¹¹⁵ Later muscle biopsies demonstrated an increase in type I and type IIB fibers, which was similar to the fiber composition of the cranial tibial muscle. These findings indicate that muscle fiber remodeling may occur and suggests that changes in muscle fiber type are manifestations of functional adaptation.

Muscle biopsy technique may affect the results of muscle fiber typing.^{109,116} For example, the percentage of type I fibers in the cranial tibial and semitendinosus muscles progressively increases from the superficial and middle regions to the deepest portion. Although the distance of the section from the origin of the muscle does not significantly affect the mean percentage of fiber types, the variation in fiber types varies less deeper within the muscle, so a needle biopsy taken from deep within muscle should provide a more consistent and reliable estimate of fiber type proportion than a superficial specimen.

Factors Affecting Muscle Contraction

Skeletal Muscle Blood Flow

The ability of muscles to contract depends on blood flow to the muscle, primarily because of the oxygen delivered to the working muscle from the red blood cells. Small decreases in blood flow are associated with reduced contraction strength. When muscles contract, muscle blood flow typically increases, depending on the metabolic rate that is established by the contraction pattern and frequency. The ability of canine oxidative skeletal muscle to maintain isometric force is limited by blood flow.¹¹⁷ This limitation implies a mismatching of blood flow and the metabolic rate. The blood flow past a muscle cell is determined by the number of capillaries per fiber, the vascular conductance, the capillaries perfused, and vascular autoregulation. The resulting mismatching of flow and metabolism accelerates the fatigue process.

The frequency of muscle contraction and tension development affects blood flow during muscle contractions.¹¹⁸ The magnitude of the blood flow response to maximal exercise in different muscles is extremely varied. For example, skeletal muscle blood flow increased progressively up to maximal oxygen uptake in the gracilis, semitendinosus, and semimembranosus muscles of dogs in one study.¹¹⁹ However, the blood flow response to maximal exercise leveled off during submaximal exercise in the gastrocnemius muscle. In addition, there are species differences in skeletal muscle blood flow.¹²⁰ Muscle blood flow can reach 300–400 mL/100 g of tissue/min in dogs, ponies, and rats, whereas it reaches only 75 mL/100 g of tissue/min in a 75-kg man. The role of the diffusion gradient of oxygen from the blood to muscle cells is also important in exercising muscle.¹²¹

Muscle capillarity decreases in aged dogs.¹²² Endurance training, however, enhances capillarity, and old rats and

humans may attain levels of capillarity comparable to their active young counterparts, even when performing considerably less exercise.¹²³ In addition, there were no age-related effects on blood flow to the triceps, deltoideus, flexor carpi ulnaris, superficial digital flexor, gastrocnemius, gracilis, semimembranosus, and semitendinosus muscles of dogs during maximal exercise on a treadmill.¹²³

Skeletal Muscle Oxidative Capacity

The ability of muscles to function depends on oxygen delivery and utilization by muscles. Oxygen uptake by the gastrocnemius muscle group of dogs appears to be linearly related to external work rate and to the load against which the muscles shorten.¹²³ The major contribution to oxygen uptake during contraction appears to be the number of contracting muscle fibers. Oxygen uptake of muscle also increases with stimulation frequency, reaching a peak at five twitches per second.¹²⁴ Further increases in stimulation frequency result in lower oxygen uptake, perhaps secondary to mechanical restraints imposed by contraction duty cycle and vascular compression. However, muscle contractions may have only a small direct effect on muscle blood flow. Their main effect may be to reduce venous pressures.¹²⁵ The limiting factors for skeletal muscle oxygen uptake vary according to the intensity of muscle contractions.¹²⁶ During transitions from rest to contractions of dog gastrocnemius muscles with 70% of peak oxygen uptake, oxygen delivery to muscle, intramuscular blood flow, and peripheral oxygen diffusion are not limiting factors for skeletal muscle oxygen uptake kinetics. An increase in capillary density (reflecting a reduced diffusion distance caused by reduced fiber size) also may not increase the muscle oxygen diffusing capacity, suggesting that diffusion distance is not a primary determinant of muscle oxygen diffusing capacity.¹²⁷ Therefore oxygen uptake may be mainly determined by intrinsic skeletal muscle oxidative metabolism.

Maximum oxygen uptake is limited by oxygen delivery as a result of a limited and uneven distribution of muscle blood flow. When oxygen delivery is reduced, hypoxia results in reduction of muscle performance. Endurance training results in an increased number of capillaries in muscle over time. Hypoxia enhances angiogenesis in canine skeletal muscle after endurance exercise, whereas lactic acid appears to inhibit vascular endothelial cells.¹²⁸ Increased oxygen delivery to muscle may decrease fatigue. Metabolic rates can be increased by increasing oxygen supply.¹²¹ During high metabolic rate isotonic tetanic contractions, muscle fatigue was diminished by polycythemia, but the effect appeared to be transient.¹²⁹ Norepinephrine can produce large increases in muscle recovery oxygen consumption, whereas β -blockers, such as propranolol, reduce the rate of oxygen uptake during exercise and postexercise recovery.¹³⁰ Catecholamines may make a significant contribution to postexercise recovery oxygen consumption.

Oxidative capacity may decline in many muscles of sedentary individuals. Endurance training can greatly improve endurance in old age with an increase in muscle oxidative capacity.¹³¹ With endurance training, old individuals may attain levels of muscle oxidative capacity similar to those in identically training young individuals.

Skeletal Muscle Carbohydrate Metabolism

Muscle fatigue may be the result of several factors, with the relative contribution of each dependent on the condition and nature of the exercise, including the intensity, duration, muscle mass, and energy substrate. Fatigue may occur as the result of accumulation of hydrogen ions, inorganic phosphate, substrate depletion, and alterations in calcium ion function.¹³² Adequate cellular oxygen content and maintenance of adenosine triphosphate (ATP) levels are critical to avoid fatigue.

An increase in muscle oxidative capacity contributes to reduced glycogen depletion. Muscle glucose transport is enhanced after a bout of exercise, regardless of age.¹³¹ Muscle glycogen depletion, creatine phosphate depletion, and lactate accumulation during contractile activity are exaggerated in older animals, apparently secondary to reduced muscle oxidative capacity and blood flow. Resting muscle glycogen concentration is diminished in older humans, probably in part because of a more sedentary lifestyle. Although several months of endurance training raise muscle glycogen concentration in older people, levels still remain below those of younger individuals.

Studies of the role of intracellular oxygen on the efflux of lactate from skeletal muscle during graded exercise from submaximal to maximal effort indicate that a constant intracellular oxygen content is unrelated to increasing lactate efflux.¹²¹ Furthermore, muscle tension development is reduced by lactate.¹³² This effect of the lactate ion appears to be independent of pH.

Aging

Muscle mass typically declines in old age, secondary to muscle fiber atrophy.¹³¹ The loss of strength in old age is predominantly accounted for by reduced muscle mass. Although there are no apparent effects of aging on muscle fiber type percentages in dogs, there is a 25% reduction of cross-sectional area of type II fibers in aged dogs.¹²² The cross-sectional area of type I fibers is relatively unchanged, however, and may actually increase in some muscles. The mechanism for age-related atrophy is unknown, but may be related to a loss in the number of alpha motoneurons.¹³¹

Heart Disease and Skeletal Muscle

Patients with chronic heart failure may be limited by exertional fatigue. Changes in skeletal muscle may contribute to this fatigue, because biochemical and histologic abnormalities of skeletal muscle may develop in patients with

chronic heart failure. Dogs with chronic heart failure have changes in skeletal muscle fiber type, fiber size, and fiber ultrastructural properties.^{133,134} Skeletal muscle weights, muscle fiber area, and the percentage of skeletal muscle type I fibers may be reduced, whereas the percentage of skeletal muscle type II fibers may be increased with heart failure. There are apparently no ultrastructural abnormalities, no differences in developed tension per gram muscle during stimulation, maximal developed tension, muscle fatigability, or preferential atrophy or hypertrophy of either muscle fiber type in skeletal muscle of dogs with heart failure. Although skeletal muscle atrophy occurs with heart failure, the remaining muscle appears to exhibit normal performance and metabolism. These changes may be the result of adaptations of skeletal muscle to anaerobic metabolism.

Response of Muscle to Disuse and Immobilization

The muscles most vulnerable to disuse atrophy are the postural muscles that contain a relatively large proportion of type I (slow-twitch) muscle fibers, extensor muscles, and muscles that cross a single joint (Box 7-6). Conversely, those least susceptible to atrophy are those not used as postural muscles (antagonist muscles), flexor muscles, muscles that cross multiple joints, and muscles that are predominantly composed of type II (fast-twitch) muscle fibers.^{58,135} Immobilization reduces the chronic load on these muscles (Figure 7-4). Muscle strength decreases rapidly during the first week of immobilization, with further losses occurring more gradually over time.¹³⁶ Loss of muscle force production is not entirely explained by muscle atrophy alone. There may be up to a 50% reduction in peak force, even when the muscle mass has been normalized to constant muscle mass. This suggests that a cellular component may be involved, such as an alteration in sarcoplasmic reticulum function. In addition, there is decreased mitochondrial function and reduced protein synthesis. Interestingly, there is often an increase in capillary density and sometimes blood flow to muscles undergoing atrophy.^{127,137,138} Removal of weight-bearing activity has less effect on type II muscle fibers because these do not perform a major function in maintaining posture. In fact, with reduced stress, there may be an increase in type II muscle fibers. This results in increased maximum muscle velocity, not only in type II muscle fibers, but also in type I muscle fibers. These changes help attenuate a decline in power output as a result of atrophy.

The length at which the muscle is immobilized also affects the degree of atrophy, with muscles immobilized in a shortened position atrophying to a greater degree, with reduced force generating capacity and oxidative activity. When a muscle is immobilized in a shortened position, the number of sarcomeres decreases, leading to reduced muscle fiber length. If the muscle is immobilized in a

Box 7-6 Summary of the Effects of Disuse and Immobilization on Skeletal Muscle

- Postural type I muscle fibers, extensor muscles, and muscles that cross a single joint are at greatest risk for atrophy.
- Immobilization with the muscle in a shortened position results in greater atrophy.
- Muscle strength decreases rapidly during the first week of immobilization, with further losses occurring more gradually.
- Loss of muscle force production not entirely explained by muscle atrophy alone.
- Alteration in sarcoplasmic reticulum function occurs.
- Sarcoplasmic reticulum calcium uptake is decreased.
- Mitochondrial function is decreased.
- Protein synthesis is reduced.
- Oxidative and glycolytic enzyme activities are reduced and are associated with reduced capacity for energy production.
- Responsiveness to insulin is reduced, which affects ability of muscle to take up glucose.
- Ca-ATPase activity is reduced following immobilization.
- Muscle atrophy is an imbalance of protein formation and degradation, being primarily a decrease in muscle protein synthesis.
- Muscle RNA synthesis is decreased up to 50%.
- Ubiquitin-proteasome pathway of intracellular protein degradation accounts for majority of nonlysosomal protein degradation and is a complex multicomponent system that selectively targets proteins for destruction.
- Pathway uses ubiquitin, a protein that covalently binds to targeted proteins and acts as a destruction signal for the proteins.
- Ubiquitin-tagged proteins degraded by a large protease complex, the 26S proteasome.
- Responsible for liberation of amino acids via cleavage of proteins such as myosin.
- Ubiquitin ligases MuRF-1 and MAFbx increase more significantly.
- Inflammatory process may be present during atrophy.
- mRNA upregulation of TNF- α , IL-1, IL-6, and leukocyte marker CD45 occurs.
- TNF- α initiates transcription of NF- κ B.
- Some NF- κ B mediators (p50 and Bcl-3) increase and are associated with disuse muscle atrophy, suggesting an alternate, noninflammatory NF- κ B.
- Inhibition of NF- κ B alone does not prevent muscle atrophy.
- NF- κ B activates MuRF-1.
- Some protein kinases activated by TNF- α or ROS, such as hydrogen peroxide, may be associated with oxidative stress and the generation of ROS, which also activate NF- κ B and FOXO.
- The IGF-1/PI3K/Akt pathway is an important intracellular regulator of muscle hypertrophy.
- IGF-1 stimulates activation of PI3K.
- PI3K creates a cell membrane binding site for a serine/threonine kinase.
- Akt (also referred to as *protein kinase B*) activates rapamycin (mTOR) kinase.
- mTOR increases protein synthesis by activating p70S6 kinase and eukaryotic translation factor 4E binding protein 1 (4E-BP-1), regulatory proteins involved in protein synthesis.
- IGF-1/PI3K/Akt pathway suppresses expression of ubiquitin ligases that are important in signaling muscle atrophy.
- The IGF-1/PI3K/Akt pathway also activates FOXO proteins, transcription factors that regulate target genes for metabolism, apoptosis, and cell cycles.
- Akt phosphorylates FOXO proteins.
- This prevents translocation to the cell nucleus, inhibiting transcription functions.
- Blockade of FOXO by the IGF-1/PI3K/Akt pathway is important to prevent muscle atrophy.

Ca-ATPase, Calcium adenosine triphosphatase; *FOXO*, forkhead box O; *IGF*, insulinlike growth factor; *IL*, interleukin; *MAFbx*, muscle atrophy F-box; *mTOR*, mammalian target of rapamycin; *MuRF-1*, muscle ring finger 1; *NF- κ B*, nuclear factor kappa B; *PI3K*, phosphatidylinositol-3-OH kinase; *ROS*, reactive oxygen species; *TNF*, tumor necrosis factor.

stretched position, the fibers become lengthened because of the addition of sarcomeres.^{136,137} Connective tissue is increased in atrophic muscles, which may result in increased muscle stiffness.

Models of reduced limb usage include hindlimb immobilization (HI), hindlimb suspension (HS), denervation, and spaceflight. There is greater muscle atrophy with HS than HI because HI muscles are able to contract against the immobilizing material. In contrast, HS muscles contract, but there is no load to contract against.

The change in muscle fiber size and fiber percentage was studied in three heads of the dog quadriceps following 10 weeks of rigid immobilization with an ESF.¹³⁵ Muscle fiber atrophy was greatest in the vastus medialis and least in the rectus femoris. The atrophic response for type I fibers was, in order from most to least atrophied, vastus medialis, vastus lateralis, and rectus femoris;

for type II fibers, atrophy of the vastus medialis was equal to vastus lateralis, and both atrophied more than the rectus femoris. In fact, vastus medialis type I and II muscle fiber areas were only approximately one third of normal following immobilization. Tetanic torque also declined by approximately 60%, and muscle fiber diameter correlated strongly with torque.¹³⁹ The rectus femoris acts as a knee extensor and hip flexor and crosses two joints, which makes it less susceptible to atrophic changes because the hip joint was not immobilized in this model. The changes in the vastus medialis suggest that this muscle performs a greater postural function than the vastus lateralis. A similar study with 10 weeks of immobilization indicated that there was a significant decrease in both type I and type II fiber area, and muscle fiber areas recovered to only approximately 70% of control values following 4 weeks of remobilization.¹⁴⁰ Tetanic torque improves

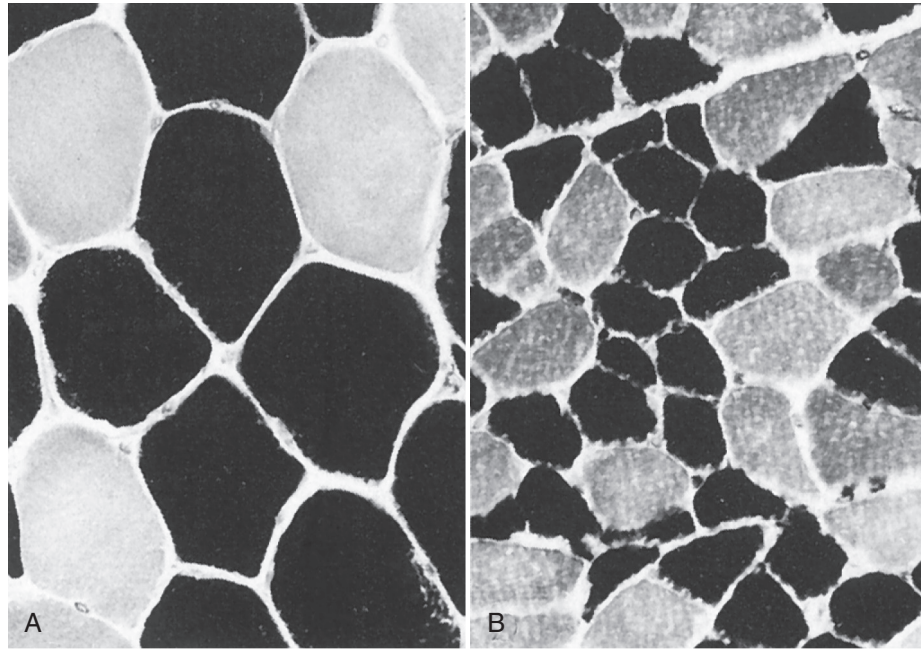


Figure 7-4 Photomicrographs of biceps femoris muscle. **A**, Normal muscle. **B**, Same muscle following 60 days of disuse. There is general atrophy of both types I (light stained) and II (dark stained) muscle fibers. (Figure 9-9 from Carlton WW, McGavin MD: *Thompson's special veterinary pathology*, St Louis, 1995, Mosby.)

during remobilization, but remains below preimmobilized levels.¹³⁹

In general the muscles most vulnerable to immobilization-induced atrophy are those that cross a single joint and have a large proportion of type I muscle fibers, such as the vastus medialis and vastus intermedius.¹³⁵ The next most susceptible muscles are antigravity muscles that cross multiple joints and have primarily type I muscle fibers. Examples of these muscles are the gastrocnemius and rectus femoris muscles. Muscles least susceptible to atrophy following immobilization are those that are intermittently activated and have predominantly type II muscle fibers, including the cranial tibial, long digital extensor, and biceps femoris muscles.

Biochemical changes also occur in muscles with immobilization. The effect of immobilization on muscle carbohydrate metabolism was investigated in dogs.¹⁴¹ Total carbohydrate and glycogen content of skeletal muscle fell during immobilization. The glycogen-degrading enzyme phosphorylase was activated 1 week after immobilization. Thereafter, enzyme activity decreased and remained significantly lower. In contrast, muscle glucose and lactate concentrations were unchanged. Oxidative and glycolytic enzyme activities may be reduced and are associated with reduced capacity for energy production. Immobilization may also result in reduced glutathione content.¹⁴² Reduced muscle activity may reduce the responsiveness of muscle to insulin, which affects the ability of muscle to take up glucose. There is also increased total muscle calcium concentration, decreased sarcoplasmic reticulum

calcium uptake, and reduced Ca-ATPase activity following immobilization.

The blood supply to the myotendinous junction is also affected by immobilization. In one study of rats, the vascular density of the myotendinous junction was decreased by 30% after 3 weeks of immobilization.¹⁴³ Vascular density returned to normal after 8 weeks of cage remobilization. Progressively increasing running resulted in greater vascular density. The effects of immobilization on peak oxygen uptake and effective oxygen diffusive conductance in skeletal muscle were studied in dogs.¹⁴⁴ Dogs were cage-confined for 8 weeks, with immobilization of a hindlimb for the last 3 weeks. Gastrocnemius muscles were electrically stimulated to elicit peak oxygen uptake at three levels of arterial oxygenation. Immobilization was sufficient to reduce muscle mass by 31% and citrate synthase activity by 68%; however, it had no effect on peak oxygen uptake or oxygen diffusive conductance. Immobilization also apparently does not affect muscle oxygen diffusing capacity.¹²⁷

Other species have similar changes in muscle with immobilization. Immobilization of a hindlimb of sheep for 9 weeks resulted in an 8% reduction in thigh circumference and a slight decrease in the area of type I fibers.¹⁴⁵ In addition, $\text{Na}^+\text{-K}^+$ pump concentration and citrate synthase activity were reduced, which may contribute to fatigue during activity. A modified rat tail suspension model, which allows the rear limbs to remain mobile without any weight bearing, was used to determine the atrophic response of muscle with unloading.¹⁴⁶ Maximal muscle atrophy occurred within 14 to 30 days. The gastrocnemius was less

severely affected by suspension than by other immobilization techniques, suggesting that muscle atrophy in the suspension model is different from immobilization atrophy. One significant response was the occurrence of decreased blood flow, which may indicate that hypodynamic nonimmobilized muscle has altered functional demands. Similar changes occur in humans. Patients immobilized with a cast after tibial fractures or ligamentous injuries had decreased limb circumference immediately after removal of the cast.¹⁴⁷ These differences persisted even after 81 days of remobilization.

Prolonged bed rest in people causes changes in muscle mass.¹⁴⁸ In humans, 4 to 5 weeks of bed rest results in a 10% loss of muscle mass in the lower limbs and 4 months of bed rest results in 20-30% loss of muscle mass. In people, 10 minutes of standing per day during bed rest attenuates muscle atrophy by 25%. If some resistance exercise is added, muscle atrophy is attenuated by up to 50%. In addition, there is an apparent shift in muscle fiber type with bed rest. After 35 days of bed rest in one study, there was a slow to fast shift in muscle phenotype.¹⁴⁹ Consistently, fiber type distribution was shifted toward type II and IIA fibers. Muscle fiber atrophy was variable among fiber types. Myosin concentration was significantly lower in type I and type IIA muscle fibers after bed rest than before bed rest, whereas the myosin/actin ratio did not vary, suggesting a disproportionate loss of myosin compared with fiber cross-sectional area and a proportional loss of myosin and actin.

Spaceflight may also result in significant loss of muscle mass because of the loss of weight bearing and loading.¹⁴⁸ Muscle volumes of the calf, thigh, and lower back were determined before and after an 8-day space shuttle mission. The soleus-gastrocnemius (−6.3%), anterior calf (−3.9%), hamstrings (−8.3%), quadriceps (−6%), and intrinsic back (−10.3%) muscles were decreased 24 hours after landing compared with baseline. After 2 weeks, the hamstrings and intrinsic lower back muscles were still below baseline.

Reflex inhibition of muscles, a situation in which sensory stimuli impede the voluntary activation of muscle, may occur with joint injury and result in muscle atrophy, and loss of strength and activation of the muscle. Reflex inhibition may be measured directly by electromyography, or the sequelae of reflex inhibition may be measured, such as thigh circumference measurement and muscle biopsy.¹⁵⁰ The most frequently cited causes of muscle reflex inhibition in joint injury are pain, joint effusion, and joint immobilization. Finally, there may be selectivity of affected muscles and muscle fibers with reflex inhibition following joint injury. In light of these findings, several suggestions have been offered for preventing reflex inhibition that can be applied to rehabilitate the most affected muscle group, the quadriceps femoris. These include cryotherapy, transcutaneous electrical nerve stimulation, electromyostimulation, traditional exercise training, joint

mobilization, rest, and proper positioning of the limb in rest and exercise.

Mechanisms of Muscle Atrophy

It is believed that muscle atrophy occurs as a result of an imbalance of protein formation and degradation, and although recent research has begun to unravel how this imbalance occurs, the exact mechanisms are unknown. Most studies suggest that decreased muscle protein synthesis is the prime cause of muscle atrophy, although there is also transient muscle protein degradation in early muscle atrophy. In people, muscle protein synthesis decreases in the first 10 days of immobilization, and then the rate of muscle synthesis appears to stabilize. RNA synthesis also decreases with immobilization.¹³⁸ Quadriceps muscle protein turnover was assessed in men immediately after the end of 37 days of leg immobilization in a plaster cast after tibial fracture.¹⁵¹ Quadriceps muscle protein synthetic rate was 0.046% per hour in the contralateral leg, but was only 0.034% per hour in the immobilized leg. Muscle RNA activity in the immobilized leg was 50% that of the nonimmobilized leg. Immobilization was associated with significant atrophy of type I muscle fibers, but no significant change occurred in type II fiber diameter. Mean quadriceps fiber volume was smaller in the injured leg by 10.6% and the difference in muscle thigh volume was 8.3%. Some studies have shown little evidence of increases in mRNA for catabolic enzymes or increases in enzyme activity during this period,¹⁵² although others suggest that such increases occur very quickly after immobilization or spinal cord transection.¹⁵³ However, it appears most likely that the process of muscle atrophy during immobilization is primarily a result of decreased protein synthesis, with continued normal muscle breakdown.¹⁵⁴

Recently, attention has focused on the role of the ubiquitin-proteasome pathway of intracellular protein degradation. It is believed that this system accounts for the majority of nonlysosomal protein degradation and is a complex multicomponent system that selectively targets proteins for destruction.¹⁵⁵ The pathway uses ubiquitin, a protein that covalently binds to targeted proteins and acts as a destruction signal for the proteins. Three distinct conjugating enzymes are involved in this process: an E1 ubiquitin-activating enzyme, an E2 ubiquitin-conjugating enzyme, and an E3 ubiquitin-ligating enzyme.¹⁵⁶ The ubiquitin-tagged proteins are then degraded by a large protease complex, the 26S proteasome. This pathway is responsible for the actual liberation of amino acids via cleavage of long-lived proteins, such as myosin. Two muscle-specific ubiquitin ligases, muscle ring finger 1 (MuRF-1) and muscle atrophy F-box (MAFbx), increase more significantly than other components of this proteolytic pathway in mice.¹¹⁴ A study of the molecular aspects of skeletal muscle atrophy demonstrated the importance of these ligases in a mouse model.¹⁵⁷ In this model, significant

skeletal muscle atrophy occurred, with the myofiber size of the cranial tibial muscle reduced by 36% compared with the untreated contralateral limb within a few days. MAFbx and MuRF-1 mRNAs were significantly upregulated by two- and sixfold, respectively. Interestingly, an early inflammatory process was present during atrophy in this study, characterized by the mRNA upregulation of TNF α , interleukin (IL) 1, and IL-6 (2-, 2.4-, and 3.4-fold, respectively), along with upregulation of the leukocyte marker CD45 (6.1-fold). Muscle rapidly recovered with remobilization, with increased levels of uncoupling protein-3 and peroxisome proliferator-activated receptor gamma coactivator-1 alpha mRNA, important components of oxidative muscle metabolism. Studies of rats in a hindlimb suspension model have also shown increased function of the ubiquitin-proteasome pathway, with increased ubiquitin mRNA levels, conjugating enzymes, and proteasome subunits, indicating the potential importance of this pathway of muscle degradation. Manipulation of this system of muscle degradation may alter the process of muscle atrophy and preserve muscle mass during periods of immobilization. However, knock-out of the MuRF1 and MAFbx genes does not completely prevent atrophy, suggesting that other ligases are involved.¹¹⁴

The IGF/phosphatidylinositol-3-OH kinase (PI3K)/Akt pathway is an important intracellular regulator of muscle hypertrophy. IGF-1 stimulates the activation of PI3K, which then creates a cell membrane binding site for a serine/threonine kinase, Akt (also referred to as *protein kinase B*). Akt activates mammalian target of rapamycin (mTOR) kinase, which increases protein synthesis by activation of p70S6 kinase and eukaryotic translation factor 4E binding protein 1 (4E-BP-1), regulatory proteins involved in protein synthesis. The IGF-1/PI3K/Akt pathway also suppresses the expression of ubiquitin ligases, which are important in signaling muscle atrophy.¹¹⁴ The IGF-1/PI3K/Akt pathway activation also activates forkhead box O (FOXO) proteins, transcription factors that regulate target genes for metabolism, apoptosis, and cell cycles. Akt phosphorylates FOXO proteins, which prevents translocation to the cell nucleus, effectively inhibiting their transcription functions. Thus blockade of FOXO by the IGF-1/PI3K/Akt pathway is important in helping to prevent muscle atrophy and may be an approach to help prevent muscle atrophy.

Chronic inflammation may also induce protein loss and muscle atrophy through the action of inflammatory mediators on the nuclear factor kappa B (NF- κ B) pathway.¹¹⁴ TNF- α is one inflammatory mediator that initiates transcription of NF- κ B. Although disuse atrophy does not result in increased TNF- α , certain NF- κ B mediators (p50 and Bcl-3) increase and are associated with disuse muscle atrophy, suggesting an alternate, noninflammatory NF- κ B pathway. Furthermore, several genes associated with p50 and Bcl-3 and muscle atrophy have been identified in a mouse model of disuse atrophy.¹⁵⁸ Although NF- κ B

activity inhibition attenuates cancer cachexia, its inhibition is not sufficient to prevent disuse muscle atrophy. Another potential mechanism of signaling muscle atrophy is the activation of MuRF-1 by NF- κ B. Mitogen-activated protein kinases may also be involved in muscle atrophy, especially when activated by TNF- α or reactive oxygen species (ROS), such as hydrogen peroxide. Immobilization of limbs is associated with oxidative stress and the generation of ROS, which may also activate the NF- κ B and FOXO systems.¹¹⁴ Inhibition of both NF- κ B and FOXO systems can reduce muscle atrophy as a result of immobilization. In one study of rat limb immobilization, control treatment resulted in a 47% decrease in muscle fiber cross-sectional area.¹⁵⁹ Fibers from immobilized muscles that had inhibition of FOXO induction had a 76% inhibition of atrophy. Inhibition of both FOXO and NF- κ B had even greater inhibition (up to 95%) of muscle atrophy. These findings suggest that the combined effects of inhibiting immobilization-induced NF- κ B and FOXO transcriptional activity has an additive effect on preventing immobilization-induced atrophy. The mechanisms of muscle atrophy may be more complex in people than in small laboratory animals.¹⁶⁰ For example, anabolic resistance may partially explain, in part, reduced muscle protein synthesis of atrophied muscle in the postprandial state, even with high levels of amino acid present. This anabolic resistance during immobilization could have implications on nutritional interventions.

Myostatin may also play a role in disuse atrophy of muscle because unloading in a rat hindlimb unloading model resulted in an upregulation of myostatin and activin IIB, the myostatin receptor.^{161,162} However, the role of myostatin in muscle atrophy remains unclear because the soleus muscle, which contains no myostatin, undergoes the greatest atrophy with unloading. It may be that myostatin inhibits hypertrophy rather than induces atrophy.¹¹⁴ Another study evaluated antibody-directed myostatin inhibition in mice during unilateral hindlimb casting.¹⁶³ Although hindlimb casting reduced muscle mass, fiber size, and function of soleus and extensor digitorum longus muscles in control animals, antibodies directed against myostatin attenuated the loss of muscle mass, fiber size, and function with greater effectiveness when given early in immobilization.

Muscle use and disuse apparently govern the number of acetylcholine receptors that are regulated through a feedback mechanism. Immobilization of skeletal muscle results in disuse atrophy and resistance to nondepolarizing muscle relaxants. Dogs undergoing cast immobilization of a hindlimb for 3 weeks had resistance to metocurine, a muscle relaxant, by the fourth day of casting.¹⁶⁴ This resistance persisted for 2 weeks after cast removal, but was normal 6 weeks after cast removal. These responses suggest that there may be upregulation of muscle acetylcholine receptors as a result of immobilization.¹⁶⁵ These changes

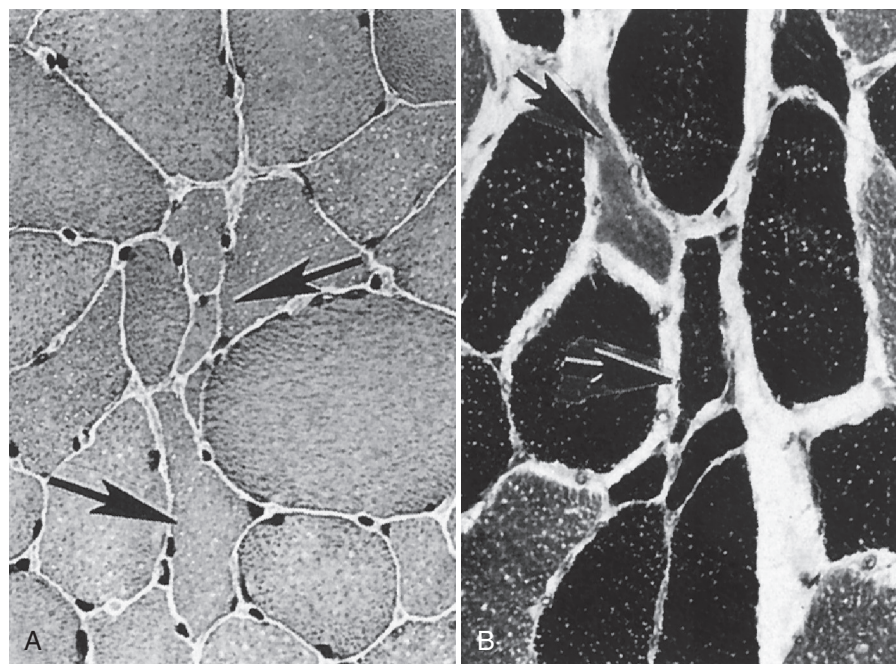


Figure 7-5 Neurogenic muscle atrophy. **A**, Early denervation atrophy as indicated by small angular fibers (arrows). **B**, Atrophic angular fibers of both type I (light) and type II (dark) muscle fibers. (From Carlton WW, McGavin MD: *Thompson's special veterinary pathology*, St Louis, 1995, Mosby.)

may also occur in patients with muscle weakness who are in the intensive care unit. A canine intensive care model that involved 3 weeks of anesthesia with pentobarbital and positive-pressure ventilation indicated that the absence of muscle tone and reflex responsiveness for 3 weeks was also associated with exaggerated resistance to neuromuscular blockade.¹⁶⁶

Neurogenic Muscle Atrophy

Muscles undergo denervation atrophy with damage to the spinal motoneurons or to the motor nerves in the ventral roots.¹⁶⁷ It has been well documented that there is an early rapid loss, followed by stabilization, of muscle mass with neurogenic muscle atrophy (Figure 7-5).^{167,168} On the other hand, disuse atrophy occurs with damage to the central nervous system. Neural activity is generally reduced with spinal cord lesions, but varies with the type of lesion and the level of spasticity.

Structural and biochemical changes occur with neurogenic atrophy, including changes in glycolytic and oxidative metabolic enzymes. Functional recovery following nerve injury and repair is directly related to the degree of muscle atrophy that takes place during the period of nerve regeneration and the ability of surviving motoneurons to sprout and reinnervate the denervated muscle fibers.^{167,169} For injuries with 85% of the motor supply to a partially denervated muscle intact, all denervated muscle fibers are reinnervated and the partially denervated muscle recovers completely. Conversely, lesions with injury to 85% of the

motoneurons result in inadequate sprouting and incomplete recovery. Motoneurons may increase the number of muscle fibers they normally supply by fivefold with sprouting. However, the distance over which sprouting occurs is limited and with extensive denervation, nerve sprouts may not grow far enough to reinnervate muscle. The extent of muscle atrophy is related to a number of factors, including the distance through which the nerve must regenerate; the age of the patient; and the type of nerve injury and other associated tendon, soft tissue, and bony damage. In animals with spinal cord injuries, the capacity of paralyzed muscles to sustain contractions is reduced, therefore reducing muscle endurance. Adequate nursing care of paraplegic or tetraplegic animals is necessary to prevent muscle atrophy from disuse.¹⁷⁰ Secondary conditions, such as disuse osteoporosis of bones and contracture of joints, must also be considered in dealing with patients having neurogenic muscle atrophy.

Computed tomography was used to evaluate neurogenic muscle atrophy in dogs with induced sciatic nerve injury.¹⁷¹ The cross-sectional area of muscles decreased 1 to 2 weeks after denervation, and the differences were significant after 3 weeks. Additional decreases were minimal after 28 days, after which time there was significant infiltration of adipose tissue into the muscle. The cross-sectional areas of the biceps femoris, semimembranosus, and semitendinosus muscles decreased by approximately 30% by 60 days, with most of the changes occurring in the first 2 to 3 weeks. In another study of denervated latissimus dorsi muscle, there

was a 20% loss of lean muscle mass after 6 six weeks, 45% loss at 6 months, and 55% loss at 9 months.¹⁶⁸ Histologic examination found atrophy of individual muscle fibers. With denervation atrophy, there is preferential atrophy of type II muscle fibers which is the opposite of what is seen with immobilization of innervated muscles.¹³⁸

Denervated dog gastrocnemius muscle has a progressive decrease in total protein content, alanine aminotransferase, aspartate aminotransferase, and glutamate dehydrogenase activity, and elevation in free amino acid, ammonia, urea, glutamine contents, and AMP deaminase activity levels following neurectomy.¹⁷² Total protein content decreased 50% by 4 weeks after muscle denervation. Catabolism of muscle protein may result in increased metabolites, including ammonia, urea, and glutamine.

Serum carbohydrate metabolism was evaluated in denervation atrophied and denervation electrical-stimulated dogs.¹⁷³ Neuromuscular electrical stimulation was applied to the gastrocnemius muscles of the latter group for 30 min/d for 15 days, using biphasic pulses of 10 V, 100 ms duration and 2 Hz. Denervation atrophy may result in hyperglycemia possibly resulting from the lack of uptake by the muscle. Electrical stimulation applied to denervated muscle may reverse the hyperglycemia, indicating that muscular work may be important in modulating serum carbohydrate metabolism.

Continuous electrical stimulation of denervated muscles during the period of nerve regeneration appears to help maintain the integrity of the muscle fibers and their potential functional capacity.¹⁶⁹ In addition, electrical stimulation may be applied for extended periods with little evidence of discomfort. This modality may be useful in patients with the potential for partial or complete recovery from nerve injury.

Changes in Muscle with Aging

Changes in muscle as people and animals age include decreased muscle mass, fiber size, and fiber number. These changes, along with increased intramuscular fat and connective tissue, result in reduced ability to generate muscle force.^{174,175}

Muscle satellite cells also decrease by the age of 70 in people.¹⁷⁶ In addition, the loss of muscle mass and muscle function may result from diminished regenerative capacity of muscle, increased preprogrammed cell death activity (apoptosis), anabolic resistance, protein degradation, and autophagocytosis occurring in skeletal muscle.¹⁷⁷ Although muscle mass declines with aging, the loss of muscle strength occurs at a greater rate than the loss of muscle mass in older people, suggesting that atrophy alone is not the cause of strength loss.

Neuromuscular changes may contribute to the loss of muscle strength, including reduced muscle activation and contractile rate.¹⁷⁸ In fact, there is a loss of spinal motor neurons with aging, which is paralleled by a reduction in

muscle fiber number and size, resulting in reduced muscle function and strength.¹⁷⁹ Aging results in reduced maximal motoneuron firing frequency and muscle activation. There is some thought that muscle atrophy with aging results, in part, from neuropathic changes that lead to motoneuron death, which leads to a decrease in the number of muscle fibers. In addition, motor unit recruitment and firing frequency are reduced in older adults.^{180,181}

Interestingly, unloading skeletal muscle in older animals leads to less disuse muscle atrophy compared with that of young animals, but skeletal muscle regrowth is slowed and less complete in older animals as compared with young during remobilization.¹⁸² Similar findings occur in humans.¹⁸³ One reason for altered recovery of muscle mass in young versus older animals after limb immobilization may be related to differences in gene expression. In one study, HI of rats resulted in a 27-37% loss in soleus muscle mass. The atrophied soleus muscle of young but not old rats recovered to the same mass as before treatment.¹⁸⁴ The mRNA levels of muscle growth potentiating factors were differentially expressed between young and older animals, and may modulate muscle growth that could explain the failure of muscles of older animals to recover lost muscle mass. In addition to changes in muscle and the nervous system with aging, tendon changes also contribute to muscle weakness with aging. The interplay between these muscle and tendon changes has an effect on the length-force and force-velocity relationships of the muscle and is likely to affect the ROM, rate of force development, maximum force development, and speed of movement of the older individual.¹⁸⁵

Recently, it has been shown that resistance training may attenuate the decline in muscle performance with aging. In people, strength training programs of 3 months' duration may increase muscle cross-sectional area by 5-17%, which is similar to changes in young adults.^{186,187}

Exercise can induce the expansion of satellite cells in skeletal muscle of older people, as well as younger people.¹⁷⁶ In addition to muscle hypertrophy as a result of strength training in older people, there also appear to be significant changes in neuromuscular function, leading to improved function.¹⁷⁹

Muscle Changes with Orthopedic Conditions

Hip Conditions

Hip conditions are common in dogs and may affect muscle mass. In some conditions, such as hip dysplasia, lameness and reluctance to exercise may result in pelvic limb muscle atrophy with compensatory shoulder muscle hypertrophy.¹⁸⁸ Excision of the femoral head and neck is sometimes performed as a treatment for severe hip dysplasia or fractures of the acetabulum. Although fewer dogs have obvious lameness or pain in the operated hip after recovery from surgery, muscle atrophy is noted in approximately 50% of dogs following femoral head and neck excision, even after

8 years.¹⁸⁹ Moderate or marked muscle atrophy is most common in larger breeds of dogs. Difficulty in jumping and in climbing stairs may be noted postoperatively in larger breeds of dog and may be related to the loss of power needed for jumping and hip extension following femoral head and neck excision. Legg-Calve-Perthes disease, or aseptic necrosis of the femoral head, is a condition affecting small breeds of dogs that is also treated with a femoral head and neck excision. In a clinical study of dogs with avascular necrosis of the femoral head, muscle atrophy was present in 25 of 35 dogs at the time of diagnosis as a result of pain and lameness.¹⁹⁰

Cranial Cruciate Ligament Injury

Changes in muscle mass are common in dogs with CCLR and following surgical treatment of the injury. In an experimental study, dogs had a CCL transected, followed by immediate stabilization with an extracapsular procedure.¹⁹¹ Results of this study revealed muscle atrophy of the surgical leg by 2 weeks, with muscle mass beginning to return between 4 and 8 weeks, but significant atrophy was still present 8 weeks after surgery. In addition, the contralateral nonsurgical limb underwent hypertrophy, possibly because of the increased loading on that limb during recovery.

A study of dogs with naturally occurring CCL deficiency evaluated patients before surgical treatment and 1.5, 7, and 13 months after surgery.¹⁹² The degree of quadriceps muscle atrophy present prior to surgery correlated significantly with the degree of cartilage fibrillation, indicating a relationship with the severity of the condition. Although there was slightly greater muscle atrophy 6 weeks after surgery, muscle mass improved 7 and 13 months after surgery, but significant residual muscle atrophy remained in many dogs even after 1 year. A measure of quadriceps atrophy may be a useful tool for assessing long-term outcome.

A long-term outcome of thigh circumference of 29 clinical patients with CCLR and stabilization with tibial plateau leveling osteotomy (TPLO) was recently reported.¹⁹³ Standing mid-thigh circumference measurements were made in triplicate. Clinical lameness was graded blindly. Although the mean thigh circumference for the operated limb was 98.5% of the contralateral limb, a significant difference existed between the limbs. Time after surgery had no apparent effect on outcome. Four dogs (14%) exhibited some lameness in the TLPO limb during evaluation (one dog was 1 to 2 years postoperative and three dogs were 2 to 3 years postoperative). The source of the lameness was not identified, and there was no difference in thigh circumference between dogs with lameness and those without lameness.

Information regarding changes in specific muscles following CCLR and stifle stabilization surgery may be useful to target specific muscles for rehabilitation. Muscle atrophy was studied in a group of dogs undergoing surgical

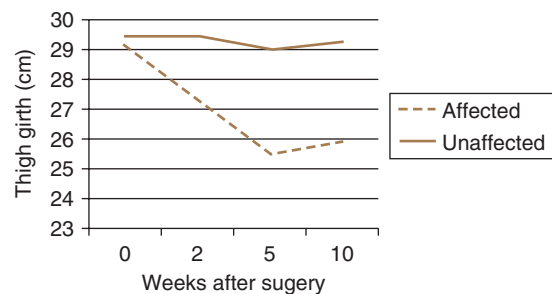


Figure 7-6 Rate of muscle atrophy following cranial cruciate ligament transection and immediate stifle stabilization with no postoperative rehabilitation.

transection of a CCL and immediate stifle stabilization using a modified retinacular imbrication technique.¹⁹⁴ Thigh circumference and body composition measured by dual-energy x-ray absorptiometry (DEXA) were used to estimate limb muscle mass before surgery, and 2, 5, and 10 weeks after surgery. Actual mass of rear limb muscles was determined at 10 weeks. Muscle mass decreased with transection of the CCL, despite immediate stifle stabilization (Figure 7-6). The changes were evident by 2 weeks, and continued until 5 weeks, followed by a slight recovery at 10 weeks. The muscles most affected at 10 weeks were the quadriceps, biceps femoris, and semimembranosus, and these muscles weighed only 60% of the intact contralateral side (Figure 7-7). The muscles of the crus are similarly affected (Figure 7-8).

Changes in thigh girth and DEXA lean tissue mass were similar over time and indicated significant muscle atrophy following cruciate transection and repair. The quadriceps and biceps femoris muscles are important in supporting weight during stance, and should be targeted during rehabilitation to improve muscle mass and strength. The time course of changes suggest that postoperative physical rehabilitation should continue for at least 5 weeks after surgery.

Another study evaluated morphometric characteristics of hindlimb muscles in Labrador retrievers with and without CCL deficiency and found that atrophy associated with CCL deficiency predominantly affected the quadriceps muscle.¹⁹⁵ The authors also suggested that dominance of the gastrocnemius muscle over active restraints to cranial tibial thrust might be associated with predisposition to CCL deficiency in Labrador retrievers. This dynamic imbalance between muscle groups of the rear limbs could serve as a basis for screening programs and preventive rehabilitation, especially considering that targeted training programs, such as plyometric training, may reduce the risk of injury 3.6 times in women.¹⁹⁶

Muscle atrophy also occurs in people following surgery for torn ACLs. Thigh and calf girth measurements of involved and noninvolved extremities were determined prior to and following knee surgery for acute and chronic

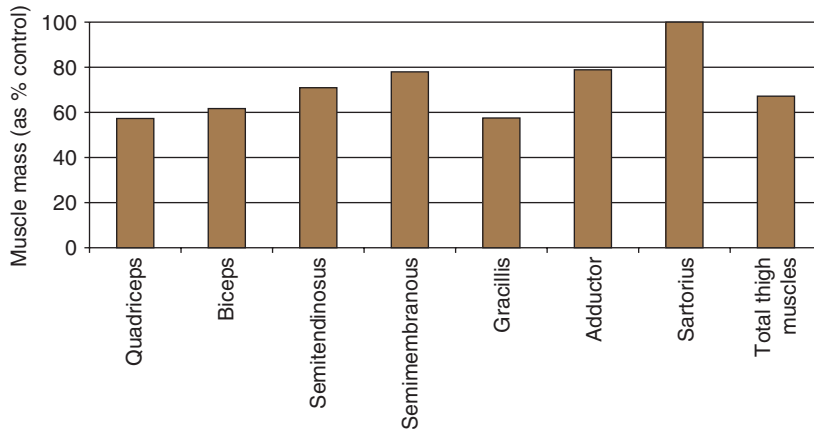


Figure 7-7 Mass of thigh muscle mass 10 weeks after cranial cruciate ligament transection and stifle stabilization with no rehabilitation. Values are percentage of the mass of the unaffected contralateral limb.

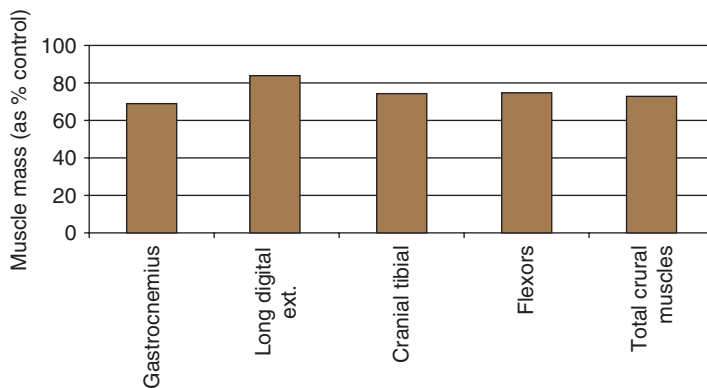


Figure 7-8 Mass of crus muscle mass 10 weeks after cranial cruciate ligament transection and stifle stabilization with no rehabilitation. Values are percentage of the mass of the unaffected contralateral limb.

knee injuries.¹⁹⁷ Although thigh and calf girth measurement differences existed between the involved and noninvolved extremities prior to and after surgery, the bulk of the girth measurement differences existed prior to surgery for both groups. The differences between involved and noninvolved limbs were in the 2-6% range, which were similar to differences observed in another study.¹⁹⁸ They concluded that altered gait and reflex inhibition contributed to muscle atrophy following injury and surgery. Muscular changes more than 5 years following ACL injury were evaluated to determine if there were differences between patients with ACL reconstruction and conservative treatment.¹⁹⁹ Thigh circumference was measured and the relationship of strength to electrical activity and muscle size was analyzed. Only minimal differences in thigh circumference were found between the uninjured and injured limb of both groups; however, isokinetic torque and electromyogram (EMG) values showed significant differences. These changes may be due to modified use of muscle fibers and altered joint receptor afferent signaling following injury.

These changes may result in chronic loss of quadriceps femoris strength in humans following ACL surgery despite aggressive rehabilitation. One study evaluated the correlation of thigh muscle size and strength with thigh circumference, muscle cross-sectional area by MRI, and isokinetic

strength.²⁰⁰ Patients were evaluated 48 months after surgery. There was a significant 1.8% decrease in thigh circumference, a 10% decrease in average quadriceps torque, and an 8.6% decrease in quadriceps cross-sectional area by MRI in the involved extremities compared with the uninvolved extremities. Persistent quadriceps weakness and decreased cross-sectional area 48 months postsurgery and rehabilitation suggest that some patients may not completely recover from surgery, despite aggressive therapy. Strength deficits may persist for long periods, even with less severe injuries, such as partial tears of the medial collateral ligament (MCL).²⁰¹ Muscle atrophy that occurs following cruciate injury and surgery may result in early muscle fatigue, which also affects EMG patterns of quadriceps and hamstring muscles.²⁰² As little as 10 minutes walking may affect the gait pattern, particularly the activity of the hamstring muscles. This suggests that endurance training, in addition to strength and stabilization training, should be emphasized during rehabilitation.

Various specific techniques can be used for strengthening of the hamstring and quadriceps muscles in people following surgery for ACL repair, with or without resistive equipment.²⁰³ In addition, electrical stimulation may help prevent muscle wasting caused by immobilization. People are generally able to return to full activity and competitive

sports after 6 to 12 months. An intense rehabilitation program following TPLO surgery for stabilization of a CCLR may improve muscle mass significantly more than standard home care. In one study, dogs with cruciate-deficient limbs had significantly reduced thigh circumference before surgery as compared with values for the contralateral limb.²⁰⁴ The rehabilitation group had significantly larger thigh circumference than the home-exercise group 6 weeks after TPLO, with no difference between the affected and nonaffected limbs.

Fractures

Muscle atrophy may occur following fractures, particularly those with severe comminution. A stable repair with anatomic reduction is vital to encourage early weight bearing, joint motion, and use of the limb. Distal femoral fractures treated with limb immobilization in extension for 3 to 7 weeks resulted in limb hyperextension, generalized muscle atrophy, abducted gait, and limited range of joint motion.²⁰⁵ Lesions in muscle biopsies included fiber size variability, increased fibrosis, and focal necrosis. Histochemical and morphometric studies showed significant type I fiber atrophy in the vastus lateralis muscle, with a reduction of the number of type I fibers. Atrophic changes in the gastrocnemius and biceps femoris muscles were not significant. The atrophy seen in the vastus lateralis muscle may be a result of immobilizing the limb in extension, which results in shortening of this muscle group. Immobilization of a muscle in a shortened position results in preferential atrophy caused by a reduced number of sarcomeres in series and reduced protein synthesis. In contrast, passive stretch of muscles in a lengthened state promotes muscle growth and may explain the relative sparing to the biceps femoris and gastrocnemius muscles.

A follow-up experimental study evaluated the effect of splinting in extension for 2 weeks following trauma to the distal portion of the quadriceps femoris muscle in dogs.²⁰⁶ Flexion of the stifle joint was limited after splinting. A reversible type I fiber atrophy of the vastus lateralis, biceps femoris, and gastrocnemius muscles occurred. Early type II fiber atrophy was seen in a few muscles. Multifocal fiber necrosis was the only irreversible change seen. Relative fiber percentages did not change appreciably during splinting or recovery.²⁰⁶

Arthritis

The relationship between arthritis and muscle mass is unknown in dogs, but we suspect that muscle mass and strength decline with the progression of OA and the subsequent reduction in activity. Muscle weakness may predispose to OA and its clinical signs in people.²⁰⁷ Reduced quadriceps strength is one of the earliest signs in people with knee OA, and the degree of strength loss has association with the severity of OA.¹⁷⁸ Muscle weakness may lead to functional deficits in joint stability and reduced physical

activity and possibly contribute to further joint deterioration. The ratio of quadriceps to hamstrings muscle strength is important in people for knee stability and for protection from excessive stress.²⁰⁸ People with knee OA have weak knee flexor and extensor strength, with relatively greater weakness in the quadriceps muscles. The mean quadriceps to hamstring strength ratio of 1.4 is less than ratios reported for young healthy adults of 2. The low ratio might be explained by reflex inhibition resulting from pain associated with disease, because the quadriceps muscle can be selectively inhibited by pain and effusion in the joint, leading to decreased production of muscle tension. In fact, joint enlargement had a negative correlation with the ratio of quadriceps to hamstring muscle strength in one study.²⁰⁹

The severity of the arthritis may also affect muscle mass. The strength and bulk of the quadriceps muscle of children with juvenile chronic arthritis were studied over a 2-year period.²¹⁰ Children with high OA severity scores had the least strength and muscle bulk. As the arthritis progressed and the severity score worsened, there was a corresponding reduction in muscle strength and bulk. The presence and intensity of local arthritis may be one important factor affecting muscle function in dogs with chronic arthritis.

Patients with knee OA had significantly weaker quadriceps strength, larger voluntary muscle activation deficits, and smaller lean muscle cross-sectional area compared with contralateral, nondiseased limbs.¹⁷⁸ Muscle activation deficits accounted for most of the variance in quadriceps strength of the OA limb, whereas lean muscle cross-sectional area explained the majority of variance in strength of the nondiseased limb. These findings suggest that small changes in muscle activation may have a profound effect on quadriceps strength in OA limbs.

Knee joint pain as a result of inflammation may also result in muscle atrophy. In one study, complete Freund's adjuvant was injected in rat knee joints to induce inflammation and pain.²¹¹ One week after injection, muscle atrophy occurred only on the injected side. Weight of the rectus femoris and soleus muscles was decreased by 20%. The reduction in cross-sectional areas was similar for fast and slow muscle fibers. The authors suggested that early treatment to inhibit joint pain and inflammation may prevent muscle atrophy.

Response of Muscle to Remobilization

Mechanical loading is essential for muscle regeneration (Box 7-7). Therefore following immobilization to protect healing muscle, early remobilization is required to induce correct growth and orientation of regenerated myofibers.²¹² Fortunately, most of the changes that occur in skeletal muscle during immobilization are reversible. One rule of thumb regarding humans states that a remobilization period of twice the duration of the immobilization period is necessary for a return of limb circumference to normal values.¹⁴⁷

Box 7-7 Considerations for Remobilization of Muscle after Immobilization

- Mechanical loading is essential for muscle regeneration.
- Regaining muscle mass during remobilization takes at least twice as long as the period of immobilization.
- There is decreased potential for muscle regeneration in older animals.
- Loss of muscle strength also occurs during immobilization.
 - Hindlimb immobilization for 10 weeks caused a 60% decline in tetanic torque that improved with 5 weeks of remobilization.
- Training, especially more intense training, results in more rapid muscle mass recovery.
 - Treadmill training and swimming are useful.
 - An adaptation period may be necessary in remobilized muscle to be able to withstand moderate training and avoid damage during early exercise.
 - Decline treadmill training may be more effective for some muscle groups.
 - Treadmill running three and six times per week is more beneficial for muscle recovery compared with no running or running once per week.
 - Resistance exercise stimulates muscle protein synthesis and activation of muscle satellite cells.
 - Eccentric muscle contractions causes greater satellite cell activation and muscle mass gains compared with concentric muscle contractions.
 - Satellite cell activation involves an initial inflammatory response.
 - Possible that NSAIDs may reduce postexercise muscle protein synthesis, but NSAIDs also decrease discomfort and muscle soreness.
 - Training before immobilization and remobilization has a positive effect on muscle mechanical behavior compared to being sedentary.
- Rapamycin (mTOR)-S6 kinase turns on muscle protein synthesis, and myostatin is rapidly suppressed with postimmobilization training.

mTOR, Mammalian target of rapamycin; *NSAID*, nonsteroidal antiinflammatory drug.

There appears to be an age-dependent decline in muscle regeneration potential, so remobilization and loading of muscle may be especially important in older animals.

The effect of remobilization following splinting the stifle joint of young dogs with and without muscle injury was studied.²⁰⁶ A rear limb was splinted in extension for 3 weeks and the biceps femoris, vastus lateralis, and gastrocnemius muscles were studied after splint removal and following 6 weeks of free activity. A reversible type I fiber atrophy occurred in most splinted muscles. Relative fiber percentages did not change appreciably during splinting or recovery. Beagle dogs undergoing HI by external fixation for 10 weeks had a 60% decline in tetanic torque, which improved somewhat during a 5-week remobilization

period.¹³⁹ The diameters of type 1 and type 2 fibers measured from the vastus lateralis muscle followed the same trend.

Similar improvements occur with remobilization in humans and other animals. Changes in limb circumference and passive resistance of the human knee caused by immobilization were studied during remobilization.¹⁴⁷ Patients immobilized with a long leg cast after tibial fractures or ligamentous injuries were evaluated immediately after cast removal and after 18, 36, and 81 days of remobilization. Although improved, differences in midthigh circumference were still present after 81 days of remobilization. During remobilization, the increased resistance to flexion found immediately after removal of the cast resolved. This may indicate a rapid readaptation of muscle length (shortened because of immobilization in a shortened position) to almost normal values. After cast immobilization for 3 weeks, rat limbs were remobilized by free cage activity or treadmill running.²¹³ Typical changes of muscle atrophy occurred during the immobilization period. Remobilization, especially treadmill running, resulted in return of muscle toward normal, particularly the fiber size, capillary number, and fiber type distribution. Another study of the same model that compared low and high intensity treadmill training after immobilization found that high-intensity activity was more effective in restoring muscles to normal than was low-intensity exercise.²¹⁴ Sheep hindlimbs were immobilized in a plaster splint for 9 weeks, followed by remobilization for 9 weeks.¹⁴⁵ There was a slight decrease in type I fiber area at 9 weeks, and a slight increase at 18 weeks. The Na(+)-K+ pump density and citrate synthase activity were reduced by 39% and 30%, respectively, after 9 weeks of immobilization. During remobilization both increased to the same level as in the control animals.

Physical condition prior to immobilization and remobilization may have an effect on muscle. One study evaluated training versus being sedentary before immobilization, and then rehabilitation. Rats were divided into seven groups: control (C); trained (T); sedentary (S); trained and immobilized (TI); sedentary and immobilized (SI); trained, immobilized, and rehabilitated (TIR); sedentary, immobilized, and rehabilitated (SIR). Swimming was used for training whereas sedentary animals were kept in cages of reduced size. Cast immobilization on a pelvic limb was used for limb immobilization, and rehabilitation was performed in water. Training improved all mechanical properties of muscle when compared with sedentarism. Training before immobilization and rehabilitation had a positive effect on muscle mechanical behavior compared with sedentarism.²¹⁵ Other studies have reported similar findings, and have suggested that training after immobilization should be carefully undertaken to avoid damage to muscle.²¹⁶ Remobilized muscle may not be able to withstand moderate training exercise before an adaptation period has occurred to avoid repeated damage during early

exercise. Another study also evaluated the effects of swimming training prior to and after a period of limb immobilization in rats. Treatment groups included C, 6 weeks under baseline conditions; T, swimming training for 6 weeks; TI, swimming training for 6 weeks and then immobilized for 1 week; and TIR, swimming training for 6 weeks, immobilized for 1 week, and then remobilized with swimming for 2 weeks. Regenerative characteristics of the soleus muscle were more significant in the groups submitted to swimming.²¹⁷

Land-based remobilization may also be important in muscle recovery. A study evaluated different remobilization protocols in muscles following HS of rats. Treatment groups included controls, suspended, suspended followed by free activity, suspended followed by training on a declined treadmill, and suspended followed by training on a flat treadmill. The soleus and anterior tibialis muscles were evaluated. The suspended soleus muscle showed a significant increase in the proportion of intermediate and hybrid fibers and a decrease in the number of type I fibers. Some of these changes were reversible after remobilization. The three remobilization programs led to the recovery of both the proportional number of fibers and their size. The anterior tibialis muscle had a significant increase in the number and size of type I fibers and a cell size reduction of type IIB fibers after suspension, which recovered after training on a declined treadmill and free movement. Especially regarding the soleus, the findings indicated that training on a declined treadmill induced greater tissue restructuring after suspension.²¹⁸

The frequency of training may also be important regarding the recovery of muscle mass after immobilization. Following 2 weeks of limb immobilization of rats, 6 weeks of exercise was instituted under different conditions: free cage activity and free remobilization, one time/week treadmill running, three times/week running, or six times/week running. Immobilization significantly reduced the cross-sectional area of soleus and gastrocnemius type I and type II muscle fibers. Immobilization also increased the ratio of type II to total fiber numbers in the muscles and induced pathologic changes in muscle fibers. Some of these changes were not corrected with free remobilization. Training one, three, and six times per week resulted in recovery toward normal levels. However, running three and six times per week was more beneficial for recovery of muscle atrophy compared with no running or running once per week.⁹⁹

Skeletal muscle regeneration is mediated by muscle stem cells, or satellite cells. Injury or stress results in activation of satellite cells, with migration to these sites and differentiation into multinucleated myofibers. Satellite cells are activated if muscle contractions are of sufficient intensity and duration. However, there are significant species differences in the magnitude of change and time of muscle protein synthesis patterns. Resistance exercise

stimulates muscle protein synthesis, and the acute increase in muscle protein synthesis with exercise is greater and longer lasting than changes after eating.²¹⁹ Eccentric muscle contractions appear to cause greater satellite cell activation and muscle mass gains as compared with concentric muscle contractions.¹⁶⁰ Nerve activity is also required to support the growth of regenerated myofibers and the development into various muscle fiber types.

Satellite cell activation also involves an initial inflammatory response.²¹² Macrophages are involved in phagocytosis of necrotic debris, and they promote myogenic differentiation by activating satellite cells and causing differentiation. During eccentric exercise, sarcomere structure is damaged and inflammatory cytokines are produced, especially IL-6.²²⁰ In addition, the NF- κ B inflammatory signaling cascade is activated during early rehabilitation following immobilization.²²¹ Because inflammation and perhaps prostaglandins may stimulate satellite cell proliferation, it is possible that NSAIDs may reduce postexercise muscle protein synthesis.¹⁶⁰ However, NSAIDs also decrease the discomfort and inflammation of muscle soreness.

Myogenic regulatory factors are a group of transcription factors that induce the transcription of skeletal muscle proteins and coordinate satellite cell differentiation.¹⁶⁰ Although further research on the mechanisms of signaling muscle protein synthesis is needed, it is known that the rapamycin (mTOR)-S6 kinase pathway is involved with turning on muscle protein synthesis. During rehabilitation from immobilization-induced atrophy in humans, myostatin is rapidly suppressed and remains at a relatively low level throughout rehabilitation and muscle recovery, suggesting that myostatin also plays a key role in the regenerative process¹⁶⁰ (Box 7-7).

Adaptations of Muscle to Increased Mechanical Stress

Mechanical stress may be induced by endurance or strength training. Endurance training consists of repetitive, low-intensity contractions that result in changes in oxidative metabolism. Strength training is characterized by a limited number of contractions that occur against a high (often maximal) mechanical load.

Endurance Training

Endurance training does not affect muscle fiber number or cross-sectional area, but it alters metabolic responses. There is a twofold increase in oxidative capacity of all muscle fiber types with endurance training. There is an increase in muscle mitochondrial density, oxidative enzyme activity, muscle glycogen, and intracellular lipids.¹⁴⁶ Fatty acid use increases while reliance on carbohydrate metabolism is reduced. Muscles become more fatigue-resistant with training. Endurance training also elevates the antioxidant and detoxicant status of muscle.¹⁴² For example, total

glutathione, glutathione peroxidase, glutathione reductase, and gamma-glutamyl transpeptidase increased in the leg muscles of beagle dogs treadmill trained for 40 km/day at 5.5-6.8 km/h, 15% upgrade, 5 days/wk for 55 weeks.

Training also results in augmentation of muscle capillarity by new capillary formation, which may increase the available surface area for oxygen diffusion and decrease the distance over which oxygen must diffuse to reach cells.¹⁴⁴ Training also appears to improve the functional blood-tissue gas exchange properties of muscle. The effects of exercise training on peak oxygen uptake and effective oxygen diffusive conductance in skeletal muscle were investigated in dogs exercised on a treadmill 1 h/day, 5 d/wk for 8 weeks.¹⁴⁴ Peak oxygen uptake by gastrocnemius muscles was 38%, 33%, and 19% greater and oxygen diffusive conductance was 71%, 75%, and 68% greater during normal levels of oxygen, moderate hypoxia, and severe hypoxia, respectively, in exercised dogs as compared with controls.

Endurance training can induce changes in muscle fiber types. The alterations initiated by long-distance running on muscle fiber type distribution was studied in young dogs run on a treadmill for 55 weeks, 5 days a week.²²² The daily running distance was gradually increased to 40 km and maintained at that level for the final 15 weeks. In the triceps brachii muscle, there was a shift from type II to type I fibers, which was also observed in the thoracic and cervical spinal muscles. In addition, there was also a shift toward a higher oxidative capacity of type II fibers. However, in the lumbar muscles, the percentage of type II muscle fibers increased significantly in the running group. Training can induce changes in fiber type composition not only in limb muscles but also in the stabilizing spinal muscles.

Although the ubiquitin-proteasome pathway has been implicated in the pathophysiologic findings of muscle atrophy, this system is also apparently involved in the response to training and conditioning. It is possible that increased proteolysis of muscle is required to repair exercise-induced damaged protein, or the general turnover of muscle proteins is increased. A field study of 44 English pointers and Labrador retrievers used for bird hunting and retrieving was conducted to examine changes in the ubiquitin-proteasome pathway. Muscle biopsy samples were collected from all dogs preseason and at peak season. There was pronounced up-regulation of ubiquitinated conjugates and the p31 subunit during the peak hunting period compared with the preseason period. The catalytic core of the proteasome (b-subunits) showed no apparent up-regulation with training, however. Increased training also resulted in increased muscle glycogen levels and citrate synthase. It appears that up-regulation of specific components of the ubiquitin-proteasome pathway was an indication that it plays a role in skeletal muscle turnover during training and conditioning.²²³

Although endurance training brings about changes in muscle metabolic properties, muscle mass may not be affected.²²⁴ Young beagle dogs trained on a treadmill for 1 year, gradually increasing running to 40 km/day at a speed of 5.5-6.8 km/hour with a 15-degree inclination during the last 15 weeks, while control dogs were kept in their cages. There was no difference in mass of cranial tibial and semitendinosus muscles between trained and control dogs.

Strength Training

Strength training uses brief, maximal contractions and requires a relatively high rate of anaerobic energy production. In addition to increased enzyme activity for glycolysis and glycogenolysis, there is an increase in muscle strength and size. Muscle fiber cross-sectional area is directly related to the tension the muscle is capable of exerting.²²⁵ Strength training by overloading muscle increases the cross-sectional area of all fiber types, especially the fast twitch fibers. New myofibrils are synthesized, and there may also be hyperplasia of muscle fibers. There is also improved synchronization of motor units.¹⁴⁶ This may help explain why the changes in muscle cross-sectional area are smaller than the changes in maximal voluntary force production.²²⁶ Resistance training may especially result in increased cross-sectional area of type II fibers.¹³¹

The contraction time is increased with muscle hypertrophy because of the greater distance that must be traveled in the sarcoplasmic reticulum during contraction.²²⁵ The increases in strength may not become apparent until the training program is well under way.²²⁶ There is little effect on oxidative metabolism by muscle cells compared with endurance training. In fact, mitochondrial density may decline with strength training because of a dilutional effect by the larger quantity of myofibrils, although the absolute volume of mitochondria are unchanged.²²⁶ In some instances, there may be a mild increase in oxidative capacity, suggested by the increase in type IIa muscle fibers with strength training seen in some studies.²²⁷ In models of muscle hypertrophy, collagen content increases, with changes in collagen metabolism beginning as early as 3 days after increased loading.⁵⁸ Muscle fiber damage may also occur during vigorous strength training. The increased collagen content may prevent skeletal muscles from contracting against excessive loads. Lactate accumulation in overworked muscles may contribute to increased collagen synthesis by stimulating proline hydroxylase.

Compensatory hypertrophy of muscles may be induced by removal of one or more muscles of a synergistic group of muscles.²²⁵ This results in overloading of the remaining muscle or muscles, with increased muscle weight and enlargement of remaining muscle fibers. In addition, there may be an increase in the number of fibers as a result of muscle fiber splitting. Weight training results in muscle fiber splitting of cats trained to lift weights.²²⁵ The degree

of fiber splitting is related to the intensity of exercise, and all muscle fiber types may be affected.

Unfortunately, there is relatively little information regarding the effect of strength training on muscles in dogs, perhaps because of the difficulty in having them perform strength training activities and the difficulty in measuring changes in strength. A study of cats trained to move a bar and lift weights for a food reward resulted in muscle hypertrophy of 7% to 34% after 40 weeks.²²⁸ A variety of studies have evaluated strength training in humans. Men participating in an exercise program designed to strengthen the quadriceps muscles had increased thigh girth, and muscle strength increased 40% with training.²²⁹ In addition, endurance and time to exhaustion while cycling and running increased, indicating some benefit to endurance exercise. The effect of adding heavy-resistance training to increase leg-muscle strength was further studied in humans undergoing endurance training.²³⁰ After 10 weeks, leg strength increased an average of 30%, but thigh girth, muscle fiber areas, and oxidative activities were unchanged. Short-term endurance was increased by 12%. Exercise to exhaustion also increased after strength training, indicating that certain types of endurance performance, particularly those requiring fast-twitch fiber recruitment, can be improved by strength-training supplementation.

Muscle strengthening may occur, even in older individuals. For example, one study of geriatric men compared the training effects of voluntary isometric contraction on the quadriceps femoris to determine if exercise changes age-related muscle weakness.²³¹ Maximal voluntary isometric contraction torque increased with even low training loads during 12 training sessions over 4 weeks. Altering dietary intake to achieve carbohydrate loading has been used to enhance muscle mass prior to competitions in humans, but one study indicated there was no advantage to carbohydrate loading to enhance muscle girth over weight-lifting alone.²³²

Although gender differences associated with strength training are unknown in dogs, female cats had greater muscle hypertrophy following weight training than male cats.²³³ Gender differences also exist regarding skeletal muscle hypertrophy subsequent to heavy-resistance training in humans. Absolute changes in muscle mass are generally greater in men than women, but the percentage changes may not be significantly different.²³⁴ For example, women completing a 20-week heavy-resistance weight training program for the lower extremity consisting of squats, vertical leg presses, leg extensions, and leg curls twice a week had a significant increase in maximal isotonic strength and hypertrophy of muscle fiber types I (15%), IIA (45%), and IIA + IIB (57%).²²⁷ These data are similar to those in men and suggest considerable hypertrophy of muscle fiber types occurs in women if exercise intensity and duration are sufficient.²³²

In people, resistance training may, both alone and in combination with feeding, be a marked stimulator of human muscle protein synthesis and, in the absence of nutrition, sometimes muscle protein breakdown.²¹⁹ When feeding is superimposed on the postexercise stimulus, there is usually a net positive protein balance and protein synthesis occurs. The question of how much loading results in muscle gain often arises regarding the practice of lifting heavier loads to induce hypertrophy and strength. Acute studies in people indicate that increased muscle protein synthesis only occurs when intensities of load lifted exceed 60% of a single repetition maximum,²³⁵ but at intensities beyond 60% and up to 90% of a single repetition maximum, there is a similar stimulation of muscle protein synthesis. Therefore it may be that resistance exercise at higher intensities has little additional value in stimulating muscle hypertrophy. However, low-intensity exercise (20% of single repetition maximum) with concurrent blood flow occlusion may stimulate muscle protein synthesis,²³⁶ with an accompanying rise in S6K1 phosphorylation. Other studies have also shown that blood flow occlusion, when practiced chronically and with low lifting intensities (30% to 40% of single repetition maximum), can result in muscle hypertrophy and strength gains.²³⁷ It is possible that venous occlusion results in local fatigue and the recruitment of type II muscle fibers, which are not normally recruited at low intensities. Recruitment of type II fibers results in activation of many signaling proteins, such as S6K1, ribosomal protein S6, extracellular signal-regulated kinase, and p38 mitogen-activated protein kinase, which are involved in regulating muscle protein synthesis.²³⁸

The Effects of Medications and Nutrition on Immobilized Muscles

Steroids

Naturally occurring or iatrogenic hyperadrenocorticism has been associated with myopathy in dogs.²³⁹ Most dogs have muscle stiffness, proximal appendicular muscle enlargement, and myotonic discharges on electromyography. Muscle weakness and muscle atrophy may be present in some dogs. Histologic changes are characteristic of non-inflammatory degenerative myopathy. Clinical signs of the myopathy may improve to varying degrees following treatment of hyperadrenocorticism. Steroids have a catabolic effect on muscle proteins, reduce protein synthesis, and alter the insulin responsiveness of muscle that is immobilized.¹³⁸ During treatment with corticosteroids or during catabolic conditions, glutamine synthetase increases and glutamine, the most abundant amino acid in muscle, is released at a high rate. On the other hand, endurance exercise reduces basal levels of glutamine synthetase.²⁴⁰ In addition, there is increased susceptibility of type II fibers to atrophy, and type IIB fibers are more susceptible to atrophy than type IIA fibers.

Resistance exercise training may help attenuate the muscle loss associated with glucocorticoid administration by up to 40%.²⁴⁰ Endurance training may also prevent glucocorticoid-induced muscle atrophy. Running attenuated muscle loss by 25-50% in rats receiving corticosteroids concurrently with the start of training. Prior endurance training does not appear to offer any advantage in preventing atrophy associated with corticosteroid administration compared with beginning training at the same time of steroid administration. Initiation of endurance exercise shortly after the onset of steroid treatment helps to prevent further muscle atrophy from that time forward, but does not prevent the atrophy that has already occurred. Atrophy sparing seems to occur to a greater extent in muscle types that are highly recruited during endurance training.

Androgens and synthetic anabolic steroids have been recommended to help attenuate loss of muscle mass because of their ability to increase protein synthesis and promote muscle growth.²⁴⁰ Although the effects of anabolic steroids on athletes and athletic performance are controversial, they do not appear to enhance aerobic metabolism. Anabolic steroids do seem to be beneficial in patients in negative nitrogen balance, such as those patients who are undernourished. The influence of oral anabolic steroids on body mass index (BMI), lean body mass, anthropometric measures, and functional exercise capacity among humans with chronic obstructive pulmonary disease was evaluated.²⁴¹ A group receiving testosterone and daily oral stanozolol for 27 weeks had increased weight, whereas the control group lost weight. Lean body mass, BMI, arm muscle circumference, and thigh circumference increased in the treated group, but there were no changes in endurance exercise capacity. Anecdotally, there may be some benefit to administration of anabolic steroids to dogs with chronic medical conditions that may result in muscle atrophy.

Antiinflammatory Drugs

The exact role of NSAIDs in the treatment of acute muscle injuries is unknown, but it has been suggested that these agents may be contraindicated, perhaps because NSAIDs may delay muscle regeneration following injury and inhibit exercise-induced satellite cell proliferation in humans. The cellular mechanisms by which NSAIDs may negatively affect muscle regeneration and inhibit the exercise-induced satellite cell proliferation are unknown. In one study, eight healthy male volunteers performed 200 maximal eccentric contractions with each leg.²⁴² Indomethacin, an NSAID, was infused into the vastus lateralis muscle of one leg to block local prostaglandin synthesis. Protein synthesis was determined from 24 to 28 hours following exercise. Myofibrillar and collagen protein synthesis were unaffected by local NSAID infusion. The expression of growth factors, such as TGF- β and IGF-1, and matrix-related genes increased after exercise and were unaffected by NSAIDs. Although NSAIDs seem to inhibit exercise-induced

satellite cell proliferation, there appear to be only limited effects on gene expression and on postexercise protein synthesis. However, COX-2 specific inhibitors apparently have little effect on muscle protein synthesis or satellite cells in people.^{243,244}

Muscle soreness may or may not be alleviated with NSAIDs or COX-2 specific inhibitors. The effects of two different NSAIDs and physiotherapy modalities on the healing of acute hamstring muscle tears of humans were studied.²⁴⁵ Patients received meclofenamate, diclofenac, or a placebo. All patients received the same intensive physiotherapy treatment, including rest, ice, compression, elevation, therapeutic ultrasound, deep transverse friction massage, stretching, and exercise over a 7-day treatment period. Pain, swelling at the site of the muscle tear, and isokinetic muscle strength were improved in all groups, but there was no difference among groups. In fact, the pain score of the more severe injuries was significantly lower in the physiotherapy group than in those receiving NSAIDs and physiotherapy at day 7. Therefore there may not be an additive effect on the healing of acute muscle injuries when NSAIDs are added to standard physiotherapeutic modalities.

Growth Hormone

Growth hormone (GH) may help to improve muscle atrophy as a result of immobilization or denervation.¹³⁸ The actions of GH are mediated by complex interactions among several hormones, receptors, and binding proteins.²⁴⁶ GH results in stimulation of protein synthesis. In addition, amino acid and glucose uptake are increased, and lipolysis is increased. Treatment with GH results in an increase in lean body mass in humans and rats, although strength is probably not improved. One of the primary effects of GH is the production of IGF-1 by the liver, which in turn also affects muscle tissue. IGF-1 may be a mediator of the response of GH because it also results in increased amino acid uptake and protein synthesis.²⁴⁷ IGF-1 stimulates muscle satellite cells to enter the cell cycle and proliferate, with some differentiating and fusing with myofibers. However, most studies indicate that IGF-1 has minimal effect on the adaptation of muscle to changes in loading. The GH response to acute aerobic exercise may be augmented with repeated bouts of exercise.²⁴⁸ During prolonged administration, resistance to the anabolic effects of GH may occur.

Administration of GH during periods of non-weight-bearing activity may not result in maintenance of muscle mass.^{194,246} However, treatment during recovery may be beneficial. Beagle dogs receiving a GH secretagogue during a 10-week immobilization and 5-week remobilization period had a 60% decrease in tetanic torque during immobilization, which was similar to untreated dogs, but had a threefold increase in tetanic torque in the treated group compared with controls during remobilization.¹³⁹

The diameters of type I and type II fibers measured from the vastus lateralis muscle followed the same trend. These data suggest that the GH secretagogue increased the size and strength of the quadriceps muscle during remobilization. GH may also attenuate the catabolic effects of corticosteroid treatment on muscle mass.²⁴⁶

Nutrition and Muscle

Amino acids, hormones, and physical activity appear to be the major short-term physiologic regulators of muscle mass, primarily through their actions on protein synthesis, whereas long-term regulation involves changes in gene expression. Amino acids are the main dietary components that stimulate muscle protein synthesis. In people, branched-chain amino acids, especially leucine, stimulate protein synthesis in a dose-dependent manner. The stimulatory effect of leucine on protein synthesis is mediated through upregulation of mRNA translation.²⁴⁹ A number of mechanisms contribute to the effect of leucine on translation initiation. The increase in muscle protein synthesis may be mediated by a key component in the signaling pathway controlling these phosphorylation-induced mechanisms, a protein kinase called the *mammalian target of rapamycin* (mTOR). There is a limit, however, to the amount of muscle protein synthesis that can occur with a single meal. Leucine, either alone or with other essential amino acids, stimulates muscle protein synthesis for a short period of up to 2 hours. Muscle protein synthesis then returns to basal rates, possibly explaining why constant delivery of leucine does not maintain elevated muscle protein synthesis.²⁵⁰ Increased dietary protein also induces enzymes of amino acid catabolism, particularly of the branched-chain and aromatic amino acids, and this may partially explain why muscle mass cannot be increased solely by eating more protein.²⁵¹ However, the effect of muscle protein breakdown by amino acid catabolism may be decreased when insulin release is stimulated, for example by glucose in a meal, because insulin apparently inhibits protein breakdown.

Branched-chain amino acid supplementation may have mild effects on muscle atrophy because of immobilization and help regulate protein turnover. In one study, short-term (6 days) HS of rats resulted in losses of myofibrillar proteins, total RNA, and rRNAs and pronounced atrophy of the soleus muscle.²⁵² Although the addition of branched-chain amino acids to the diet did not prevent muscle atrophy and had no apparent effect on regulators of proteasomal protein degradation, they reduced the loss of myofibrillar proteins and RNA, and attenuated several signal transduction proteins that act as regulators of protein synthesis and cell growth in muscle.

Leucine is especially important in the stimulation of muscle protein synthesis and can be a sole stimulus for muscle protein synthesis, but a full complement of amino acids are necessary for muscle synthesis.²¹⁹ Younger people

seem to have greater ability to synthesize muscle protein than older people with the same level of amino acids available, suggesting a potential reduced signaling response for muscle synthesis in older people.

Providing appropriate nutritional components after exercise may affect muscle protein synthesis. When branched chain amino acids, including leucine, isoleucine, and valine were administered to people during and after one session of quadriceps muscle resistance exercise, an increase in mTOR, p70 S6 kinase, and S6 phosphorylation was found in the recovery period after exercise.²⁵³ Exercise without branched-chain amino acids intake led to a partial phosphorylation of p70 S6 kinase without activating the enzyme. These results suggest that branched-chain amino acids administered after exercise may stimulate muscle protein synthesis.

Although insulin has little effect on protein synthesis in human muscle, it has an inhibitory effect on protein breakdown. In the immediate postexercise period there are marked increases in muscle protein breakdown.^{254,255} When food containing protein and carbohydrate is given in the postexercise period, the increase in protein synthesis is exacerbated and the increase in the protein breakdown is blunted in an insulin-related effect, so there is a net increase in muscle protein synthesis.²⁵⁶ Whereas exercise stimulates protein synthesis in muscle, disuse atrophy results in a marked decrease of muscle protein synthesis.¹⁵²

Much discussion has occurred regarding protection of muscle from injury as a result of fatigue and excessive contraction in people. *N*-acetylcysteine has been suggested as a possible muscle protectant from overuse injury. Repeated electrical stimulation or eccentric exercise in which muscle lengthens while being loaded may cause injury.²⁵⁷ Inflammation and muscle damage may result in delayed-onset muscle soreness.²⁵⁸ Muscle-derived ROS may be involved in the development of muscle fatigue and may result in muscle damage as a result of excessive contractile activity.²⁵⁹ The lipoxygenase pathway may promote the production of ROS. Although study results of other antioxidants to prevent postexercise muscle soreness have been mixed, *N*-acetylcysteine possesses antioxidant activity, increases glutathione synthesis, reduces ROS during muscle contractions, and scavenges hydrogen peroxide and hydroxyl free radicals. One study investigated the effects of pretreatment with *N*-acetylcysteine on fatiguing contractile activity by electrical stimulation of muscles.²⁵⁹ Muscle fatigue was delayed, tetanic force was preserved, and plasma creatine kinase and lactate dehydrogenase activities were decreased in *N*-acetylcysteine-treated rats. The content of muscle-derived ROS in both gastrocnemius and soleus muscles was also decreased in *N*-acetylcysteine-treated rats, suggesting that pretreatment has a protective effect against injury induced by fatiguing contractile activity in skeletal muscle.

Skeletal muscle recovery from disuse atrophy requires the recruitment of insulin signaling for muscle growth, which is driven by protein synthesis. There is some belief that dietary fish oil may enhance insulin signaling and protein metabolism. One study was performed to evaluate whether dietary fish oil facilitates muscle recovery during remobilization after disuse atrophy.²⁶⁰ Immobilization for 10 days, followed by 3 or 13 days of remobilization, was applied to the hindlimbs of rats fed corn oil (control group) or fish oil as a source of dietary fat. The immobilization-induced reductions in soleus muscle weight and myosin heavy-chain content were significantly restored by 3 days of remobilization in the group receiving corn oil. However, muscle recovery in the fish oil group did not significantly change until 13 days of remobilization. Throughout the remobilization period, levels of prostaglandin $F_{2\alpha}$ and COX-2 mRNA were significantly increased, although the increase of prostaglandin $F_{2\alpha}$ was less in the fish oil group until day 13. These findings suggest that dietary fish oil may inhibit early muscle recovery after disuse atrophy by suppressing signaling pathways and prostaglandin $F_{2\alpha}$ synthesis.

Recently, vitamin D receptors have been found in muscle tissue.²⁶¹ Vitamin D deficiency has been associated with muscle weakness and changes in EMGs, which improve with vitamin D supplementation. There may also be atrophy of type II muscle fibers with vitamin D deficiency. Other studies have shown a relationship between 25-hydroxycholecalciferol levels and various measures of muscle strength and function with aging. Vitamin D receptor expression decreases with age and is unaffected by vitamin D levels. Vitamin D may have a role in transcription and regulation of IGF binding protein, which binds IGF in tissues, including muscle. Vitamin D levels also have an effect on membrane calcium channels of muscle cells in numerous species.²⁶² Because calcium is critical for skeletal muscle function, vitamin D levels may have a significant effect on muscle function, performance, and injury risk.

Ligaments and Tendons

Ligament and Tendon Structure and Biomechanics

Ligaments and tendons consist primarily of type I collagen that forms parallel fibers. Ligaments extend between bones to stabilize joints, whereas tendons permit movement of bones by connecting muscles to bones.⁵⁸ Both are relatively pliant and flexible to allow natural movements of the bone to which they attach. They are also strong and relatively inextensible to offer resistance to applied forces, especially tensile forces. Ligament and tendon structure and chemical composition are very similar in humans and many animal species, including rats, rabbits, dogs, and

monkeys. They are composed of dense connective tissue with primarily parallel fibers of collagen, with a small number of collagen fibers that run perpendicular to the predominant direction of stress. Cross-links that form between collagen chains are important to collagen tensile strength and resistance to proteases.²⁶³ Reducible cross-links are found in newly formed collagen, while nonreducible cross-links are found in mature collagen and account for stronger and stiffer collagen. The concentration of cross-links and their ratio varies between species and location. For example, the rabbit CCL has three to five times more nonreducible cross-links than the patellar tendon.²⁶³ Ligaments also have more cells per mass than tendons.²⁶³

Most ligaments do not function as a unit structure because at a particular joint position, one portion of a ligament may be taut while another part is relaxed.²⁶⁴ This is relevant because certain joint positions may need to be avoided to reduce stress on healing ligaments. Ligament insertions may be direct or indirect.²⁶⁴ With a direct insertion, collagen fibrils course directly into the bone, with some fibers blending with the periosteum. In contrast, indirect insertions have a zone of fibrocartilage and mineralized fibrocartilage between the ligament and cortical bone. These zones are important when considering ligament insertions and their responses to stress and rehabilitation from injury.

The biomechanical responses of ligaments and tendons are nonlinear because of progressive fiber recruitment and viscoelastic properties. The structural properties of the entire bone-ligament-bone complex that must be considered depend on the mechanical properties of the ligament, macroscopic and microscopic ligament geometry, the cross-sectional area and length of the ligament, and the material properties of the fibrous and ground substance constituents.

When initially loaded, ligaments and tendons stretch with a slight increase in load. On histologic examination, the fibers change from a relaxed wavy appearance to a parallel alignment as the fibers become oriented in the direction of loading. This region on a load-deformation curve is known as the *toe region* (Figure 7-9). As the load is increased, the curve becomes more linear, corresponding to full parallel orientation of the fibers. The linear portion of the curve also indicates elastic deformation, with return to normal structure when the load is released. If loading continues, the load-deformation curve exits the linear region and enters the region of plastic deformation. This point is termed the *yield point*. From a histologic perspective, some microfailure may occur even before the yield point is reached, resulting in damage to the ligament or tendon. After the yield point is reached, gross failure of the ligament occurs and permanent structural damage occurs and remains even if the load is released. When the ligament or tendon fails and can support only a negligible load, the ultimate failure point has been reached. It is important to

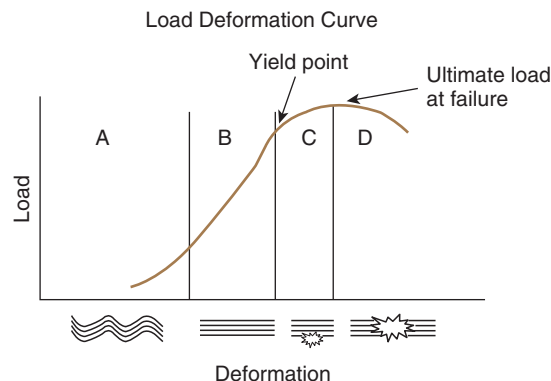


Figure 7-9 Load-deformation curve of ligament biomechanical testing. **A**, Toe region. Note the bottom portion of the diagram indicating the wavy appearance of the collagen fibers. **B**, Linear physiologic loading and zone of elastic deformation. The collagen fibers become taut. **C**, Early permanent damage and zone of plastic deformation. There is damage to some of the fibers. **D**, Ultimate load at failure. There is gross disruption of the ligament.

note that the ligament may retain continuity despite micro- and macrofailure.

The mechanical properties of ligaments and tendons are a reflection of their material characteristics, allowing for comparison between other ligaments and tendons. They also reflect the orientation and organization of the collagen fibers. High strain-rate techniques for studying knee ligament failure in people have replaced the previous low strain-rate methods and distinguish the failure mechanism of ligaments from that of bone. For example, anterior cruciate bone-ligament-bone preparations of primates failed at a higher load and absorbed more energy at a faster rate of deformation than at a slower rate.²⁶⁵ Specimens failed by ligament disruption at faster rates, and by tibial avulsion fractures at slower rates. Ligament function is further defined by measuring the restraining force provided by specific ligaments. Advances in analytic techniques and models permit precise analysis of joint position, motion, and laxity.²⁶⁶

Recent studies have indicated that, in addition to tendon and ligament support, neuromuscular reflexes may play a large role in joint stability.²⁶⁴ Impulses from mechanoreceptors in the joint capsule, menisci, and ligaments transmitted to the spinal cord result in instantaneous reflex regulation of muscles acting on a joint to provide coordinated joint function and stability. Canine lateral collateral ligaments have nerve fibers and Golgi-Mazzoni and Ruffini corpuscles, some of which resemble Golgi tendon organs.²⁶⁷ Mechanoreceptors are more numerous at the ends than in the middle of ligaments, and they are present in both the subsynovial and the interfascicular connective tissue and are surrounded by vessel networks. The number of corpuscles may be related to the type of stress to which a ligament is subjected. ACL-deficient humans have disruption

of an ACL-hamstring rapid reflex arc, which normally helps to limit anterior motion of the tibia.²⁶⁴ Instead, a slower reflex originating from the knee joint capsule limits anterior tibial motion. Rehabilitation techniques to help restore ligament-muscle reflexes should be considered.

Knowledge of the blood supply to ligaments and tendons is necessary to understand how healing and rehabilitation may be affected. Blood flow to the proximal, middle, and distal parts of the CCL were 3, 2.2, and 3.5 mL per 100 g of tissue per minute, respectively in one study.²⁶⁸ The respective values in the caudal cruciate ligament were 2.8, 2.8, and 3.2 mL per 100 g of tissue per minute. The cruciate ligaments are relatively hypovascular, with blood flow only 50% of that of the synovium of the suprapatellar pouch.²⁶⁹ Application of a cranial stress diminishes the blood flow to the ligament to one fifth of the baseline value, an effect that is reversible. Division of the infrapatellar fat pad causes a twofold decrease in perfusion to the ligament, whereas dissection of the enveloping synovium results in a complete cessation of blood flow.

Many factors affect the ligament-bone unit response. Increasing age weakens the tissue-bone preparation.²⁷⁰ Tendons also respond biochemically to the stresses placed on them.²⁷¹ In compression regions of rabbit tendons, the GAG content is 15 times greater than in tension-transmitting regions. Dermatan sulfate predominates in regions of tension, whereas chondroitin sulfate is the major GAG in regions of compression. The relative GAG content changes with alterations in the stresses placed on tendons induced by transposing the tendon, and the content reverts back to normal when the tendon is replaced to its original position. Other factors affecting tendons and ligaments include the amount of activity the animal undergoes and medications the patient may receive.

Response of Ligaments and Tendons to Disuse and Immobilization

Immobilization is commonly used to treat injuries to ligaments and tendons. There are many models of ligament and tendon disuse, including internal and external fixation methods, casts and slings, denervation, disarticulation, and bed rest. These models have allowed the study of histologic, biochemical, and biomechanical changes that occur with disuse. There is an adverse decline in structural and material properties of ligaments and tendons with immobilization of the joints that they cross. Even if joint motion is allowed, stress deprivation rapidly reduces the mechanical properties of the tendon and ligament tissues (Box 7-8).²⁷² The bone-tendon/bone-ligament complex is especially affected by immobilization. The effects of stress deprivation appear to be time and dose dependent, and differences may exist in the responses of various ligaments and tendons within the same animal because of differences in cellular, structural, and biochemical properties. For example, the CCL may be more resistant to stress

Box 7-8 Summary of the Effects of Disuse and Immobilization on Ligaments and Tendons

- Stress deprivation rapidly reduces mechanical properties of tendons and ligaments.
- Bone-tendon and bone-ligament complexes are especially affected.
- Time and dose of immobilization and differences in cellular, structural, and biochemical properties affect degree of changes.
- Decreased cross-sectional area, disorganization of parallel structure of fibrils and cells, a net decrease in collagen mass, and reduced GAG, HA, chondroitin sulfate, dermatan sulfate, and water content with immobilization.
- May have a 0.2%/week loss of ligament strength and energy to failure with cage confinement.
- May have a 45% loss of load to failure, and up to 75% decrease in cranial cruciate ligament stiffness with limb immobilization for 12 weeks.
- Stifle medial collateral stiffness may decrease 40% with 6 weeks of immobilization and take 18 weeks to return to normal biomechanical properties after remobilization.
- Continued joint motion during reduced weight bearing helps preserve ligament properties.
- Effects of immobilization on tendon are less severe compared with ligaments.
- Recovery of bony insertion sites is prolonged compared with ligament/tendon midsubstance, up to 1 year.
 - Bony resorption occurs at ligament bone interface.

GAG, Glycosaminoglycan; HA, hyaluronic acid.

deprivation than the MCL.^{272,273} Typically, there is reduction of cross-sectional area; disorganization of the parallel structure of fibrils and cells; increased collagen turnover with increased collagen synthesis and degradation; a net decrease in collagen mass; and reduced GAG, HA, chondroitin sulfate, dermatan sulfate, and water content.²⁷² Remobilization returns the mechanical properties to nearly normal over time, but the recovery of the bony insertion sites is prolonged compared with the ligament-tendon midsubstance. Knowledge of the detrimental effects of immobilization may influence the management of ligament and tendon injuries.

It is well recognized that a reduction of activity, such as confinement to a cage, may affect ligament and tendon properties.²⁷³ A study of related rhesus monkeys showed a 0.2% loss in strength and energy to failure per week of captivity in the CCL.²⁷⁴ More profound effects may be expected with complete immobilization of a limb. In a classic study, the effect of body cast immobilization on the biomechanical properties of bone-CCL-bone units of rhesus monkeys was studied.²⁷⁵ After 8 weeks of total-body plaster immobilization, there were significant decreases in

maximum failure load and energy absorbed to failure (39% and 32%, respectively). There was also a 69% decrease in ligament stiffness. These changes likely indicate an alteration in the functional capacity of the ligament to resist loading and elongation, factors that relate to the ligament's ability to provide joint stability. The immobilized ligament units frequently failed by avulsion fractures of the bone, which correlated with bone resorption in the cortex immediately beneath the ligament insertion site. The effect of immobility on knee ligaments depend, in part, on the histologic characteristics of the ligament-bone attachment. The ligament-bone junctions with zones of fibrocartilage were little affected. In contrast, the MCL, which inserts directly into the tibia and periosteum without well-defined zones of cartilage, showed marked interruption after immobility as the result of bone resorption in the subperiosteal and outer cortical regions. In some places the ligament was attached only to the overlying periosteum. After 5 months of remobilization following total body immobilization, there was only partial recovery in ligament strength, although ligament stiffness and compliance parameters returned to control values. Up to 12 months were required for nearly complete recovery of ligament strength.

Similar effects of immobilization have been noted on canine ligaments. The effect of immobilization on the CCL of dogs was studied using a model of internal skeletal fixation for 12 weeks.²⁹ Failure of the femur-ligament-tibia complex occurred through the tibial insertion of the ligament for both experimental and control limbs. The load at failure and stiffness of the immobilized limbs were 45% and 73%, respectively, of the nonimmobilized CCL. Even though there was a 13% reduction in soft tissue mass, the loss of collagen was greater in the tibia and femur than in the CCL, and correlated with mechanical failure at the bony insertion. Bone atrophy was the result of increased resorption of bone rather than decreased bone formation. This study demonstrated that there was significant atrophy of the CCL and bone in immobilized joints of dogs.

Other studies have confirmed the importance of bone resorption and the reduction of bone collagen resulting from disuse and its relationship to strength of the medial collateral and CCL complexes.^{273,276} In a similar study of canine stifles immobilized in a cast for 12 weeks, changes in the tibial insertion of the ligament were apparent, including the presence of many osteoclasts, numerous large fibroblasts, and replacement of bone by loosely arranged fibrous tissue.²⁷³ Interestingly, in this study the femoral attachments of the medial and lateral collateral ligaments and the origin of the CCL showed little bone resorption, indicating that the response to immobilization may be site-specific. Even so, MCL strength decreased 40% following 6 weeks of immobilization.²⁷⁷ Restriction of activity to cage confinement also causes significant bone atrophy of the tibial insertion sites of the collateral ligaments if adequate activity is not allowed. Additionally, there is decreased

thickness of ligament and tendon fiber bundles, greater extensibility per unit load, and unchanged collagen content after immobilization.

Although stress deprivation with immobilization appears to reduce the mechanical properties of ligaments and tendons in a time- and dose-dependent manner, stress deprivation with continued mobility may have different effects. Maintaining joint motion and reducing the period of immobilization may help preserve ligament properties. The effects of stress deprivation with joint motion on the CCL were studied in dogs by freeing the tibial insertion of the ligament and moving it caudally and securing it with a screw.²⁷⁸ Stifle joints were allowed complete mobility. The cross-sectional area of the ligament was 115% larger than controls at 6 weeks. The tensile strength was unchanged at 6 weeks, but was only 67% of control values at 12 weeks. This model of stress deprivation with continued joint motion on the CCL suggests that there may be an increase in ligament cross-sectional area and a concurrent decrease in biomechanical properties after a time, although the changes are less severe than immobilization without joint motion.²⁷²

Other models of stress deprivation with joint motion for relatively short periods have shown different results. The effect of active joint motion on ligament and meniscus mass in a non-weight-bearing model of disuse was studied. A unilateral tarsal disarticulation model of disuse that maintained active knee motion without weight bearing for 8 weeks was studied in dogs.³⁰ A large and similar loss of collagen and calcium mass occurred in the femur and tibia, indicating net bone resorption. No loss of tissue mass occurred in the collateral and cruciate ligaments or menisci. The strength of the femur-CCL-tibia complex was not qualitatively or quantitatively different between control and experimental limbs. So although immobilization for greater than 8 weeks results in atrophy of bone, ligament, tendon, and meniscus in some canine studies, the absence of weight bearing for 6 to 8 weeks with maintenance of joint motion appears to help preserve femur-ligament-tibia complex mechanical strength, despite the occurrence of bone atrophy.

The effects of immobilization on the biomechanical and morphologic properties of the femur-MCL-tibia (FMT) complex and its components have also been investigated in rabbits.²⁷⁹ After 9 weeks of immobilization, FMT stiffness was significantly reduced, and the ultimate load to failure and energy absorbing capacity were reduced by 69% and 82%, respectively, and an increased number of failures occurred by tibial avulsion. On histologic examination, the femoral and tibial insertion sites showed increased osteoclastic activity, resorption of bone, and disruption of the normal attachment of the bone to the ligament. The cross-sectional area of the CCL also decreased with immobilization of the stifle joint.²⁸⁰ Diminished stress also results in increased collagen turnover, with formation of immature collagen.²⁶⁴

Compared with ligament, relatively less is known regarding the effects of immobilization and subsequent recovery on tendon. In a study of healthy young men, the nondominant leg was immobilized for 2 weeks.²⁸¹ Immobilization caused a 6% decrease in muscle size and 9% decrease in muscle strength, although Achilles tendon size and collagen turnover were unchanged. Another study evaluated people with ankle fractures that were immobilized for 7 weeks.²⁸² Achilles tendon cross-sectional area did not change significantly during immobilization, whereas calf muscle cross-sectional area and strength decreased 15% and 54%, respectively. Local collagen turnover increased in the immobilized leg after 7 weeks of immobilization. Therefore tendon seems to be more resistant to short-term immobilization as compared with muscle.

Small animals may have some loss of tendon properties with immobilization. There is a dose-dependent effect of stress shielding on the mechanical properties of rabbit patellar tendons.²⁷² In addition to decreased ultimate stress of tendons with stress shielding, there is a significant increase in the cross-sectional area of tendons, despite fewer numbers of collagen fibrils per unit area. There was a significant loss of ultimate stress on tendons after 1 to 3 weeks.²⁷²

Tenascin-C, a protein with limited expression in normal musculoskeletal tissues, provides elasticity in mesenchymal tissues subjected to heavy tensile loading. Tenascin-C is expressed in normal rat myotendinous and myofascial junctions, and the Achilles tendon, but is absent in normal skeletal muscle. Following cast immobilization for 3 weeks, tenascin-C was decreased or was completely absent in the regions expressing tenascin-C normally.²⁸³

As the previously discussed studies have indicated, ligaments and tendons are not metabolically inert structures, although their oxygen consumption is approximately 10 times lower than liver tissue and 7.5 times less than skeletal muscles. Immobilization of rat limbs causes an additional 36% reduction in oxygen consumption and decreased aerobic enzyme activity of ligaments and tendons, suggesting that the metabolic activity is further lowered with decreased levels of physical activity.²⁸⁴

Response of Ligaments and Tendons to Remobilization

Although the mechanical properties of immobilized ligaments return to normal relatively quickly,²⁷² the load to failure of the bone-ligament-bone complex lags behind, indicating that there is asynchronous healing of the bone-ligament-bone complex. After 6 weeks of immobilization of the lower limbs of dogs, 18 weeks of remobilization was necessary for return of the normal structural properties of the FMT complex.²⁷³ In fact, up to 1 year of remobilization may be required for normalization of the ligament-tibia complex in some instances, whereas the mechanical properties of the ligament return to normal in a relatively short

period of time, as newly synthesized collagen fibers gradually mature and strengthen with subsequent stress resumption.

In the study of immobilized stifles of rhesus monkeys, 5 or 12 months of remobilization was allowed after casting for 8 weeks.²⁷⁵ Following 5 months of rehabilitation, there was still a 20% deficit in load to failure and energy stored at failure, but only a 7% deficit in stiffness. After 12 months of rehabilitation, there was only a 9% deficit in load to failure and an 8% deficit in energy stored to failure.

The effects of remobilization following 9 weeks of immobilization were studied in the rabbit FMT complex.²⁷⁹ With 9 weeks of remobilization, the ultimate load and energy-absorbing capabilities of the bone-ligament complex improved to approximately 80% of normal. Following 12 weeks of immobilization, 1 year was required to recover the functional properties. Failure by tibial avulsion became less frequent, and the stress-strain characteristics of the MCL returned to normal. Histologic examination found that the sites of insertion of the ligament also showed evidence of recovery.

Little is known about the effect of stretching on the biochemical and morphologic features of tendons that have been immobilized. One study evaluated the response of rat tendons to stretching after immobilization.²⁸⁵ Immobilization followed by stretching resulted in greater chondroitin sulfate at the compression region of tendons. Immobilized animals exhibited the greatest disorganization of collagen bundles, which was partially recovered after stretching or with remobilization. Although the results of this study did not result in recovery with stretching to the degree that occurred with free activity, a revision in stretching procedures, such as duration and frequency of the sessions, may be beneficial in cases of immobilized tendons.

In the study of healthy young men previously reviewed, the nondominant leg was immobilized for 2 weeks, followed by a recovery period of 2 weeks. Although there was a decrease in muscle size and mass with immobilization, there was no change in Achilles tendon size or collagen turnover. Remobilization resulted in increased muscle size, but strength was unchanged. No differences in tendon size could be detected between the two legs after 2 weeks of recovery, although collagen synthesis was increased in the previously immobilized leg.²⁸¹ A further study evaluated people with ankle fractures that were immobilized for 7 weeks, followed by a 7-week remobilization period.²⁸⁶ Again, Achilles tendon cross-sectional area did not change significantly during either immobilization or remobilization, but calf muscle size and strength decreased during immobilization; although there was some muscle size and strength recovery after 7 weeks of remobilization, it was still not complete.

Tenascin-C expression increased in rats with free cage activity for 8 weeks following immobilization, although not to normal levels.²⁸³ More strenuous loading after

immobilization in the form of 8 weeks of low- and high-intensity treadmill running resulted in greater tenascin-C expression in myotendinous and myofascial junctions and the Achilles tendon, reaching the level of normal tissues, suggesting that some level of increased loading is beneficial in recovery of tendon structure after immobilization.

Postoperative Healing of Tendons and Ligaments

The function of tendons and ligaments is to transmit tensile forces. Early studies suggested that healing ligaments should be immobilized for proper healing. Immobilization of surgically repaired tendons or ligaments for 6 weeks results in a wound with approximately 50% the normal tensile strength for tendons and ligaments.²⁸⁷ Recently, it has been suggested that early, controlled mobilization of healing tendon and ligament repairs may be beneficial. Earlier tension across a ligament or tendon wound orients the healing fibers and results in stronger healing. For example, MCLs of canine stifles sutured with polyester suture in a locking loop pattern and immobilized for 3 weeks, followed by active motion, resulted in valgus-varus laxity of 150% and ultimate strength of 92% compared with intact controls. In comparison, doubling the length of the immobilization period to 6 weeks resulted in valgus-varus laxity of 300% and only 14% of the ultimate strength of controls. Repaired canine tendons immobilized for 3 to 6 weeks after surgery also apparently have sufficient strength to allow protected active motion. Current knowledge of healing tendons and ligaments suggests that postoperative immobilization for 3 weeks will allow acceptable return to function while minimizing the risk of damage if appropriate suturing techniques are used. The immobilization splint should be designed to allow some active motion of the joints if possible, but should limit the amount of weight bearing (Figure 7-10).²⁸⁸ After immobilization is completed and coaptation devices are removed, the amount and type of exercise allowed should be severely limited and increased gradually over time. Consideration should be made to protective coaptation if the owner is unable to adequately restrict the animal. By 6 weeks after repair, tendons have approximately 50% of tensile strength as compared with normal. It is believed that tendons need only 25-33% of their original strength to withstand the forces of normal physiologic muscle contractions, indicating that some light activity should not be detrimental to healing.²⁸⁷

Medial Collateral Ligament

Several studies have evaluated early mobilization following injury to the MCL. In one study, injury to the MCL was induced by making incisions through the superficial and deep portions of the ligament, and then repair was performed by suturing the incision.²⁷⁷ Ligaments that were not immobilized during the 6-week healing period were



Figure 7-10 Coaptation following repair of a superficial digital flexor laceration. Note the external skeletal fixator is configured to prevent flexion of the tarsus and weight bearing on the digits to limit stress on the healing repair.

stronger than those that were immobilized with a cast, suggesting that ligament strength is sensitive to physical activity. The effects of early mobilization and exercise on healing of injured MCL were further studied.²⁸⁹⁻²⁹¹ Transected canine MCL were subjected to either no surgical repair with 6 weeks of mobilization, surgical repair with 3 weeks immobilization at 90 degrees followed by 3 weeks remobilization, or surgical repair with 6 weeks of immobilization. Knees were tested in tension to determine the structural properties of the FMT complex and the mechanical properties of the healing MCL. Laxity increased in all knees, but early mobilization resulted in the best joint stability. The structural properties of the FMT complex and the MCL substance were also improved in the early mobilization groups. Stiffness of the MCL in early mobilization groups was approximately 60% of controls, whereas stiffness of prolonged immobilization stifles was only 36% of control stifles. The load at failure of the FMT complex was approximately 40% and 20% of normal stifles in early mobilized versus immobilized stifles, respectively. Similarly, the tensile strength of the MCL was approximately four times greater in stifles that underwent early mobilization, but values of immobilized limbs were only 20% of normal stifles. These studies indicate that early mobilization (less than 3 weeks of immobilization) is the treatment of choice in cases with isolated MCL injuries. It is important to realize that an intact CCL provides significant stability to the stifle joint and allows the MCL to heal. If the CCL is intact, implementation of early motion is recommended.

The long-term effects of surgical repair and immobilization on the healing of the transected MCL have also been studied in a canine model.²⁹² In one group, the transected MCL of the left knee was not repaired, and the joint was not immobilized, whereas in the other group the MCL was repaired and the joint was immobilized for 6 weeks. Types I and III collagen, reducible collagen cross-links, and total collagen concentration for both groups returned to normal levels by 48 weeks postoperatively. Biomechanically, the nonimmobilized group achieved better results than the group that was immobilized in terms of varus-valgus knee motion and ultimate load of the FMT complex. However, the healing MCLs had tensile strength of only 62% and 45% of controls for the two groups at 48 weeks. These results suggested that conservative treatment with early mobilization might be better than surgical treatment with immobilization for an isolated grade III MCL injury.

Healing ligaments may be less able to resist excessive strain during the repetitive loading experienced during daily activity. The ability of ligaments to resist repetitive low cyclic loading (<10% of the normal ligament failure strength) was studied in normal and healing rabbit MCLs.²⁹³ MCLs were surgically gapped in either a unilateral or bilateral fashion with immobilization of the right hindlimb in the bilateral group. MCL scars were allowed to heal for 3, 6, and 14 weeks and were cyclic creep tested at increasing forces. The cyclic modulus of the nonimmobilized scars was less than that of normal ligaments. The percent increase in modulus during cycling was greater for scars than for normal ligaments. Furthermore, immobilization significantly decreased the ability of scars to resist strain, with a majority of immobilized scars failing during repetitive loading. Failures were preceded by a reduction in cyclic modulus indicating damage to the healing ligaments that resulted in eventual total failure. These results suggest that joints with healing ligaments may have increased strain in joint structures while they are under stress, potentially leading to joint instability. Although immobilization could be used temporarily to maintain joint stability, remobilization would likely lead to total failure of the previously immobilized ligament.

Injuries to multiple stifle ligaments may affect healing of the MCL. Although the previous studies have suggested that during normal stifle motion the functional deficit of the MCL in valgus rotation is compensated for by the remaining structures, especially the CCL, MCL strength was just 14% of normal after 14 weeks of healing when both ligaments were transected, compared with transection of the MCL alone, which was 52% of control values.²⁹⁴ There were also marked arthritic changes in the joint when both ligaments were transected.

Thermal shrinkage has been used in other ligaments to improve healing. The effects of immobilization on healing collagenous tissue treated with thermal shrinkage were investigated in a rabbit model.²⁹⁵ Thermal shrinkage was

performed on the medial collateral ligaments of rabbit knees. The joints were immobilized in half of the knees. Animals were evaluated 0, 3, 6, and 9 weeks postoperatively. The failure load of shrunken ligaments was significantly lower than that of the intact ligaments on the day of operation. Shrunken ligaments reached their highest failure loads in the third week. The immobilized ligaments remained weaker than the mobile group, but this difference was statistically significant only in the ninth week. Immobilization seems to have negative effects on the healing ligament.

Lateral collateral ligament healing is similar to healing of the MCL.²⁶⁴ Repaired ligaments heal faster with more organized tissue than nonrepaired ligaments, but both are still weaker than control ligaments after 10 weeks of healing.²⁹⁶

Cranial Cruciate Ligament

Transected ends of the CCL retract and do not join together, as other ligaments may. Repair of transected ligaments results in resorption of 40% of the ligament 24 months after repair, and weakness persists after 4 years of healing.²⁹⁷ The CCL is not normally in contact with synovial fluid because of an overlying synovial membrane. However, when the ligament is damaged, synovial fluid may bathe the damaged ligament, which may inhibit healing because of the inhibitory effects of synovial fluid on fibroblast proliferation.²⁶⁴ In addition, there are increased levels of collagenase with cruciate ligament and meniscal injuries. A study of experimentally created CCL and MCL transection indicated that postoperative immobilization for 6 weeks resulted in greater arthritic changes in articular cartilage than in dogs undergoing early stifle mobilization.²⁹⁸

Human patients undergoing rehabilitation consisting of immediate knee motion and early weight bearing do not have an increased incidence of abnormal knee displacement in the early phases after undergoing bone–patellar tendon–bone allograft or iliotibial band extraarticular stabilization of ACL injuries.²⁹⁹ Although similar studies have not been conducted in dogs, our clinical impression is that there is not an increased incidence of stifle instability in dogs undergoing rehabilitation, and, in fact, there may be less displacement because of the additional stability afforded by greater muscle mass of patients undergoing rehabilitation.

Flexor Tendons

Immediate immobilization of flexor tendons following repair results in diminished ultimate tensile strength in the early postoperative period. Using an early passive motion protocol, ultimate tensile strength and ultimate load are increased 2–3 times more than if the limb is immobilized.³⁰⁰ In addition, joint motion is significantly greater than if the joints are immobilized. Early protected passive motion augments biomechanical properties and excursion of

healing flexor tendons. These results were confirmed in a study that compared different suturing techniques in dogs undergoing early mobilization following repair of lacerated deep digital flexor tendons. The mechanical and histologic healing of tendon repairs using early active motion was evaluated with three flexor tendon repair techniques over a 21-day period.²⁸⁸ Adhesions occurred in 15% of the tendon splint repairs. Smooth tendon gliding was obtained in the other specimens in which repairs were successful. Histologic examination demonstrated evidence of progressive healing without surrounding adhesions in repaired tendons. Improved suture techniques have the potential to withstand the stress produced by active digital motion protocols. Immobilization may also result in adhesion formation, reducing the excursion of healing tendons and reduced ROM. In one study, 10 minutes of daily passive motion, with 1.5 to 1.7 mm of tendon excursion prevented adhesions in a sharp transection model of the deep digital flexor tendon.^{301,302} Doubling the excursion resulted in no additional improvement. Therefore less aggressive passive-motion exercises may be sufficient to prevent adhesion formation. In fact, overly aggressive ROM rehabilitation may be deleterious. Tendons undergoing excessive motion with formation of a large gap between tendon ends are at risk for rupture during early healing.³⁰³ Increased rate of passive motion also has beneficial effects. In one study of healing canine digital flexor tendons, passive motion at a rate of 12 cycles/min for 5 minutes daily resulted in similar gliding function with significantly greater tensile properties than dogs receiving 1 cycle/min for 60 minutes.³⁰⁴ Repair of tendons should occur within 7 days for optimum restoration of gliding function.³⁰⁵ Although repair of tendons as long as 21 days after injury results in similar strength and stiffness as tendons undergoing immediate repair, there is significant loss of gliding function with delayed repair.

There may be some benefit to delaying rehabilitation for several days after tendon repair. One study compared the short-term outcome following flexor tendon repair and postoperative rehabilitation beginning on day 1 versus day 5 in dogs.³⁰⁶ Initiating postoperative mobilization at day 5 resulted in no tendon ruptures compared with tendon ruptures in four of the dogs (33%) in the group subjected to mobilization starting at day 1. There was a trend toward lower resistance of flexion in the day 5 start group, but there were no differences between the groups by day 10. Starting rehabilitation on day 5, when initial gliding resistance is lower, may have an advantage over earlier starting times, when surgical edema and other factors increase the initial force requirements to initiate tendon gliding.

Rehabilitation of partial flexor tendon lacerations may be different than for complete lacerations. In one study of partial lacerations of the deep digital flexor tendon, dogs not undergoing surgical repair had higher ultimate loads and stiffness than repair groups.³⁰⁷ In addition, passive

rehabilitation of nonrepaired tendons resulted in less gap formation than unrestricted active mobilization of repaired tendons.

Both passive and active rehabilitation are superior to immobilization following partial tendon laceration, but few studies have directly compared the two. Gap formation, adhesions, gliding function, and structural properties of repaired and unrepaired tendons were evaluated following 3 weeks of unrestricted active rehabilitation versus passive mobilization for partial laceration of canine deep digital flexor tendons.³⁰⁸ There were no differences in tendon excursion, total joint rotation, or adhesions between any groups. Gap size was higher with active mobilization. There was no effect of rehabilitation protocol on the strength or stiffness of healing tendons at 3 weeks. Tendon repair did not affect tendon strength, but did produce higher stiffness in healing tendons at 3 weeks. The authors concluded that active rehabilitation appears safe for partial lacerations less than 60%. Although repair appears to weaken the tendon in the early stages of healing, it may provide some biomechanical benefit by the middle stages of healing.

Retaining vascularity does not seem to prevent significant reduction in tendon properties that occur postoperatively, nor does it accelerate the return in strength and stiffness.³⁰⁹ The medial half of the patellar tendon was placed as a vascularized graft to replace the ACL on one side and as a nonvascularized graft on the other in monkeys. Grafts had only 57% of control ACL stiffness and 39% of control maximum force by 1 year. Elastic modulus and maximum stress also increased over time, but at 1 year were only 34% and 26% of normal values, respectively.

Little is known about the localized changes in bone mass that occur following tendon or ligament injury. Interruption of normal load transfer at the insertion site may lead to localized loss of bone. In a canine study, the deep digital flexor tendon was transected from its insertion and either repaired or left unrepaired.³¹⁰ Postoperatively, limbs in the repair group were cast immobilized except for 10 minutes of daily passive mobilization rehabilitation, whereas the laceration-only group dogs were allowed full weight bearing. At 5-42 days postinjury, bone mineral density (BMD) was assessed. There was significant bone loss in the distal phalanx after tendon laceration, repair, and postoperative passive mobilization, with BMD decreases of 20%, 40%, and 41% at 10, 21, and 42 days. In contrast, BMD was reduced by 28% in bones subjected to tendon laceration and full weight bearing. Osteoclastic resorption occurred in both groups. Bone loss may affect healing by impeding restoration of a strong tendon-bone interface.

Modalities, such as therapeutic ultrasound, may aid tendon healing. The Achilles tendon of five dogs was severed proximal to its insertion.³¹¹ Tenorrhaphy was performed using locking-loop sutures. A cortical screw was

used to immobilize the tibiotarsal joint, and it was removed 4 weeks after tenorrhaphy. Ultrasound therapy was given to the treated group starting the third day postoperatively using 0.5 W/cm² for 10 minutes daily for 10 days. The dogs showed significant lameness for the first 4-5 days, which disappeared earlier in the ultrasound-treated dogs than the controls. By day 40, the echogenicity was improved in the treated group compared with controls. However, the tendon showed near normal echoic texture in both groups by day 120. The Achilles tendon in treated dogs showed comparatively fewer adhesions than in controls. Histologic features of tendon healing were improved in treated dogs throughout the study, and by day 120, the tendon tissue was comparatively acellular and looked like a normal tendon.

Rotator Cuff Tendons

The response of healing tendons in different areas of the body may differ from flexor tendons. In studies of animal models of rotator cuff repair, immobilization may improve mechanical properties more than passive motion. The effects of daily passive motion or continuous immobilization on joint mechanics in a rat rotator cuff repair model following injury to the supraspinatus muscle were evaluated.³¹² Continuous immobilization or passive motion was performed for 2 weeks before all underwent a remobilization protocol for 4 weeks. Total ROM was less with passive motion than with continuous immobilization at 2 weeks and remained significantly less following 4 weeks of remobilization. Joint stiffness at 2 weeks was increased with passive motion in comparison with continuous immobilization. There were no differences between the groups in terms of collagen organization or mechanical properties. The authors concluded that in this model, immediate postoperative passive motion was detrimental to passive shoulder mechanics, perhaps because passive motion increased scar formation in the subacromial space, resulting in decreased motion and increased joint stiffness.

A similar study concurred with these findings. Gimbel and colleagues investigated the effect of short and long durations of various activity levels on healing supraspinatus tendon to bone insertion sites in rats.³¹³ The postoperative activity level was controlled through shoulder immobilization, cage activity, or moderate exercise for 4 or 16 weeks. Decreased activity level had a positive effect on stiffness and modulus of elasticity of the insertion site at 16 weeks (immobilization > cage activity = exercise). Collagen disorganization was decreased with a decrease in activity level at 4 weeks, but was similar between groups at 16 weeks. Decreasing activity level by immobilizing the shoulder improved tendon to bone healing in this study.

Thermal shrinkage of the joint capsule is occasionally performed in mild cases of medial glenohumeral instability. It is important to immobilize the shoulder joint after surgery, but the optimal time of immobilization is unknown.

Ideal timing for remobilization is still controversial, and the length of time that the canine shoulder joint should be immobilized after thermal shrinkage is unknown, but abandoning immobilization protocols for longer than 3 weeks should be considered based on healing of stifle medial collateral ligaments with thermal shrinkage.²⁹⁵

Adaptations of Ligaments and Tendons to Increased Mechanical Stress

Much like other musculoskeletal tissue, ligaments and tendons also adapt to applied stress and motion. Ligaments and tendons are morphologically, biomechanically, and biochemically sensitive to stress deprivation as well as stress enhancement. Although testing procedures vary among studies, most have consistently demonstrated that training increases the strength of ligaments and tendons. Compared with other tissues, such as muscle, the metabolic activity of ligaments and tendons is lower, probably because of poorer vascularity and circulation.³¹⁴ The responses of ligaments and tendons to increased training stresses are therefore slower and less pronounced than in muscles. It is also important to consider the conditions under which laboratory animals are housed when interpreting the results of experimental studies because ligaments and tendons are very sensitive to stress deprivation, and limiting normal activity with cage confinement may affect strength of these tissues.³¹⁵

Ligaments retain 80-90% of their baseline mechanical properties with minimal stimulation. The effects of exercise on ligaments and tendons are less profound than the effects of immobilization. In a healthy patient, ligament exercise is not likely to increase the strength and stiffness by more than about 10-15%. Sprint training results in increases in ligament weight but not ligament-bone junction strength.³¹⁵ However, endurance activity with regular mechanical loads results in increases in junction strength and ligament stiffness. Low-duration (30 minutes) and high-frequency (6 days/week) endurance exercise appears to have the most benefit to ligament strength. A program of low-intensity exercise combined with short bouts of high-intensity exercise in young horses resulted in thicker and stronger superficial digital flexor tendons as compared with horses that were stall rested or stall rested with a single daily bout of high-intensity exercise.³¹⁶

The effects of exercise on morphologic characteristics and mechanical properties of the canine stifle medial collateral ligament have been investigated.^{277,317} Dogs undergoing treadmill exercise for 6 weeks had a 12% increase in failure load of the medial collateral ligament, and a 15% increase in hydroxyproline content (indicative of collagen content) of the lateral collateral ligament compared with control dogs. In addition, collagen fiber diameter was greater following training, although GAG content, water content, collagen concentration/dry weight or collagen concentration per weight/length unit were unchanged.

Another study evaluated the effect of exercise on the structural properties of swine FMT complex. Animals were trained on a treadmill daily for 12 months. Exercised pigs had a significant increase in the ultimate load to failure of their ligaments when normalized for body weight, but there were no significant changes in deformation, or energy absorbed of ligaments. Results found in young rabbits that were trained on running wheels indicated that training increased the breaking strength of many ligaments and tendons, including the tibia-CCL-femur complex.³¹⁷ Studies in rats and mice have shown similar results and have suggested a possible additive role of testosterone in enhancing ligament and tendon strength with training.⁵⁸

Studies in other species have not always demonstrated a corresponding increase in breaking strength, fiber diameter, and collagen content, however. For example, there was no increase in hydroxyproline content despite an increase in breaking strength in rats undergoing training.⁵⁸ In addition, there was an increase in collagen turnover with training. More of the newly formed collagen may be in the soluble form because of fewer cross-links. Metabolic changes have been noted in tendons and ligaments of rodents undergoing training, including increased tendon nitrogen content; malate dehydrogenase; reduced nicotinamide adenine dinucleotide oxidase; lactate dehydrogenase; isocitrate dehydrogenase; and succinate dehydrogenase early in the training period.⁵⁸ The water content does not change with training, but the number of cell nuclei per unit volume of tendon are greater in trained rats. Training results in an increase in oxygen consumption of less than 10% in ligaments and tendons, although there is a 58% elevation in muscle tissue. It is unclear why chronic exercise does not produce a greater effect.²⁸⁴ Changes in the content of an elastic extracellular matrix protein, tenascin-C, have also been noted with immobilization and remobilization of ligaments and tendons.³¹⁸ Although rat tenascin-C was present at the normal rat bone-patellar tendon junction, it was almost completely absent after 3 weeks of cast immobilization. Remobilization in the form of cage activity for 8 weeks resulted in some increase, but not to normal levels. Tenascin-C activity around the chondrocytes and fibroblasts of the bone-tendon junction and the collagen fibers of the tendon belly was normal following remobilization with 8 weeks of treadmill training, however.

Stretching is frequently recommended in dogs in training or during rehabilitation to help improve joint ROM. In a study of rat tail tendons, stretching to 108% of the original length resulted in return to 104% of the original length when the stretch was released.³¹⁹ A second stretch resulted in more stretch per unit load (less stiff). Histologic examination found evidence of damage to some strands of collagen. Despite the changes in stiffness, the maximum breaking load was not significantly altered after the second stretch.

Effects of Medications on Ligaments and Tendons

Medications are sometimes administered locally or systemically in the treatment of musculoskeletal conditions. In particular, corticosteroids may be administered intraarticularly and have effects on ligaments and tendons. The effect of intraarticular corticosteroid injections on the mechanical properties of anterior cruciate bone-ligament-bone units was determined in rhesus monkeys.³²⁰ Alterations in ligament strength and elongation properties depended on corticosteroid dosage and time after injection. Fifteen weeks after three weekly injections of large doses of methylprednisolone acetate (6 mg/kg), there were significant decreases in the maximum failure load (20%), energy absorption prior to failure (11%), and stiffness (11%) of the ligament unit. In contrast, only minimal alterations occurred after 6 weeks suggesting effects on the ligaments was not immediately seen but was long lasting. A third group of animals received intraarticular injections (0.6 mg/kg) spaced 2 weeks apart. Although decreases occurred in maximum failure load (9%) and energy absorption (8%) by 15 weeks, the magnitude of changes was less. There was microscopic evidence of collagen fiber failure at multiple levels throughout the ligament. Although high and frequent doses of intraarticular corticosteroids produce the greatest alterations in ligament strength and function, even low doses given infrequently affected ligament properties. A single intraarticular injection or injections repeated at intervals of several months is recommended to minimize ligament damage. Furthermore, direct injection of corticosteroid into the ligament should be avoided because fibrocyte death and alterations in ligament strength and stiffness may occur for up to 1 year.³²⁰

Injection of corticosteroid inside a tendon has deleterious effects, with collagen necrosis and decreases in tensile strength occurring.³²¹ Peritendinous injections are generally safer, and with proper indications, there may be fewer complications. Systemic administration of corticosteroids also has fewer adverse effects on tendons than intratendinous injection. However, administration of very high doses (10 mg/kg) may result in reduced breaking strength of sutured canine tendons.³²¹ In rabbits receiving corticosteroids, bony insertion sites became osteoporotic and failure tended to occur by fracture.³²² Tissues with shorter turnover times, such as bone, tend to weaken faster than tissues with longer turnover times, such as tendons and ligaments, when corticosteroids are administered. In fact, tendons may initially become slightly stiffer early in the course of steroid administration because fibroblast proliferation and collagen production are reduced, resulting in decreased insoluble collagen. Long-term steroid use eventually results in weakening of ligaments and tendons because of the continued inhibition of collagen synthesis.

Treatment with topical dimethyl sulfoxide (DMSO) has been recommended for the treatment of local inflammatory conditions, including injuries to tendons and muscles. However, topical treatment with DMSO of the Achilles tendons of mice indicated that the tensile strength of the tendon decreased in relation to the length of time the mice were treated.³²³ The maximum decrease of 20% was seen 7 days after treatment began. These data suggest that vigorous muscle activity should be avoided during treatment with DMSO.

Fluoroquinolone antibiotics may contribute to tendinopathies. The incidence of tendinitis and tendon rupture in human patients treated with ciprofloxacin suggests that these antibiotics alter tendon fibroblast metabolism. There was a 67% decrease in cell proliferation compared with control cells, and a 36% to 48% decrease in collagen synthesis of canine Achilles tendon, paratenon, and shoulder capsule fibroblasts incubated with serum levels of ciprofloxacin in cell culture.³²⁴ Ciprofloxacin also caused a 14% to 60% decrease in PG synthesis and significant increases in matrix-degrading proteolytic activity. These changes suggest that ciprofloxacin stimulates matrix-degrading protease activity from fibroblasts and inhibits fibroblast metabolism, which may contribute to the clinically described tendinopathies associated with ciprofloxacin therapy. In fact, there is a black box warning with the use of any fluoroquinolone antibiotics for possible tendinopathy, especially in patients older than 60 years.

Certain pharmacologic agents may positively affect ligaments and tendons. For example, treatment with bone morphogenetic protein-2 results in more extensive bone formation around tendons placed through bone tunnels with closer apposition of new bone to the tendon in treated limbs, resulting in higher tendon pull-out strength.³²⁵ Bone morphogenetic protein can accelerate the healing process when a tendon graft is transplanted into a bone tunnel.

Bone

Normal Bone

Bone consists of organic and inorganic components. The organic component consists of collagen fibers embedded in a matrix of GAGs, among other components of ground substance. The inorganic component consists primarily of calcium and phosphorus in the form of hydroxyapatite crystals. These crystals are embedded within the organized collagen fibers. Bone provides the musculoskeletal tissues with compressive strength during weight bearing, and the organic component provides tensile strength.

Long bones have a compact portion of lamellar bone found in the outer cortex, and loosely woven trabecular bone found in the inner medullary bone. The inner medullary bone is typically more metabolically active because of the relatively large surface area, but it is also relatively



Figure 7-11 Radiographs of the metacarpal bones of a 10-month-old dog with fractures of the distal radius and ulna. **A**, At the time of fracture. **B**, Four weeks after cast placement. Note thinning of the cortices and generalized bone atrophy.

weaker as compared with the more dense outer cortical bone.

Effects of Immobilization on Bone

Many models exist to study the effects of immobilization on the skeleton, including local or systemic immobilization. Local immobilization involves casting, splinting, denervation, paralysis, disarticulation, and limb suspension, whereas systemic immobilization is accomplished with body casting, bed rest, or spaceflight. Reduced stress on bones may also result from stress shielding following application of implants for repair of fractures. Care must be taken in employing the various models. They have similar patterns of bone loss, but they respond slightly differently in the location and the amount of bone loss, depending on the degree of unloading.

Models of forelimb and HI that prevent weight bearing on the limb induce a reduction in cortical and cancellous bone mass, reduce cortical bone density and stiffness, and increase turnover in cancellous bone.^{326,327} The changes that occur following immobilization vary depending on the length of immobilization and remobilization, age of the animal, and the bone involved. The effects appear to be more profound in younger dogs (Figure 7-11). In fact, immobilization during growth may result in permanent changes in bone mass and the relation of bone mineral content (BMC) to body weight.³²⁸ Older animals may have less preexisting bone mass as a result of senile osteoporosis, and these dogs may lose less bone following immobilization than those with higher initial bone mass.^{83,329} Trabecular bone is affected to a greater degree than cortical bone, and the effects of immobilization are more extensive in the more distal weight-bearing bones (Figures 7-12 to 7-14).^{329,330} Diaphyseal bone loss during immobilization

affects the cranial and caudal cortices to a greater extent than the medial and lateral cortices.³³¹ These changes are consistent with those seen in other species, including rats.³³²

Immobilization models may produce loss of 60% of the trabecular bone mass.³²⁹ This level may suggest a physiologic threshold at which basal hormonal and cellular activity determined by genetics is reached in the absence of mechanical stimuli.³²⁹ During the acute phase of immobilization, the greatest bone loss occurs in medullary bone from an increase in bone resorption and a decrease in formation. During the chronic phase, bone mass plateaus and cellular activities may revert back to normal levels.

The response to cast immobilization of bone in young adult dogs may be divided into three stages of histologic response.³³³ In stage I, there is rapid initial loss of bone, reaching its maximum rate of loss at 6 weeks, with relatively equal contributions of the periosteal, Haversian, and endosteal regions to the loss. The rate of bone loss slows during stage II, which consists of 24 to 32 weeks of immobilization. The majority of the bone loss during this stage comes from the periosteal envelope. Stage III, characterized by greater than 32 weeks of immobilization, results in bone mass that has been reduced to 30% to 50% of original values. This pattern is qualitatively similar in all bones of the forelimb, but the distal bones are affected to a greater degree. Another study of immature dogs demonstrated a 55% decrease in bone mass of the distal tibial metaphysis following 4 weeks of unilateral hindlimb cast immobilization.³³⁴

The extent of resorption surface and the increase in the total histologically “active” periosteal envelope parallel the phases of bone loss. The type of metabolic bone activity also varies depending on the length of immobilization. In

Figure 7-12 Trabecular bone volume of the radius following various periods of immobilization and remobilization. Note the relatively greater loss of bone volume and trabecular bone compared with the diaphyseal bone in [Figure 7-13](#).

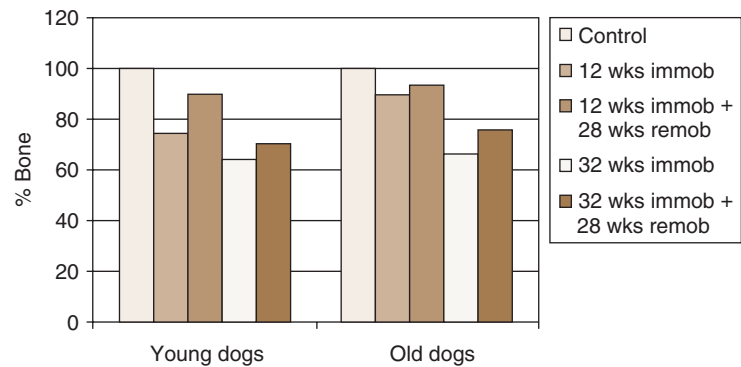


Figure 7-13 Bone volume of the diaphysis of the radius after varying periods of immobilization and remobilization. Bone loss is greater in young dogs immobilized for long periods.

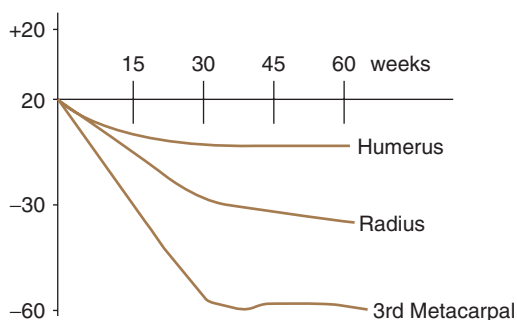
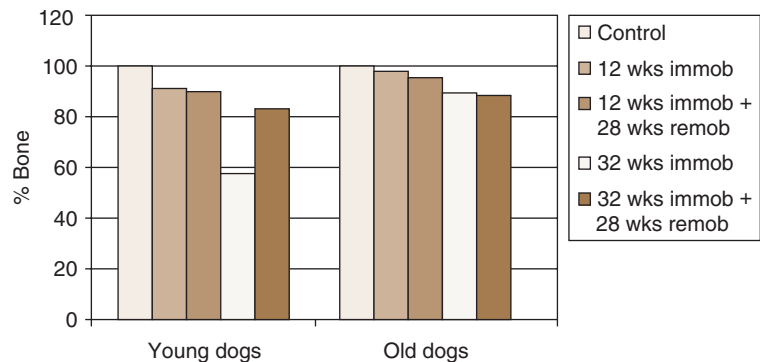


Figure 7-14 Effects of immobilization on density of various bones.

cast-immobilized young adult beagle dogs, the total active metacarpal bone surface (resorption and formation) was reduced 60% below control values during the first 2-4 weeks, increased 50% above controls at 6 weeks, and subsequently returned to control values at 8-12 weeks.³³³ The response of bone to immobilization differs in mature dogs and is related to the length of immobilization.⁸³ In one study, bone volume was estimated in the third metacarpus, radius, ulna, and humerus at the mid-diaphysis and the distal metaphysis of dogs with forelimb immobilization for up to 32 weeks. In both young and mature dogs, bone volume decreased with the duration of immobilization and the types of changes were similar in the metaphysis and

diaphysis. The old dogs, which began the study with 10% less bone than the younger dogs, showed smaller proportional losses than younger dogs, most evident in the diaphysis. In both age groups, the distal, weight-bearing bones tended to show greater losses. Following 32 weeks of immobilization, young dogs had 54% less bone volume in the third metacarpal diaphysis, whereas older dogs had 38% less. Metacarpal trabecular bone decreased 50% in young dogs and 47% in older dogs. Adult dogs had a 45% decrease in bone mass of the distal radial metaphysis following 16 weeks of cast immobilization in another study.³³⁵

The amount of bone loss associated with the length of immobilization apparently has limits. Although immobilization studies have demonstrated bone loss of up to 60% of the trabecular bone mass, there appears to be a maintenance bone turnover level that occurs with continued immobilization.³²⁹ The remaining maintenance level suggests a physiologic threshold at which basal hormonal and cellular activity is reached in the absence of mechanical stimuli. During the acute phase of immobilization, bone loss occurs in bone adjacent to marrow, and from an increase in bone resorption and a decrease in bone formation.³³⁰ There may be more erosion of the inner cortical surface in older dogs. Young growing dogs also have an interruption of periosteal bone deposition, resulting in smaller bone diameter. Immobilization also results in

increased regional blood flow and bone marrow pressure. The flow of the entire long bone is approximately four times greater than that of cortical bone, indicating that cancellous bone receives much of the blood flow. There is no significant difference in flow rate between the endosteal and periosteal cortices, however. There are variations of blood flow among long bones, which might be due to differences in their structure and function, and which may help to explain the differences in bone loss following immobilization.³³⁶ Secondary factors, such as low dietary calcium or fluctuating hormone levels may also be involved. At steady state, or chronic phase, bone mass plateaus to a maintenance level with cellular activities returning back to normal levels.

Biomechanical properties of cortical and cancellous bone are also significantly affected by immobilization. Forelimbs of dogs immobilized for 16 weeks had decreases in cortical load, yield, and stiffness as well as cancellous bone failure stress, yield stress, and modulus, compared with control limbs.⁸² In general, immobilized limb cancellous bone mechanical properties were 28-74% of control values, and cortical bone mechanical properties were 71-98% of control values. Similar to the differences in bone loss that occur in different bones, biomechanical properties of different bones in the same immobilized limb may also vary. Torsional stiffness of radiuses and metacarpal bones immobilized for 12 weeks decreased approximately 12% and 30%, respectively, as compared with controls.^{329,337} Increasing the length of immobilization results in further deterioration of biomechanical properties. Metacarpal bone immobilized for 26 weeks retained only 50% of normal stiffness.

There is increased urinary and fecal excretion of calcium and phosphorus in people with cast immobilization of a limb, and similar findings may be expected in dogs.⁵⁸ Excretion of hydroxyproline, an indicator of collagen content, also increases with disuse. Relatively equal proportions of bone matrix and bone mineral are lost during bone atrophy, suggesting that the organic and mineral composition of bone is similar following a period of immobilization. Histologic examination reveals an increase in osteoclast activity and vascularization with disuse, suggesting an increase in bone resorption. Prostaglandin E (PGE) may also be involved in the development of immobilization osteoporosis.³³⁴ Osteoporosis induced by fiberglass-cast immobilization of the hindlimb of immature dogs for 4 weeks resulted in significant loss of bone mass and a twofold increase in bone PGE, suggesting that PGE plays a role in immobilization osteoporosis.

Bone resorption also occurs at insertion sites of ligaments as a result of disuse, and there is decreased mechanical strength and bone collagen associated with disuse.²⁷⁵ An internal skeletal fixation model applied for 12 weeks resulted in significant atrophy of bone and loss of collagen in the tibia and femur at the insertion and origin of the CCL.²⁹ When a fast rate of deformation was applied, failure of the femur-ligament-tibia complex occurred through the

tibial insertion of the ligament. Bone atrophy was the result of increased resorption of bone rather than decreased bone formation. Preservation of active joint motion during immobilization may have a protective effect on the bone-ligament-bone complex.³⁰ A unilateral tarsal disarticulation model of disuse that maintained active knee motion without weight bearing for 8 weeks maintained the strength of the femur-CCL-tibia complex despite loss of collagen and calcium mass in the femur and tibia, suggesting that active joint motion helps to preserve the mechanical strength of the femur-ligament-tibia complex.

Bone loss may be reduced in paraplegic human patients who are ambulatory as compared with those who are non-ambulatory, who develop osteoporosis of the pelvis and limbs.⁵⁸ A relatively low degree of physical training may inhibit the development of osteoporosis. Slow oscillation of bedridden human patients with intact neuromuscular function from a horizontal position to a 20 degrees foot-down position every 2 minutes may help preserve bone mass. The contraction of antigravity muscles during the downward tilt may be partially responsible for maintenance of bone mass by adding stress to bones during muscle contraction. There also appears to be an association of bone mass with muscle mass, with similar degrees of bone and muscle loss with disuse. Non-weight-bearing exercise of bedridden patients is relatively less effective in reducing calcium loss, but standing for 3 hours reduces urinary excretion, suggesting that weight bearing rather than physical activity is primarily responsible for reducing bone atrophy during bed rest.

In summary, young dogs tend to lose bone more quickly following immobilization as compared with older animals, even though older dogs may have less bone mass to begin with. The degree of bone loss increases with the length of immobilization, with the rate of loss slowing as the time of immobilization increases. Bone loss is greater in the more distal weight-bearing bones as compared with proximal minimally weight-bearing bones (Box 7-9).

Bone Changes Resulting from Disease Conditions

Bone changes may be detected after total hip replacement and complications of this procedure. In one study, dogs with unilateral primary cementless total hip arthroplasty had no right-left difference in tibial BMC or cortical bone cross-sectional geometry after 6 months, but after 2 years there was a 5-6% difference in BMC, perhaps because of subclinical disuse of the operated limb.³³⁸ In dogs with failed cemented prostheses, tibial BMC may be more than 20% lower in the operated limb. Successful revision of the total hip replacement results in improved BMC, suggesting that improved limb function after revision surgery may result in gain of previously lost bone.

Bone loss may also occur following knee surgery as a result of relative disuse of the affected limb. Following

Box 7-9 Summary of the Effects of Disuse and Immobilization on Bone

- Models of limb immobilization decrease cortical and cancellous bone mass and cortical bone density and stiffness and increase turnover of cancellous bone.
- Limb immobilization results in decreases in cortical load, yield, and stiffness as well as cancellous bone failure stress, yield stress, and modulus.
- The changes that occur during immobilization vary depending on the length of immobilization and remobilization, age of the animal, and the bone involved.
 - Longer periods of immobilization result in greater bone loss.
 - Generally no further bone loss after 32 weeks of immobilization.
 - Effects are more pronounced in younger dogs.
 - Immobilization during growth may result in permanent reduction of bone mass.
 - Older animals may have less preexisting bone mass.
 - Trabecular bone is affected to a greater degree than cortical bone.
 - Effects of immobilization are more extensive in the more distal weight-bearing bones.
- Recovery of mechanical and morphologic properties of bone following immobilization depends on the length and type of

immobilization, the type and intensity of remobilization, and the age of the animal.

- Longer periods of immobilization result in longer recovery times.
 - Immobilization for 6 weeks requires 8 to 12 weeks for near full recovery of bone mass.
- Immobilization for 24 to 28 weeks requires longer for near full recovery of bone mass.
- With greater than 32 weeks of immobilization, bone mass remains at 30% to 50% of original values, despite remobilization.
- Younger animals recover lost bone mass more quickly than older animals.
- Training after immobilization results in more complete and rapid recovery of lost bone mass.
 - An initial period of mild treadmill training three times per week is recommended, because bone biomechanical properties may initially worsen during the initial remobilization period.
- This can be followed by a period of higher intensity training.
- Jump training may result in recovery of more normal trabecular bone mass.

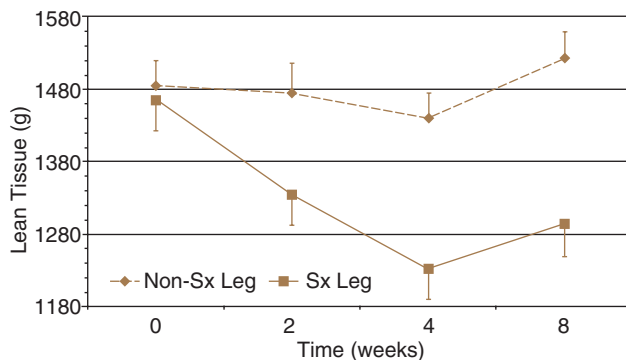


Figure 7-15 Changes in mean lean tissue content over time for the surgical and nonsurgical legs following transection of the cranial cruciate ligament and immediate stifle stabilization. (From Francis DA, Millis DL, Head LL: Bone and lean tissue changes following cranial cruciate ligament transection and stifle stabilization, *J Am Anim Hosp Assoc* 42(2):127-35, 2006. ©From the Journal of the American Animal Hospital Association (jaaha.org) 2006. All rights reserved.)

CCL transection and extracapsular stabilization, DEXA was used to analyze BMC and lean tissue mass of the entire surgical and nonsurgical legs.¹⁹¹ Regional BMC and BMD of the proximal, mid-, and distal tibia of the surgical and nonsurgical legs was also determined. There was significant loss of BMC and lean tissue in the entire surgical leg compared with the nonsurgical leg. Interestingly, the pattern of tissue loss was different. There was an immediate loss of lean tissue mass for the first 4 weeks, followed by slight recovery of lean tissue mass between 4 and 8 weeks (Figure 7-15). In contrast, there was little change in BMC for the first 2 weeks (Figure 7-16). This was followed

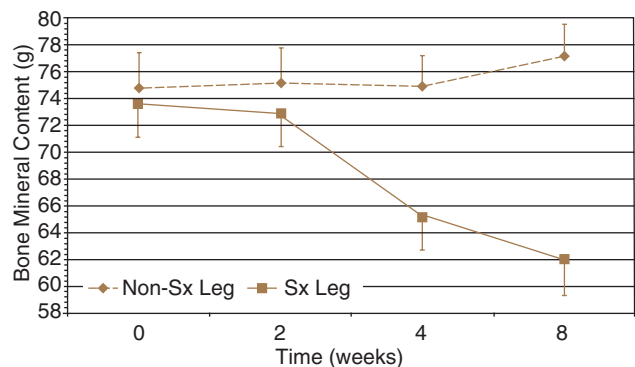


Figure 7-16 Changes in mean bone mineral content over time for the surgical and nonsurgical legs following transection of the cranial cruciate ligament and immediate stifle stabilization. (From Francis DA, Millis DL, Head LL: Bone and lean tissue changes following cranial cruciate ligament transection and stifle stabilization, *J Am Anim Hosp Assoc* 42(2):127-35, 2006. ©From the Journal of the American Animal Hospital Association (jaaha.org) 2006. All rights reserved.)

by a rapid loss of bone mineral from 2 to 8 weeks, with no apparent recovery of bone mineral during that period. Regarding regional changes in bone, significant loss of BMC and BMD was detected in the tibia of the surgical leg and was most pronounced in the metaphyseal regions (Figure 7-17).

Another study of dogs with surgically created CCL transection demonstrated that there was a dramatic 22-34% decrease in BMD of the affected distal femur 12 weeks after surgery.³³⁹ Studies of humans indicate similar decreases in BMD of approximately 20% in the distal femur, patella, and proximal tibia up to 1 year after surgery

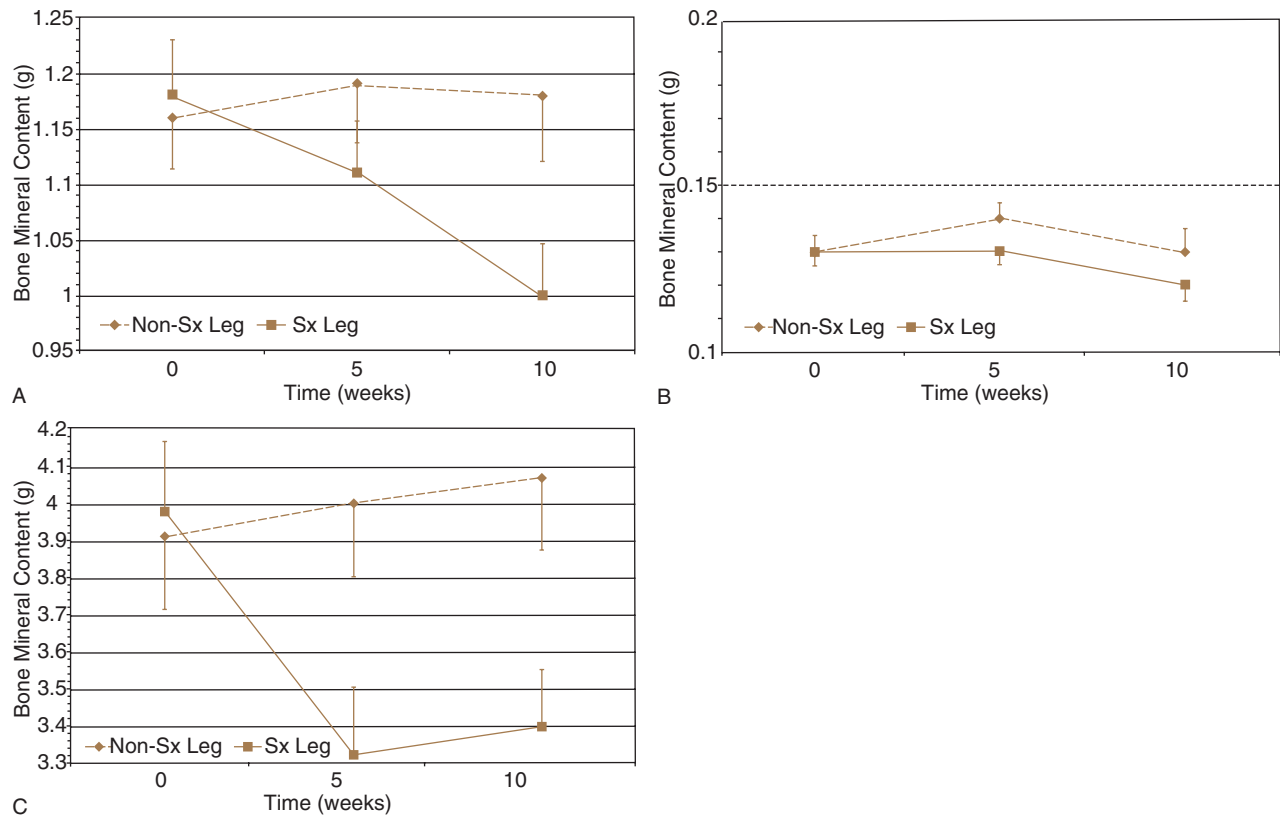


Figure 7-17 Changes in bone mineral content over time at the proximal tibial metaphysis (A), midtibial diaphysis (B), and distal tibial metaphysis (C) of the surgical and nonsurgical legs.

for ACL injury.^{340,341} In addition, 10–11 years after surgery bone density may be decreased by 3% to 9%.³⁴² Bone density had significant correlation with functional scores of the knee in people, but not with knee stability.

In addition to bone changes following ligament injury, it is possible that there may be decreased bone density following fracture. In humans, 2% to 7% decreases in bone density were found in the distal femur, patella, proximal tibia, and calcaneus 10 years after femur fracture.³⁴³ Relative bone density had correlation with pain and functional scores of the affected limb, but not with fracture type, fracture location, or non-weight-bearing time after injury. Similar decreases in bone density were seen 9 years after tibial fractures in another study.³⁴⁴

Denervation of a limb of a growing dog results in decreased lengths of the humerus and radius.³⁴⁵ Furthermore, loss of innervation to a region may result in decreased bone density.³⁴⁵ However, active muscle contraction, either through reflex walking or electrical stimulation, helps attenuate loss of bone density.³⁴⁶ A functional neuromuscular stimulation-induced knee extension exercise system was used in human patients with spinal cord injuries up to three times per week for 36 sessions using a progressive resistance load.³⁴⁷ Maintenance of bone density in some subjects suggests that training may retard the rate of bone loss that typically occurs with spinal cord injuries.

Stress Protection

Stress shielding or stress protection occurs when metal implants, such as bone plates and screws, are used to repair fractures or in joint replacement surgery. Although rigid metal plates stabilize the fracture site, help maintain contact between bone fragments, and allow early weight bearing and patient mobility, the higher stiffness of the implant results in bone loss as a result of decreased physiologic loading of the bone. Calculations were made of the alterations in cyclic bone stresses caused by the application of bone plates on canine femurs.³⁴⁰ The magnitude of the reduction in the loads borne by the bone and the degree of shift in the bone stress neutral axis during the stance phase of gait was influenced by plate geometry, plate stiffness, and plate location. Bone remodeling is very sensitive to small changes in cyclic bone stresses. Changes in cyclic bone stresses of less than 1% of the ultimate strength can cause measurable differences in bone remodeling after a period of a few months.

The pattern of cortical atrophy induced by plate fixation for 7 months on canine femurs was evaluated in another study.³⁴⁸ A pronounced reduction of cortical bone (approximately 15% less cortical area) was observed in the diaphysis of the plated bone at the end of the study period. The loss of cortical bone was mainly caused by endosteal resorption with enlargement of the medullary cavity.

Neither periosteal resorption nor formation of woven bone under the site of the plate was observed. Osteoporosis caused by rigid plate fixation may occur by thinning of the cortex rather than by reduction of the mechanical properties of the osseous tissue.³⁴⁹ In addition to enlargement of the medullary canal, there may also be decreased cortical thickness under the plate as a result of endosteal resorption. The stiffer the implant and the longer the implant is in place, the greater the bone loss and the lower the recovery potential of the bone. For example, 6 weeks after screw application, some resorption of bone is evident and by 14 weeks, more significant resorption occurs.

Implants may also interfere with blood circulation, and the weakened bone can refracture after plate removal.³⁵⁰⁻³⁵² Disturbance of blood supply may be the main cause of osteopenia in the early stage, whereas disuse of the affected limb may be the most important factor in later stages of fracture healing. Osteopenia may be treated, in part, by active exercises and normal weight bearing before plate removal.³⁵²

Stress shielding may also occur following total hip arthroplasty.³⁵³ Stiffness of the femoral stem is a design variable that is partially responsible for femoral remodeling. The basic pattern of bone remodeling is characterized by proximal cortical atrophy, and distal cortical and medullary bone hypertrophy. Low-stiffness stems may alter this pattern, leading to reduced proximal bone loss, increased proximal medullary bone hypertrophy, and no distal cortical hypertrophy, suggesting that stem stiffness has a profound effect on stress shielding.

Effects of Remobilization on Bone

The recovery of mechanical and morphologic properties of bone following a period of immobilization depends on the length and type of immobilization, the type and intensity of remobilization, and the age of the animal.³⁵⁴

The potential for recovery of bone lost during disuse, both in the diaphyseal cortical and metaphyseal cancellous bone, was evaluated in young adult and old beagle dogs.⁸³ Following immobilization of a forelimb for up to 32 weeks, there was considerable recovery of the original bone loss during remobilization. In both age groups, the residual deficits increased with the duration of immobilization and were similar in the metaphysis and in the diaphysis. In addition, the distal, weight-bearing bones tended to show greater losses and also greater recovery both in diaphyseal and metaphyseal bone. The older dogs had greater residual deficits, most evident in the diaphysis. Following 32 weeks of immobilization and 28 weeks of remobilization, a 50% loss in the third metacarpal diaphysis of younger dogs immediately following the immobilization period decreased to 15% (a 70% recovery), whereas older dogs had a 38% loss that decreased to 23% (a 40% recovery). In contrast, immobilization for 6 or 12 weeks resulted in complete

recovery of bone after remobilization of 10 or 28 weeks, respectively. In fact, bone volume of third metacarpal trabecular bone exceeded the original control values in some dogs.

The stages of bone loss that young adult dogs undergo in response to cast immobilization have been discussed.³³³ The response to remobilization may be similarly divided into stages. In stage I, in which there is a rapid initial loss of bone following 6 weeks of immobilization, there is near full recovery of bone mass within 8 to 12 weeks of cast removal. Radiographic evidence of periosteal stress reactions of the metacarpal bones, indicated by external bone formation in the distal metacarpal metaphysis, was present in 60% of younger dogs, but none of the older dogs following 12 weeks of immobilization and up to 28 weeks of remobilization.³³¹ In stage II, bone loss slows, but recovery is also slower following 24 to 32 weeks of immobilization. During the remobilization period, young and old dogs immobilized for 32 weeks may have periosteal stress reactions of the metacarpal bones, indicated by external bone formation in the distal metacarpal metaphysis.³³¹ Although there was no evidence of a break in bone continuity, micro-damage (mechanical fatigue) may have occurred at these sites, and periosteal stress-induced reactions may occur at sites of increased stress in the absence of actual fractures. Continued stress to bones during the remobilization period may indeed lead to stress reactions. Immobilization for longer than 32 weeks characterizes stage III, in which the loss of bone mass remains at 30% to 50% of original values, despite remobilization.

Complete recovery of trabecular bone following immobilization may be limited by changes in bony architecture.³⁵⁴ Following 18 weeks of immobilization, trabecular area of rat tibias increased by 42% and 51% following 10 and 20 weeks of remobilization. Remobilization resulted in increased bone area by decreasing porosity during the first 10 weeks of remobilization, but little further increase occurred between 10 and 20 weeks of remobilization. The recovery of bone loss was mainly the result of an increase in trabecular width. In fact, there was no recovery of the number of trabeculae lost during immobilization. Remobilization added bone on existing surfaces, but was not anabolic enough to reconnect trabeculae. This may explain why there may be a persistent loss of bone following long periods of immobilization, despite a long remobilization period.

Although it is clear that a simple period of free exercise during the remobilization period may not result in complete recovery of bone following immobilization, high-intensity training following immobilization may result in more complete recovery. One study of rats undergoing limb immobilization and subsequent remobilization indicated that high-intensity treadmill running resulted in restored BMC and BMD of the affected limb, whereas animals undergoing free activity had BMC and BMD

values less than those of controls.^{355,356} The expression of osteocalcin, a marker of osteoblastic activity, corresponded to bone formation during the remobilization period.³⁵⁷ When the training period ceased, however, the beneficial effects of training were lost. It appears that intensified remobilization is necessary to restore bone mineral after disuse, but the benefits may be lost if activity is terminated.

A period of mild treadmill activity following free remobilization may also be beneficial for dogs with immobilized limbs. Forelimb immobilization of 1- to 2-year-old dogs for 16 weeks, followed by a remobilization period of 16 weeks of kennel confinement and 16 weeks of treadmill exercise administered three times per week, resulted in the return of cortical and cancellous BMD and mechanical properties to essentially normal levels.⁸²

Other activities may also be beneficial after a period of immobilization. The effects of jump exercise during remobilization were examined in rats undergoing immobilization by tail suspension.³⁵⁸ Comparisons were made with nonsuspended controls, suspended controls, and suspended controls with spontaneous recovery. The jump exercise protocol consisted of 10 jumps/day, 5 days/week, and the jump height was 40 cm. Tail suspension induced a 13.6% decrease in total femoral BMD and marked deterioration of trabecular architecture. After 5 weeks of free remobilization, femoral BMD, calf muscle weight, and body weight returned to age-matched control levels, but trabeculae remained thinner and less connected. On the other hand, rats undergoing jump training after immobilization showed significant increases in trabecular thickness, number, and connectivity compared with suspended rats with spontaneous recovery.

Despite the positive effects of training on bone recovery after a period of immobilization, some caution is necessary when instituting a training program. One experiment evaluated the effects of immobilization and remobilization on mechanical properties of femoral shafts of rats.³⁵⁹ Rats had the right hindlimb immobilized for 3 weeks by taping and were compared with controls. Immobilization was followed by 4 weeks of free remobilization. Immobilization followed by remobilization resulted in lower maximum bending moment, yield moment, stiffness, and work to failure compared with controls. Remobilization resulted in worsening of properties of femora. The results suggested that the strength of bone decreases during the first period of free remobilization and may be associated with increasing brittleness of cortical bone. Similar findings have been obtained in another study of rats.³⁶⁰

Adaptation of Bone to Increased Stress

The bone response to mechanical loading is proportional to an applied load of given magnitude and frequency of application. There are also proportional increases in second messengers, growth factors, bone matrix, and bone strength

with loading.³⁶¹ Animal models using dogs, turkeys, mice, rats, chickens, and pigs have demonstrated a threshold-response and dose-response relationship to mechanical loading of bone, either by mechanical means or exercise. In general, relatively low-intensity training may have no or only mild effects on bone density or bone growth, whereas higher-intensity training results in greater bone density, but may inhibit growth in bone length and girth in growing animals.⁵⁸

Despite much knowledge regarding the influence of physical activity on bone, precise information regarding the type, intensity, frequency, and duration of exercise on bone remains elusive. The remodeling of bone is slow, taking several months to complete. The age at which training begins is important. Initiation of activity early in life appears to be the most beneficial.³⁶² Within limits, loading of growing bone leads to a better result than loading of adult bone. The production of prostaglandins and growth factors may modulate the process of bone modeling. Although the optimal pattern of bone loading is unknown, evidence suggests that bone should be loaded with high peak forces and strain rates (high impacts), create strain throughout the bony column, consist of relatively few repetitions, and be long term and progressive. In humans, weightlifters have the greatest bone mass, followed by throwers, runners, soccer players, and swimmers. This suggests that weight and strength training appears to provide the greatest effects on bone, and swimming the least, perhaps because the buoyancy of water alters the load placed on bone as compared with weight lifting on land.

Resistance exercise training may also affect bone of animals. Rats conditioned to press levers, facilitating full extension and full flexion of the hindlimbs while wearing a weighted vest, had an increase in cancellous bone area of the proximal tibia after 6 weeks.³⁶³ In addition, osteocalcin, trabecular number, thickness, and osteoid covering cancellous bone were greater in trained animals, suggesting that resistance training increases cancellous bone area by stimulating bone formation. A study of rats undergoing exercise with or without sudden impact loading of the femur indicated that cortical wall thickness increased with any activity, and breaking strength of the affected bones was greater in bones undergoing impact loading as compared with sedentary controls.³⁶⁴ In spite of the enhanced biomechanical properties of impact-loaded bones, there was little difference in BMC in the exercised groups, suggesting that significant improvements may occur in biomechanical properties with little change in bone mass with impact loading.

In dogs, mild increases in loading of bones have little effect on bone mass or bone density. Impact loading or chronic moderate increases in loading result in increased bone mass and bone density. A study of mature beagle dogs with an ostectomy of a radius showed that there was little change in bone mass and density of the contralateral